Lens2Mesh

Complete User Manual

Version 1.0.5 | English edition

From first project to final export

All menus and submenus, processing dialogs, dropdown choices, reference settings and export controls - with practical guidance on what each option does and when to choose it.

Use this manual in two ways

Start with Getting started and A complete production workflow if you are new. Use the clickable contents and PDF bookmarks to look up an exact menu, window or setting when you are working.

Screenshot quality

Interface screenshots are native HiDPI PNG captures embedded losslessly. Long dialogs are divided into readable groups. They document actual controls rather than a synthetic reconstruction result.

Version scope

Updated for Lens2Mesh 1.0.5. Changed controls are captured from the actual Windows Qt application; unchanged reference content is retained from the approved complete edition. File dialogs and styling can vary. Controls dependent on missing data or hardware may be disabled.

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Install, launch and prepare the first project

The distributed Windows Setup includes its Python environment, packages and reconstruction engine. Users of the installer do not need to install Git or Python separately. Download the complete corrected 1.0.5 installer from the official release, rather than copying an executable out of an installed folder.

System and drivers

The distributed installer targets 64-bit Windows 10/11. A working OpenGL-capable driver is needed for 3D display. CUDA dense reconstruction additionally requires compatible NVIDIA hardware, driver and engine. Memory needs depend heavily on photo count/resolution and chosen products; no single RAM value guarantees every survey.

Download and verify

Download Setup and its matching SHA-256 checksum from the release. In PowerShell, Get-FileHash -LiteralPath followed by the quoted Setup path and -Algorithm SHA256 computes its hash. Compare the complete value to the matching checksum. A matching hash checks integrity, not a geometric or security certification.

Install and launch

Install Lens2Mesh_Setup_1.0.5.exe first; it includes Python, engines and offline models. Close Lens2Mesh before installing/updating and choose the existing folder to upgrade. Without the optional GPU add-on, ALIKED and LightGlue automatically use CPU. For their GPU acceleration, install Lens2Mesh_NeuralGPU_Addon_1.0.5.exe into the same folder with the app closed. Standard COLMAP GPU processing remains in the base package and requires a compatible NVIDIA driver. No CUDA Toolkit installation is needed. Keep surveys outside the program directory; launch from the Start menu or desktop shortcut.

First launch check

Open About and confirm 1.0.5. Open Preferences to inspect theme, engine detection and GPS policy. A CPU-only system can align and mesh existing points, but the CUDA depth-map route remains unavailable without suitable hardware.

Working offline

After the complete installer and any required compatible model/runtime bundles are available, processing can operate offline. Optional neural routes need their required offline weights; the program does not create missing model bundles simply by selecting a feature method.

Create the first survey

Make a dedicated local project folder, choose New Project, add the intended photos and Save. Keep originals separate from generated .files data. Start with a manageable photo subset that actually covers one connected scene, not unrelated images.

Update and uninstall

Preserve project files and originals outside the application directory. To update, close the app and run the intended complete Setup. Removing/replacing application files is separate from deleting a survey; keep project backups and confirm the newly installed version before continuing.

Acquire photos that can reconstruct

Software settings cannot recover geometry that photographs do not constrain. Before experimenting with expensive processing, inspect whether each surface is sharp, textured and observed from sufficiently different positions.

Overlap and viewpoint change

Each important patch needs repeated visibility in overlapping images with useful baseline. Turning a camera around a fixed optical centre supplies little triangulation depth. Move between viewpoints while keeping shared features visible.

Sharpness and stable camera settings

Avoid motion blur and large exposure/focus/zoom changes within a calibration group. Preserve original files and metadata. Do not resize/crop photos after marking or alignment without rebuilding the corresponding geometry/observations.

Texture and reflections

Blank painted walls, glossy floors, glass, mirrors and repeated tiles can create missing or false matches. Capture surrounding distinct features and additional oblique views. Higher processing quality alone cannot make a blank or view-dependent reflection a reliable 3D feature.

Scene consistency

Avoid people, moving objects and large illumination changes. Different photos should describe the same stable surface. Ghosting filters are a limited mitigation, not a replacement for a consistent acquisition.

Reference distribution

Distribute measured control and independent checks across the scene and height range rather than clustering them in one corner. Use clearly identifiable targets. More observations with almost parallel rays do not necessarily improve depth precision.

Inspect an initial alignment

Look at camera layout and sparse geometry before building dense products. A complete registered count can still contain a folded room, false loop closure or wrong connection. Fix those causes first.

Getting started: choose your deliverable

Lens2Mesh reconstructs overlapping photographs into camera poses, tie points and surface products. Choose the deliverable first: a textured model for 3D viewing, a point cloud for geometry analysis, a DSM for surface heights, or an orthomosaic for a plan image. These products have different inputs and accuracy requirements. This manual describes the published, corrected 1.0.5 application; it does not assume that every workflow needs every processing step.

New user: a first project

Create a project, add a small sharp overlapping photo set, save it, run Align Photos and inspect the cameras and tie points. Continue to a dense cloud only after the reconstruction is coherent. Build a mesh and texture for 3D output. Save after each useful result.

Aerial mapping

Use overlapping nadir photographs with cross-strip coverage where possible. Inspect GPS availability and coordinate/height references. Align, optimize with appropriate measured reference, build a dense cloud, then a DEM or mesh and orthomosaic. Validate with independent check points before calling the result survey-grade.

A room or walk-around object

Photograph from different positions, not only by turning at one spot. Include textured features and views of corners joining the surfaces. Incremental camera estimation is a useful starting choice. GPS inside a room is generally unsuitable as a strong reference; review its validity before enabling it.

A textured 3D model

Use mesh plus Build Texture, then export a textured format. A clean texture can hide poor geometry: inspect shaded and wireframe views too. Orthophoto mapping is mainly for surfaces that can be projected without vertically overlapping parts.

A height raster or CAD contours

Establish a valid orientation and independently measured scale first. Build DEM produces a surface model, not automatically bare earth. Mesh contours follow model Z, so an incorrectly oriented or unscaled model gives unsuitable contour elevations or intervals.

Hardware route

Alignment can run on CPU. The Build Point Cloud depth-map route requires a compatible NVIDIA CUDA engine. Point-cloud meshing, texturing and other stages have different dependencies; a GPU preference for feature extraction is not a general CPU/CUDA switch for every operation.

Project files, units and safe iteration

A project is more than its .pmsx file. The metadata file points to photos and processing products, normally stored beside it in a matching .files folder. Keep an untouched baseline when comparing settings; rebuilding products is different from changing their display.

Recommended folder layout

Keep a project folder containing Survey.pmsx, Survey.files and a separate Photos folder. Use a local disk with enough free space for temporary and final data. Source photos must remain reachable; the project file is not a universal archive of every input.

Save and Save As

Save writes current project metadata. Save As can offer to copy working data to the new project location. Copy results only omits recomputable depth maps; Copy everything includes them. Keeping data in its existing location leaves an external dependency. Review the exact copy dialog before choosing.

Local coordinates

A reconstruction without independent reference has arbitrary scale, orientation and origin. Values displayed as model units are not automatically metres. Measuring a distance in that same model cannot independently calibrate its scale.

Geographic and projected coordinates

For WGS 84 longitude/latitude, X is longitude and Y is latitude in degrees. UTM uses an appropriate zone/hemisphere and metric easting/northing. Use mutually compatible horizontal and vertical references for cameras, markers and imported shapes. The app does not supply a general geoid-height conversion.

Stale products

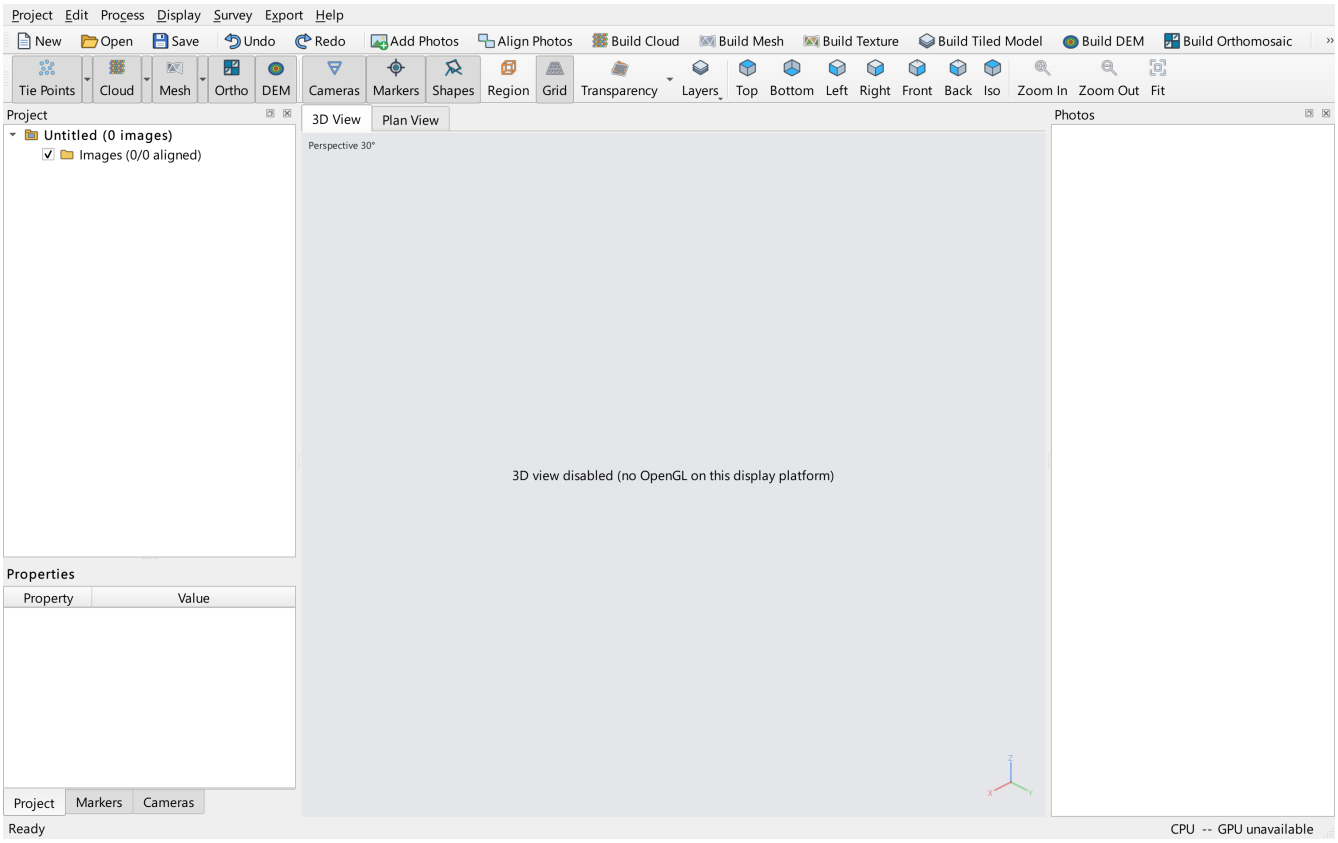
Moving observations or optimizing cameras can change geometry. Read Update required or stale-source messages. Rebuild dependent depth maps/clouds, meshes, textures and rasters as applicable; simply redisplaying an old result does not update its calculations.

What Undo covers

The Edit Undo/Redo handlers restore marker edits. Do not treat them as a global recovery system for processing, data removal or product replacement. Save a separate baseline for destructive filtering or rebuilding experiments.

Workspace, navigation and toolbar

Project, Markers and Cameras share the left dock area; Photos is normally at the right. The centre provides 3D View and Plan View. A visible layer checkbox is a display choice, not automatically a processing constraint.



Actual interface reference: Workspace, navigation and toolbar. See the option explanations below; screenshot values are not universal recommendations.

Project tree and Properties

Expand Images, Markers and generated products to inspect the project. Select an item to read its Properties, such as point/face count, processing settings, file location, time or stale status. A product display checkbox shows/hides it; it does not recompute it.

3D View

Inspect cameras, tie points, dense clouds, meshes, tiled models and shapes. Use the toolbar orientation and Fit commands to recover the view. The renderer supports orbit/zoom navigation; mouse interaction can depend on platform and modifiers. The explicit view commands are the reliable reference.

Plan View

A shared view for source photographs and raster products. Select a photo to inspect or mark it; select Ortho or DEM to return to that raster. Choosing a photograph clears the active Ortho/DEM selection, so choosing the raster again brings it back.

Photos

Thumbnails open source photos. Multi-selection is used for photo actions. A marker filter can restrict visible photos; Reset Filter restores them. A missing thumbnail alone does not establish that alignment failed: inspect the photo path and logs.

Processing toolbar

New, Open, Save, Undo, Redo, Add Photos, Align Photos, Build Cloud, Build Mesh, Build Texture, Build Tiled Model, Build DEM, Build Orthomosaic and Settings call the corresponding menu operations. Their prerequisites and options are described in the reference chapters.

Tie Points / Cloud / Mesh split buttons

Click the main part to show or hide the layer. Click its arrow to choose a rendering style. Turning a layer off retains the selected style. Display choices are shared with the Display menu and Project tree.

Ortho / DEM / Cameras / Markers / Shapes

These toolbar buttons select raster products or toggle overlays. Marker visibility is separate from whether marker measurements enter optimization. Hiding Cameras does not remove or disable source photographs.

Region / Grid / Layers

Region and Grid control visual aids. Layers contains additional tiled model, labels, trackball and projection/display options. Output-region bounds in DEM/orthomosaic dialogs are separate numeric settings.

Transparency

Toggle a translucent surface view to inspect overlaid shapes. The arrow opens Surface opacity (5-100%, default 40%): lower values are more transparent. The mode can affect models, clouds, photos and orthomosaics. It changes presentation, not reconstructed geometry.

Top / Bottom / Left / Right / Front / Back / Iso

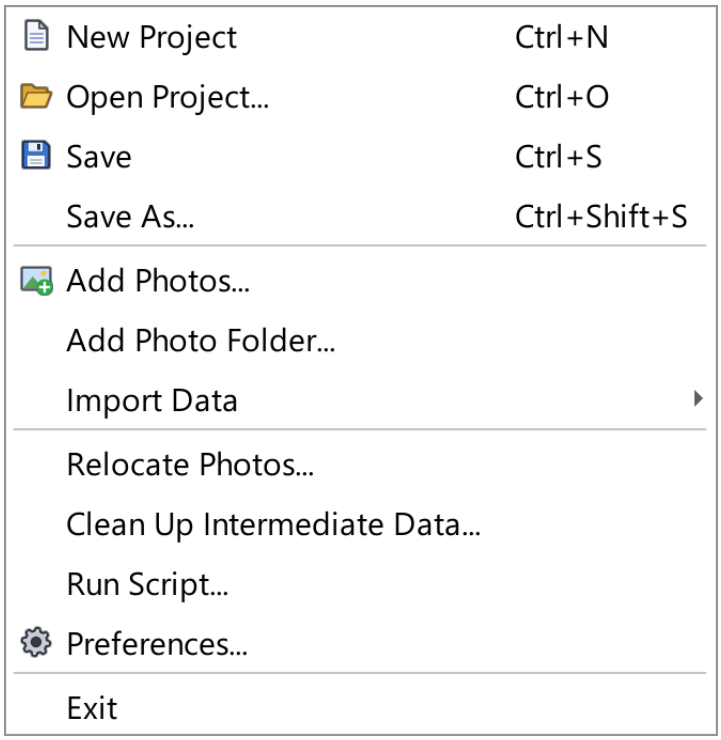
Choose a fixed 3D viewing direction. Zoom In, Zoom Out and Fit adjust framing. View orientation alone does not rotate the saved survey coordinate system.

Console / Jobs / status

Console shows log output and provides a Python command input; Jobs records work entries. They can open as floating panels. CPU/GPU status values describe current resource use, not geometric quality.

Menu: Project

Visible 1.0.5 commands. Availability depends on the selected data, processing prerequisites and compatible hardware.



Actual interface reference: Menu: Project. See the option explanations below; screenshot values are not universal recommendations.

New Project

Create a new empty project. Save or discard current edits when prompted. Use a separate project for a separate survey; this release exposes one survey data group per project.

Open Project

Open an existing .pmsx project. The associated processing files and original photos must still be reachable. If paths changed, use Relocate Photos rather than creating a second set of duplicate images.

Save

Write current project metadata. Processing requires a saved project location. A successful save does not prove that every external photo or product has been copied into the project folder.

Save As

Save under a new name/location. Read the project-data copy choices. Copying results and depth maps can require substantial disk space; keeping existing locations leaves links to those locations.

Add Photos

Select individual supported image files. Use it when the desired photographs are mixed with unrelated files. Review sharpness, overlap and camera/lens changes before alignment.

Add Photo Folder

Add photos from a selected folder. Organize the intended survey set first. This is an input-selection shortcut, not an automatic guarantee that the images cover one reconstructable scene.

Import Data

Open the Import Data submenu. Each command is explained in its own entries immediately following this menu.

Relocate Photos

Repair moved source-photo links by selecting the new location. Review missing or ambiguous matches. Relocation preserves the intended photo identities; it is not a replacement for re-aligning edited/resized originals.

Clean Up Intermediate Data

Choose recomputable working-data categories to remove and inspect the estimated size. Finished products and source photos are protected by this operation. Deleted intermediates may have to be rebuilt for later processing.

Run Script

Choose a Python script to execute with application/project access. Use scripts you understand and trust; they can change project data and access files. This is an advanced automation facility, not a processing preset. Read, syntax and execution failures include a traceback. Scripts run on the GUI thread; avoid blocking it with long custom processing.

Preferences

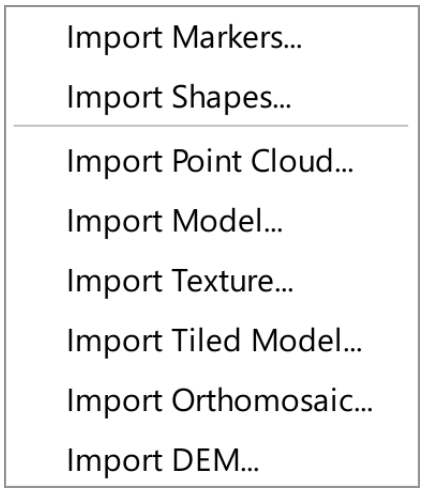
Open General, GPU and Advanced settings. The project-wide Use Photo GPS policy affects later matching/reference use; theme and default view affect presentation. See the complete preference reference.

Exit

Close the application, responding to unsaved-change prompts. Wait for or cancel active work first. Closing a window is not a substitute for saving a finished project.

Menu: Project > Import Data

Visible 1.0.5 commands. Availability depends on the selected data, processing prerequisites and compatible hardware.



Actual interface reference: Menu: Project > Import Data. See the option explanations below; screenshot values are not universal recommendations.

Import Markers

Read marker/reference XML, TXT or CSV and merge markers by identity. A declared coordinate system may update the project reference. Check labels, axis order, units and photo observation links after import.

Import Shapes

Read supported vector data and choose its coordinate system, layer and height treatment. Use this for boundaries or annotations from GIS/CAD; the shape import chapter explains each transformation/draping option.

Import Point Cloud

Import a PLY point cloud as a project product. This does not create image alignment or camera poses. Confirm scale and frame match the project before combining it with other products.

Import Model

Import a PLY mesh. The application reads its vertex/face counts and displays it. Replacing the mesh invalidates an older texture atlas that belonged to a different surface.

Import Texture

Attach an existing compatible texture product to the model. Use the expected texture directory and matching geometry/UV data; an unrelated JPEG alone is not a calibrated texture product.

Import Tiled Model

Load a supported existing tiled-model product. Keep its index and referenced content together. Importing a tileset does not reconstruct the original source photos.

Import Orthomosaic

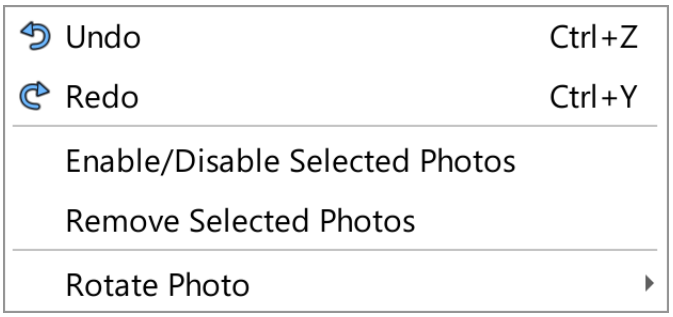
Load a supported raster orthomosaic. Verify its coordinate system, bounds and pixel size. A plain picture without placement information cannot establish reliable world coordinates.

Import DEM

Load an elevation raster. Confirm that its heights and coordinate reference match the project. Colour imagery and numerical elevation values are different data products.

Menu: Edit

Visible 1.0.5 commands. Availability depends on the selected data, processing prerequisites and compatible hardware.



Actual interface reference: Menu: Edit. See the option explanations below; screenshot values are not universal recommendations.

Undo

Undo a supported marker edit, including the stored marker snapshot. It is not a universal reversal for alignment, filtering or file deletion. Keep saved baselines for processing experiments.

Redo

Reapply a marker edit that was undone. A new edit may replace the available redo history. Verify the marker coordinates and observations after restoring a change.

Enable/Disable Selected Photos

Toggle inclusion of selected photo entries. This changes processing participation, not merely camera-icon visibility. Reconsider the selection before a new alignment; disabling GPS is a separate project setting.

Remove Selected Photos

Remove selected photo entries from the project. Read the confirmation and save a baseline. This is different from hiding camera overlays and should not be treated as undoable processing cleanup.

Rotate Photo

Open the Rotate Photo submenu. Each command is explained in its own entries immediately following this menu.

Menu: Edit > Rotate Photo

Visible 1.0.5 commands. Availability depends on the selected data, processing prerequisites and compatible hardware.



Actual interface reference: Menu: Edit > Rotate Photo. See the option explanations below; screenshot values are not universal recommendations.

Rotate Right

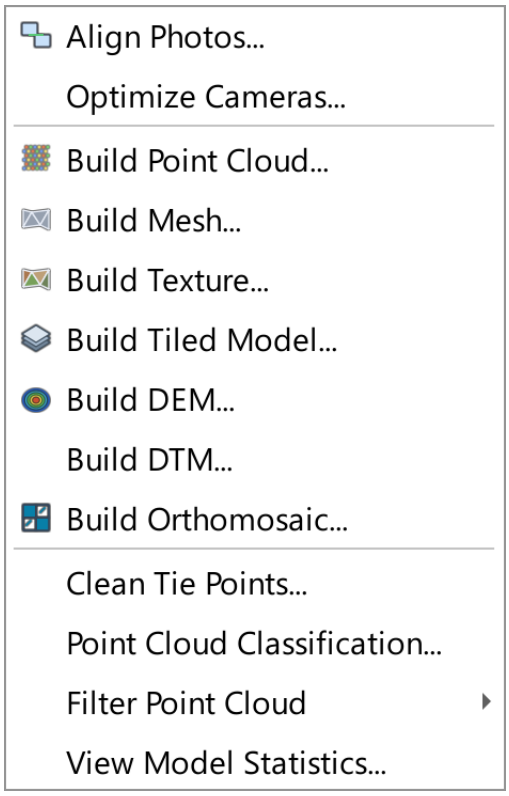
Rotate the displayed photo by 90 degrees clockwise. Use for comfortable viewing/marking; it is not a new camera calibration or a geometric rotation of the reconstructed scene.

Rotate Left

Rotate the displayed photo by 90 degrees counterclockwise. Check source-photo observations in the displayed orientation; the reconstruction uses the appropriate original photo coordinates.

Menu: Process

Visible 1.0.5 commands. Availability depends on the selected data, processing prerequisites and compatible hardware.



Actual interface reference: Menu: Process. See the option explanations below; screenshot values are not universal recommendations.

Align Photos

Detect features, match image pairs and estimate cameras/tie points. A successful run replaces the alignment; failure preserves the previous valid result. Marker constraints are handled later through Optimize Cameras.

Optimize Cameras

Apply chosen marker/GPS reference constraints and optionally bundle adjustment/calibration fitting. Use realistic uncertainty values and compatible coordinate systems. Rebuild downstream products when camera geometry changes.

Build Point Cloud

Create CUDA depth maps and fuse a dense cloud. Requires aligned photos and a compatible CUDA engine. Quality, support and isolation filters have separate effects; see all options in the dense-cloud chapters.

Build Mesh

Build a mesh from a point cloud or the Delaunay fused-cloud/visibility route. Choose surface trimming and target faces according to geometry, memory and output needs. The menu says Build Mesh; the dialog title can say Build Model.

Build Texture

Sample aligned photographs onto the mesh texture atlas. Choose mapping, blending, page size/count and source-photo resolution. Texturing does not fix incorrect camera geometry or holes in the mesh.

Build Tiled Model

Build spatial tiles from a model or cloud. Model tiles support b3dm/glb and point tiles use pnts. Tile limits affect viewing granularity and memory; preserve the complete hierarchy when exporting.

Build DEM

Rasterize a mesh or point cloud into a surface-height grid. Choose source, resolution, interpolation and output bounds. It is a DSM workflow, not an automatic ground-classification/DTM tool.

Build DTM

Build a separate terrain raster from classified ground (class 2). Inspect the classification first. This does not replace the DSM; see the DTM settings reference.

Build Orthomosaic

Project photos onto a suitable mesh or DEM to form a plan raster. Choose surface, resolution, blending and bounded filling. Validate the reference frame and actual image detail before treating it as map data.

Clean Tie Points

Open Clean Tie Points / Gradual Selection. Select a metric and threshold, inspect the number selected, then Delete selected. Removing too many tie points can weaken coverage; work on a saved baseline.

Point Cloud Classification

Estimate ground and non-ground in a separate cloud, retaining the original. Review up direction, source units and terrain thresholds in the classification reference.

Filter Point Cloud

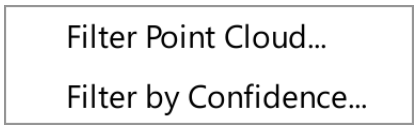
Open the Filter Point Cloud submenu. Each command is explained in its own entries immediately following this menu.

View Model Statistics

Display summary information about the current model. Counts and statistics help inspect complexity; they do not establish independent dimensional accuracy or completeness.

Menu: Process > Filter Point Cloud

Visible 1.0.5 commands. Availability depends on the selected data, processing prerequisites and compatible hardware.



Actual interface reference: Menu: Process > Filter Point Cloud. See the option explanations below; screenshot values are not universal recommendations.

Filter Point Cloud

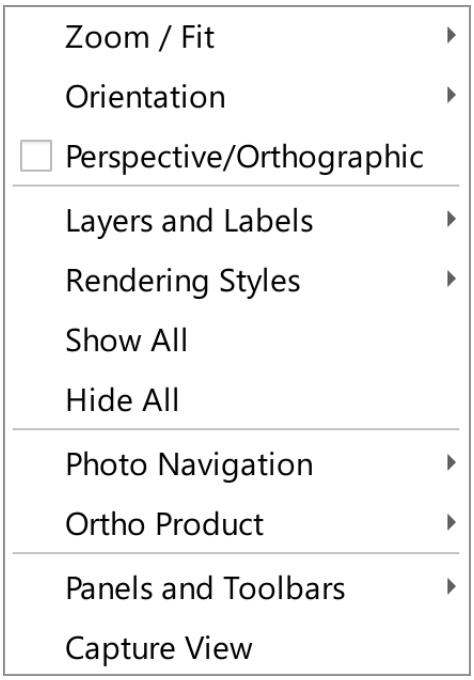
Filter an existing cloud by confidence, statistical neighbourhoods, radius and optional terrain assumptions. This modifies the cloud. Compare removals with the geometry you need to preserve.

Filter by Confidence

Keep points within the selected supporting-photo confidence interval. This simplified dialog hides the other filters. Check whether the imported/generated cloud actually contains confidence information.

Menu: Display

Visible 1.0.5 commands. Availability depends on the selected data, processing prerequisites and compatible hardware.



Actual interface reference: Menu: Display. See the option explanations below; screenshot values are not universal recommendations.

Zoom / Fit

Open the Zoom / Fit submenu. Each command is explained in its own entries immediately following this menu.

Orientation

Open the Orientation submenu. Each command is explained in its own entries immediately following this menu.

Perspective/Orthographic

Switch the 3D projection. Perspective gives depth cues; orthographic is useful for visual comparison without perspective size changes. Neither mode establishes a measured scale.

Layers and Labels

Open the Layers and Labels submenu. Each command is explained in its own entries immediately following this menu.

Rendering Styles

Open the Rendering Styles submenu. Each command is explained in its own entries immediately following this menu.

Show All

Turn on available display layers. This can make a dense scene visually cluttered or slower to draw; disable unnecessary layers after inspection.

Hide All

Turn off display layers without deleting their data. Use individual layer controls to show the specific product or overlay you need.

Photo Navigation

Open the Photo Navigation submenu. Each command is explained in its own entries immediately following this menu.

Ortho Product

Open the Ortho Product submenu. Each command is explained in its own entries immediately following this menu.

Panels and Toolbars

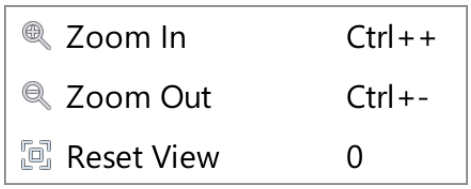
Open the Panels and Toolbars submenu. Each command is explained in its own entries immediately following this menu.

Capture View

Save an image of the current view. Use for illustration or communication; use the data-export commands for coordinates, full raster resolution or mesh geometry.

Menu: Display > Zoom / Fit

Visible 1.0.5 commands. Availability depends on the selected data, processing prerequisites and compatible hardware.



Actual interface reference: Menu: Display > Zoom / Fit. See the option explanations below; screenshot values are not universal recommendations.

Zoom In

Magnify the active view. This is navigation only and does not add resolution to the model, image or raster.

Zoom Out

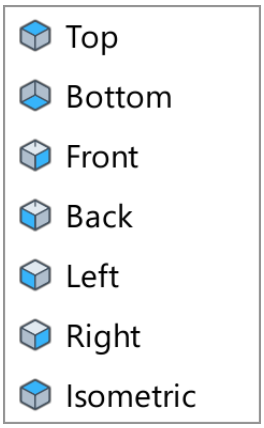
Reduce magnification of the active view. Use Fit/Reset View if the scene is difficult to find.

Reset View

Fit/reset the active view to recover useful framing. It does not reset alignment, camera calibration or survey reference coordinates.

Menu: Display > Orientation

Visible 1.0.5 commands. Availability depends on the selected data, processing prerequisites and compatible hardware.



Actual interface reference: Menu: Display > Orientation. See the option explanations below; screenshot values are not universal recommendations.

Top

Set the 3D view to top orientation. Use it to inspect shape, coverage or surface alignment. It changes the viewing direction, not the stored coordinate frame.

Bottom

Set the 3D view to bottom orientation. Use it to inspect shape, coverage or surface alignment. It changes the viewing direction, not the stored coordinate frame.

Front

Set the 3D view to front orientation. Use it to inspect shape, coverage or surface alignment. It changes the viewing direction, not the stored coordinate frame.

Back

Set the 3D view to back orientation. Use it to inspect shape, coverage or surface alignment. It changes the viewing direction, not the stored coordinate frame.

Left

Set the 3D view to left orientation. Use it to inspect shape, coverage or surface alignment. It changes the viewing direction, not the stored coordinate frame.

Right

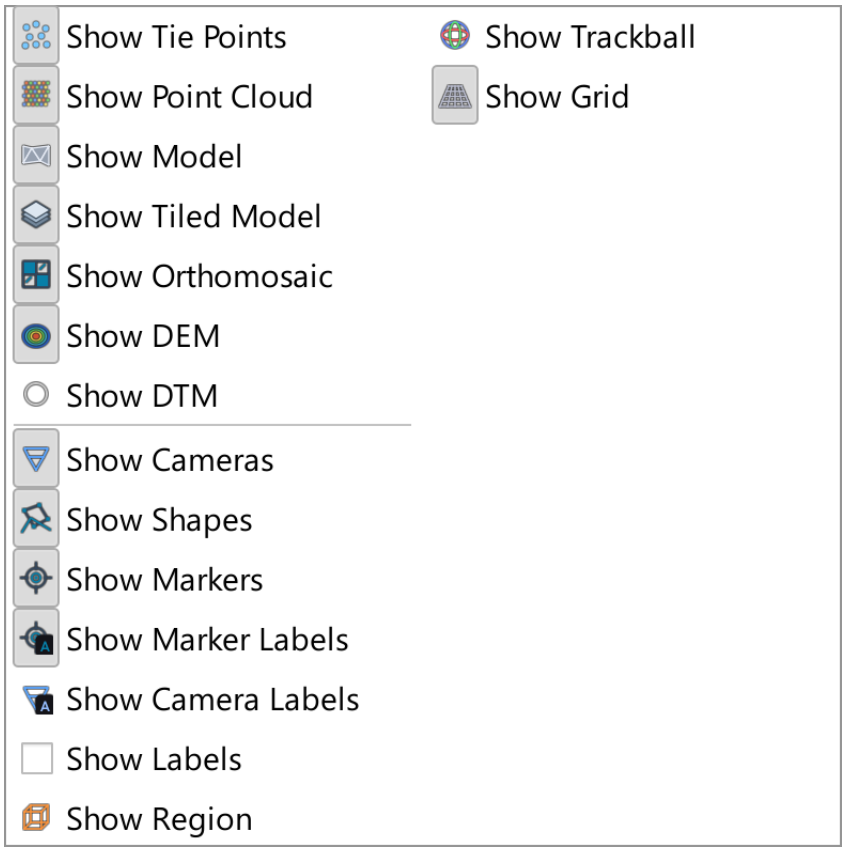
Set the 3D view to right orientation. Use it to inspect shape, coverage or surface alignment. It changes the viewing direction, not the stored coordinate frame.

Isometric

Set the 3D view to isometric orientation. Use it to inspect shape, coverage or surface alignment. It changes the viewing direction, not the stored coordinate frame.

Menu: Display > Layers and Labels

Visible 1.0.5 commands. Availability depends on the selected data, processing prerequisites and compatible hardware.



Actual interface reference: Menu: Display > Layers and Labels. See the option explanations below; screenshot values are not universal recommendations.

Show Tie Points

Toggle display of sparse tie points. Use it to isolate the information needed for inspection. Hiding this layer does not delete it or remove its measurements from optimization.

Show Point Cloud

Toggle display of dense/imported cloud. Use it to isolate the information needed for inspection. Hiding this layer does not delete it or remove its measurements from optimization.

Show Model

Toggle display of mesh. Use it to isolate the information needed for inspection. Hiding this layer does not delete it or remove its measurements from optimization.

Show Tiled Model

Toggle display of tiled product. Use it to isolate the information needed for inspection. Hiding this layer does not delete it or remove its measurements from optimization.

Show Orthomosaic

Toggle display of orthomosaic. Use it to isolate the information needed for inspection. Hiding this layer does not delete it or remove its measurements from optimization.

Show DEM

Toggle display of height raster. Use it to isolate the information needed for inspection. Hiding this layer does not delete it or remove its measurements from optimization.

Show DTM

Show or hide the separate ground-only raster. Build DTM first; DSM and DTM remain separate products.

Show Cameras

Toggle display of camera frusta. Use it to isolate the information needed for inspection. Hiding this layer does not delete it or remove its measurements from optimization.

Show Shapes

Toggle display of vector overlays. Use it to isolate the information needed for inspection. Hiding this layer does not delete it or remove its measurements from optimization.

Show Markers

Toggle display of marker symbols. Use it to isolate the information needed for inspection. Hiding this layer does not delete it or remove its measurements from optimization.

Show Marker Labels

Toggle display of marker names. Use it to isolate the information needed for inspection. Hiding this layer does not delete it or remove its measurements from optimization.

Show Camera Labels

Toggle display of camera names. Use it to isolate the information needed for inspection. Hiding this layer does not delete it or remove its measurements from optimization.

Show Labels

Toggle marker and camera label overlays together. Use individual label toggles if only one type is needed. Label visibility changes readability, not reference use.

Show Region

Toggle display of region outline. Use it to isolate the information needed for inspection. Hiding this layer does not delete it or remove its measurements from optimization.

Show Trackball

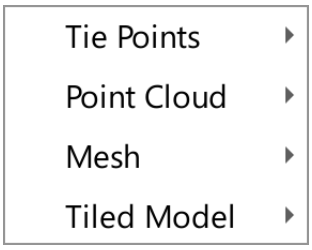
Toggle display of rotation guide. Use it to isolate the information needed for inspection. Hiding this layer does not delete it or remove its measurements from optimization.

Show Grid

Toggle display of 3D reference grid. Use it to isolate the information needed for inspection. Hiding this layer does not delete it or remove its measurements from optimization.

Menu: Display > Rendering Styles

Visible 1.0.5 commands. Availability depends on the selected data, processing prerequisites and compatible hardware.



Actual interface reference: Menu: Display > Rendering Styles. See the option explanations below; screenshot values are not universal recommendations.

Tie Points

Open the Tie Points submenu. Each command is explained in its own entries immediately following this menu.

Point Cloud

Open the Point Cloud submenu. Each command is explained in its own entries immediately following this menu.

Mesh

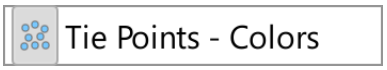
Open the Mesh submenu. Each command is explained in its own entries immediately following this menu.

Tiled Model

Open the Tiled Model submenu. Each command is explained in its own entries immediately following this menu.

Menu: Display > Rendering Styles > Tie Points

Visible 1.0.5 commands. Availability depends on the selected data, processing prerequisites and compatible hardware.



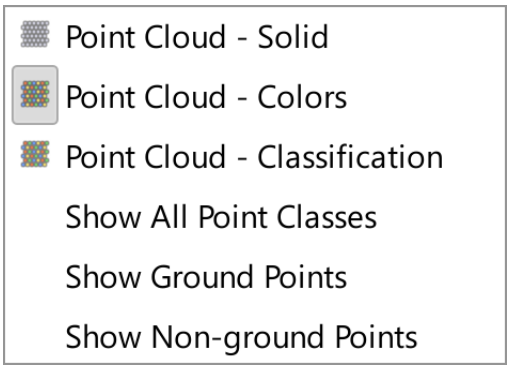
Actual interface reference: Menu: Display > Rendering Styles > Tie Points. See the option explanations below; screenshot values are not universal recommendations.

Tie Points - Colors

Display sparse points with their recorded colours. Choose it for inspection or presentation; style selection is not a rebuild. Textured modes require suitable colour data; elevation colours are not a new height measurement.

Menu: Display > Rendering Styles > Point Cloud

Visible 1.0.5 commands. Availability depends on the selected data, processing prerequisites and compatible hardware.



Actual interface reference: Menu: Display > Rendering Styles > Point Cloud. See the option explanations below; screenshot values are not universal recommendations.

Point Cloud - Solid

Display cloud points with a uniform colour. Choose it for inspection or presentation; style selection is not a rebuild. Textured modes require suitable colour data; elevation colours are not a new height measurement.

Point Cloud - Colors

Display cloud points with their recorded colours. Choose it for inspection or presentation; style selection is not a rebuild. Textured modes require suitable colour data; elevation colours are not a new height measurement.

Point Cloud - Classification

Colour the cloud by its stored classes. This changes display, not classification.

Show All Point Classes

Display all stored classes without reclassification or deletion.

Show Ground Points

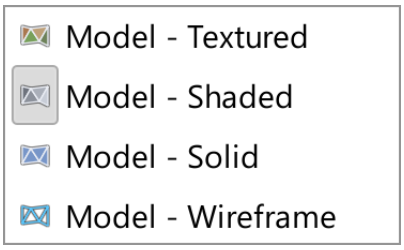
Display ground points (class 2). Hidden points remain in the source.

Show Non-ground Points

Display non-ground points (class 1). This is a visibility filter, not deletion.

Menu: Display > Rendering Styles > Mesh

Visible 1.0.5 commands. Availability depends on the selected data, processing prerequisites and compatible hardware.



Actual interface reference: Menu: Display > Rendering Styles > Mesh. See the option explanations below; screenshot values are not universal recommendations.

Model - Textured

Display the mesh with its current texture. Choose it for inspection or presentation; style selection is not a rebuild. Textured modes require suitable colour data; elevation colours are not a new height measurement.

Model - Shaded

Display the mesh with lighting that reveals surface shape. Choose it for inspection or presentation; style selection is not a rebuild. Textured modes require suitable colour data; elevation colours are not a new height measurement.

Model - Solid

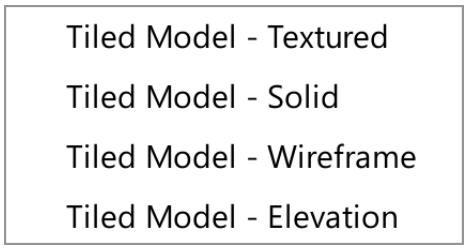
Display the mesh as a uniform solid. Choose it for inspection or presentation; style selection is not a rebuild. Textured modes require suitable colour data; elevation colours are not a new height measurement.

Model - Wireframe

Display mesh triangle edges. Choose it for inspection or presentation; style selection is not a rebuild. Textured modes require suitable colour data; elevation colours are not a new height measurement.

Menu: Display > Rendering Styles > Tiled Model

Visible 1.0.5 commands. Availability depends on the selected data, processing prerequisites and compatible hardware.



Actual interface reference: Menu: Display > Rendering Styles > Tiled Model. See the option explanations below; screenshot values are not universal recommendations.

Tiled Model - Textured

Display tiled mesh with texture. Choose it for inspection or presentation; style selection is not a rebuild. Textured modes require suitable colour data; elevation colours are not a new height measurement.

Tiled Model - Solid

Display tiled geometry with a uniform appearance. Choose it for inspection or presentation; style selection is not a rebuild. Textured modes require suitable colour data; elevation colours are not a new height measurement.

Tiled Model - Wireframe

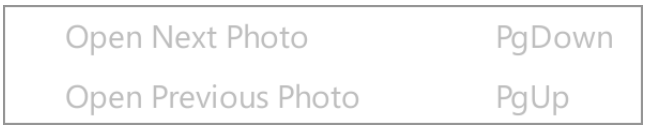
Display tile triangle edges. Choose it for inspection or presentation; style selection is not a rebuild. Textured modes require suitable colour data; elevation colours are not a new height measurement.

Tiled Model - Elevation

Display tiled geometry coloured by elevation. Choose it for inspection or presentation; style selection is not a rebuild. Textured modes require suitable colour data; elevation colours are not a new height measurement.

Menu: Display > Photo Navigation

Visible 1.0.5 commands. Availability depends on the selected data, processing prerequisites and compatible hardware.



Actual interface reference: Menu: Display > Photo Navigation. See the option explanations below; screenshot values are not universal recommendations.

Open Next Photo

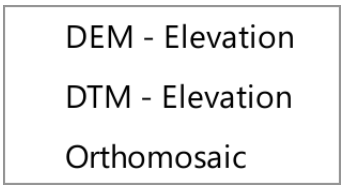
Open the next photo in the current navigation/filter context. Use for checking overlap or reviewing a marker across images.

Open Previous Photo

Open the previous photo in the current navigation/filter context. Reset a marker filter if expected photos appear missing.

Menu: Display > Ortho Product

Visible 1.0.5 commands. Availability depends on the selected data, processing prerequisites and compatible hardware.



Actual interface reference: Menu: Display > Ortho Product. See the option explanations below; screenshot values are not universal recommendations.

DEM - Elevation

Show the DEM as an elevation product in Plan View. A colour ramp visualizes heights; it is different from the photo colours in the orthomosaic.

DTM - Elevation

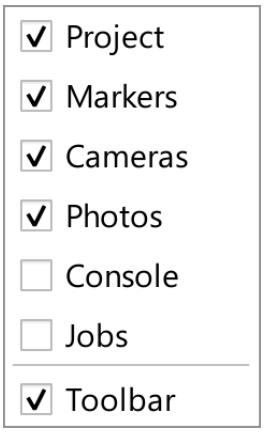
Select the DTM elevation raster in Plan View without rebuilding it.

Orthomosaic

Show the orthomosaic in Plan View. If a photo is currently displayed, this command returns the shared view to the raster product.

Menu: Display > Panels and Toolbars

Visible 1.0.5 commands. Availability depends on the selected data, processing prerequisites and compatible hardware.



Actual interface reference: Menu: Display > Panels and Toolbars. See the option explanations below; screenshot values are not universal recommendations.

Project

Show or hide the product tree and properties. Reopen it here when missing from the layout. Console and Jobs normally open as floating panels. Changing visibility preserves the underlying project data.

Markers

Show or hide marker coordinates, observations and residuals. Reopen it here when missing from the layout. Console and Jobs normally open as floating panels. Changing visibility preserves the underlying project data.

Cameras

Show or hide camera reference/estimated values. Reopen it here when missing from the layout. Console and Jobs normally open as floating panels. Changing visibility preserves the underlying project data.

Photos

Show or hide source-photo thumbnails. Reopen it here when missing from the layout. Console and Jobs normally open as floating panels. Changing visibility preserves the underlying project data.

Console

Show or hide logs and a Python command input. Reopen it here when missing from the layout. Console and Jobs normally open as floating panels. Changing visibility preserves the underlying project data.

Jobs

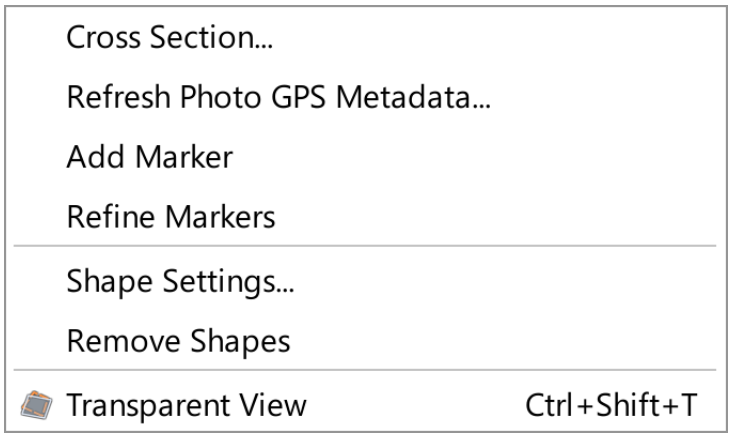
Show or hide processing entries. Reopen it here when missing from the layout. Console and Jobs normally open as floating panels. Changing visibility preserves the underlying project data.

Toolbar

Show or hide the two main toolbar rows. Reopen it here when missing from the layout. Console and Jobs normally open as floating panels. Changing visibility preserves the underlying project data.

Menu: Survey

Visible 1.0.5 commands. Availability depends on the selected data, processing prerequisites and compatible hardware.



Actual interface reference: Menu: Survey. See the option explanations below; screenshot values are not universal recommendations.

Cross Section

Open the closable bottom panel for a live section line, distance and mesh profile. Draw in 3D View, a compatible orthophoto or an aligned photo. Edit with endpoint/centre handles.

Refresh Photo GPS Metadata

Reread original-photo GPS/receiver metadata and report unreadable files. Existing model coordinates remain unchanged; apply updated references deliberately with Optimize Cameras.

Add Marker

Create a marker entry. Then place the same marker on source photos and supply measured reference coordinates when needed. A marker name alone is not a control constraint.

Refine Markers

Refine usable source-photo marker placements using image information. Review the resulting matches, especially with repeated texture or weak viewpoint separation. Predicted hints alone are not confirmed observations.

Shape Settings

Edit layer/selected-shape display: colours, line style, symbols, fill, labels and depth behavior. These settings change appearance, not source vector coordinates.

Remove Shapes

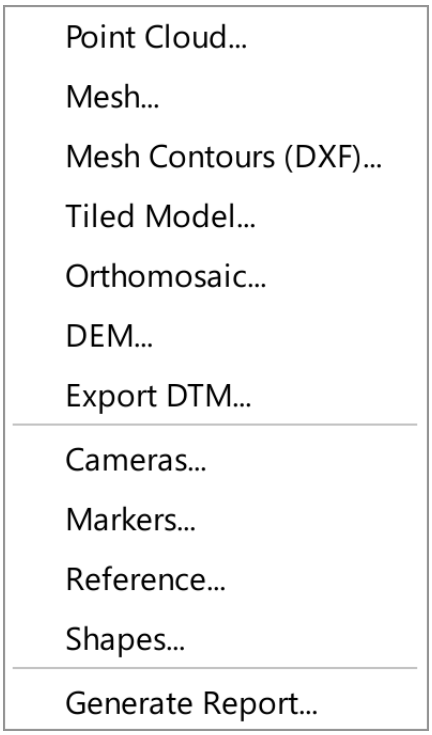
Remove imported shapes after confirmation. Use layer visibility when you only want to hide them temporarily.

Transparent View

Toggle translucent model display for shape inspection. The toolbar provides opacity control. This does not recalculate the model or alter exported geometric coordinates.

Menu: Export

Visible 1.0.5 commands. Availability depends on the selected data, processing prerequisites and compatible hardware.



Actual interface reference: Menu: Export. See the option explanations below; screenshot values are not universal recommendations.

Point Cloud

Export the dense cloud, or tie points if no dense cloud exists, as PLY or DXF points. Verify the actual source. PLY is more suitable than enormous DXF point sets for many downstream workflows.

Mesh

Export mesh geometry or a compatible textured model. Available choices depend on whether a current texture exists. See the format reference for PLY, DXF, OBJ and GLB packaging/coordinate behavior.

Mesh Contours (DXF)

Cut the current mesh at a chosen model-Z interval and write 3D DXF contour polylines. Major contours occur every five intervals; the four between are minor. Validate scale and orientation first.

Tiled Model

Copy the tileset to a folder or package it as ZIP. Keep all referenced tiles and texture files. A single top-level index without its content is incomplete.

Orthomosaic

Export the orthomosaic using resolution, format, compression, transparency, overviews and block-size choices. Keep world/projection/placement sidecars when produced.

DEM

Export numerical surface elevations as GeoTIFF. Select appropriate resolution, compression and BigTIFF behavior. A JPEG screenshot of a coloured DEM is not a numerical height raster.

Export DTM

Export an existing DTM with the raster export controls. Verify frame, cell spacing, nodata and interpolation. This command does not turn a DSM into bare earth.

Cameras

Write aligned estimated camera information to CSV, including the appropriate world offset. This is estimated reconstruction output, distinct from the input GPS-reference CSV.

Markers

Write marker/reference information as a text report or marker XML. XML preserves marker structure and observations; text is useful for reviewing coordinates and available residuals.

Reference

Write input photo reference longitude, latitude and altitude to CSV. These are original photo-reference values, not the optimized camera-centre output of Cameras export.

Shapes

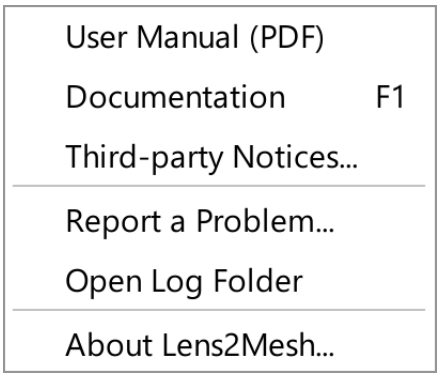
Export selected shape layers as Shapefile, DXF R12 or GeoJSON in the chosen available coordinate system. Preserve required sidecars and inspect mixed-geometry outputs.

Generate Report

Create an HTML report and a companion view image, using the project name, camera status, counts and processing log. Keep both files together. This release report is a summary, not a substitute for independent accuracy assessment.

Menu: Help

Visible 1.0.5 commands. Availability depends on the selected data, processing prerequisites and compatible hardware.



Actual interface reference: Menu: Help. See the option explanations below; screenshot values are not universal recommendations.

User Manual (PDF)

Open the bundled complete English reference offline in the system PDF reader. A PDF reader/file association must be available.

Documentation

Open the application documentation. Help > User Manual (PDF) opens the complete bundled offline reference.

Third-party Notices

Open bundled component licence and replacement information. Component licences are distinct from the application licence.

Report a Problem

Open a saved diagnostic report with an optional description and reviewable details. Report to GitHub opens a browser form; nothing is submitted until you submit it there.

Open Log Folder

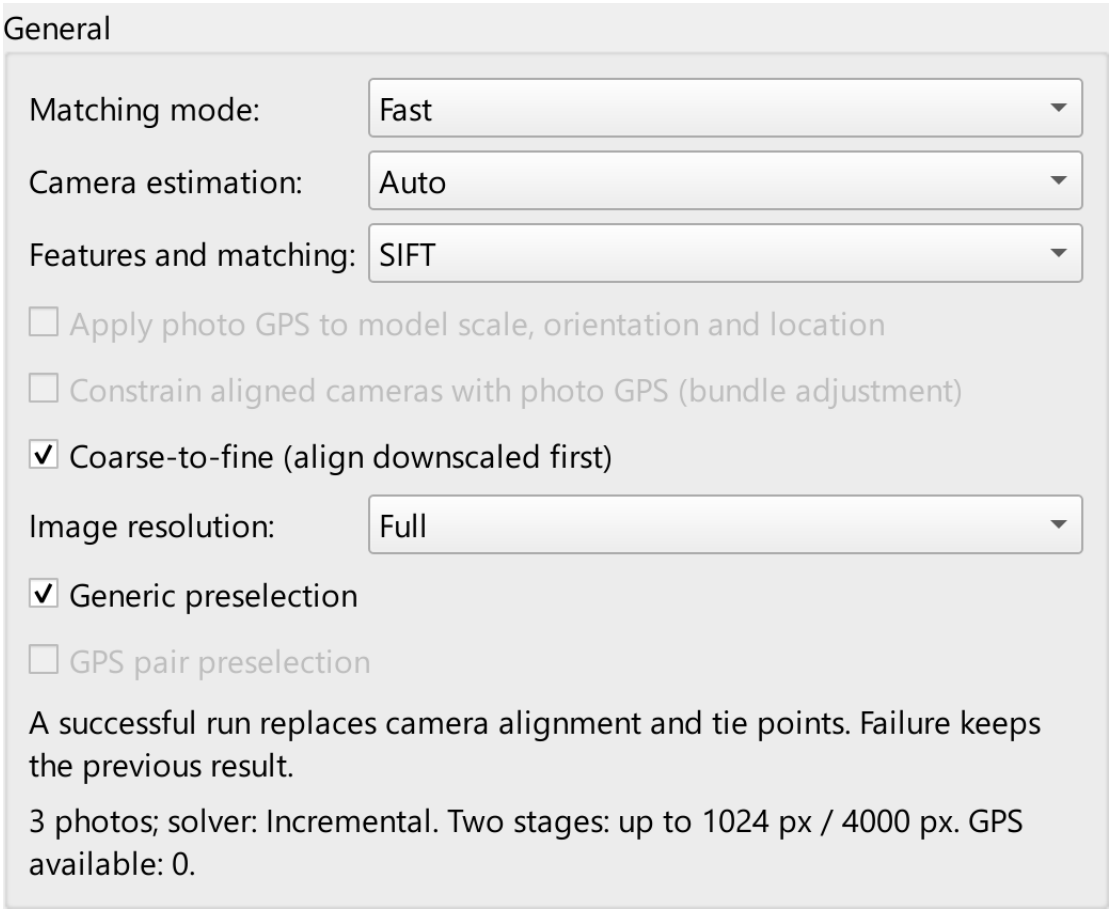
Open the application log directory. Include the relevant run/operation and full error text when seeking support, and review report contents before sharing.

About Lens2Mesh

Show the installed version, component information and licence summary. Open View full license for the actual permitted-use terms.

Align Photos: matching and resolution

Use Process > Align Photos after adding and saving the intended photos. Feature extraction finds repeatable image features; matching links photos; camera estimation constructs camera poses and tie points. Inspect the result before spending time on dense products. Defaults are starting points, not a universal optimum.



Actual interface reference: Align Photos: matching and resolution. See the option explanations below; screenshot values are not universal recommendations.

Matching mode: Fast

Combines candidate-pair preselection using image descriptors, sequence and valid GPS where enabled. Useful for larger overlapping sets where all-pairs matching is costly. It may miss links in unusual layouts; inspect coverage and try Exhaustive on a manageable problematic set.

Matching mode: Exhaustive

Tests all photo pairs rather than relying on preselection. Suitable for small or moderately sized unordered sets and diagnosis of missed candidate links. Pair count grows roughly with the square of photo count, so cost rises quickly.

Matching mode: Sequential

Uses ordered-photo matching for a capture sequence. Choose it when adjacent images overlap and file/order progression represents the capture path. It is less suitable for a randomly mixed photo collection.

Camera estimation: Auto

Uses the supported integrated global mapper for sufficiently large sets (100+ photos) when available, otherwise Incremental. Read the solver summary: Auto is a dispatch choice, not a separate geometric algorithm.

Camera estimation: Global

Solves camera relationships together. It can be efficient for connected sets, but repeated patterns or weak constraints can produce a plausible-looking incorrect arrangement. Compare with Incremental when a room folds or parts rotate.

Camera estimation: Incremental

Adds registered photos progressively. A useful robust starting choice for walk-around and interior sets. It can take longer on large projects and still depends on sufficient overlap and non-ambiguous features.

Camera estimation: Hierarchical

Splits the problem into parts and solves/merges them using a compatible COLMAP engine. Consider it for large sets; weak connections between parts remain a data-quality problem, not something a solver name guarantees to fix.

Features and matching: SIFT

The tested default feature/matching route. Start here for predictable operation and to isolate dataset problems before introducing neural dependencies. Supports the Fast coarse-to-fine route.

SIFT + LightGlue

Uses SIFT features with LightGlue matching. Requires the compatible engine and offline neural model bundle. May be slow on CPU or large sets; availability is checked before processing.

ALIKED + LightGlue

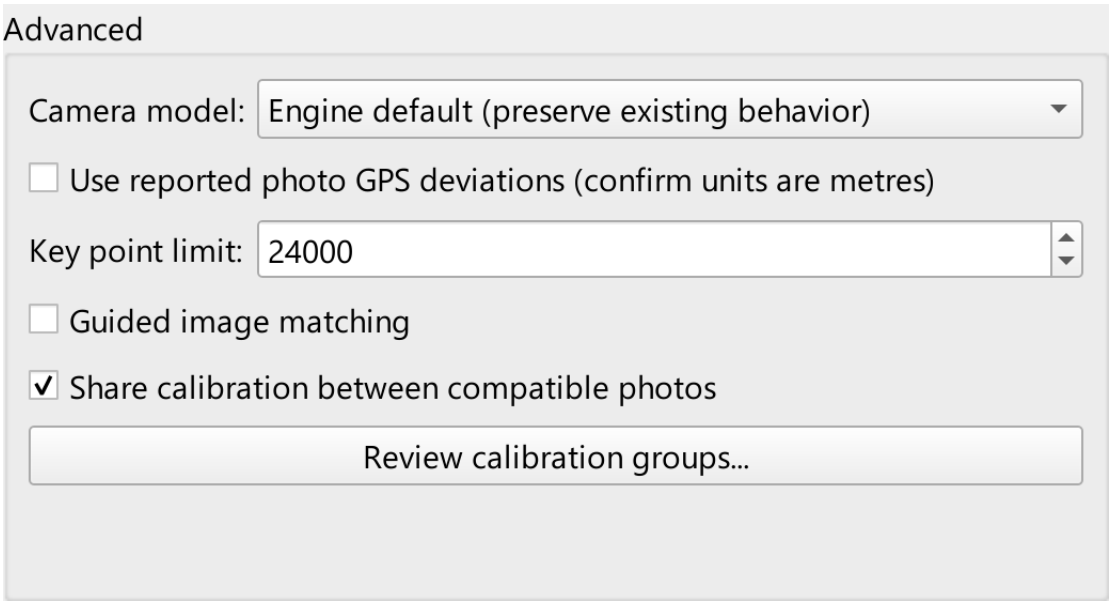
Uses learned ALIKED features and LightGlue matching. Requires the compatible engine/model bundle. The app bounds extraction resolution to 2048 pixels and uses one extraction worker for memory control. More complex is not automatically more accurate.

Image resolution: Full / Half / Quarter / One eighth

Uses dimensions at 1, 1/2, 1/4 or 1/8 of the source size respectively. Full retains small features but costs more work. Half is a useful initial comparison; use lower settings for speed only if sufficient features remain. These are alignment resolutions, separate from dense-cloud Quality.

Align Photos: reference and advanced options

Pair selection, initial GPS placement and later reference adjustment are separate operations. A greyed-out choice may be inapplicable to the selected matching method or unavailable because reference data/hardware are missing.



Actual interface reference: Align Photos: reference and advanced options. See the option explanations below; screenshot values are not universal recommendations.

Apply photo GPS to model scale, orientation and location

When valid GPS and the project state permit it, initializes the reconstructed model frame from photo positions. It is not the same as selecting matching pairs and does not substitute for measured control. The checkbox may be disabled for an already aligned/referenced state or insufficient usable GPS.

Coarse-to-fine

In Fast + SIFT, align a downscaled stage first, then use the requested finer resolution when a second stage is justified. Useful to reduce expensive fine matching. The status note indicates one or two stages; a small target resolution may need only one.

Generic preselection

In Fast mode, use image appearance/global descriptor similarity to choose candidate pairs. Useful for broad coverage beyond adjacent file order. Repeated-looking surfaces can confuse similarity, so inspect the resulting reconstruction.

GPS pair preselection

In Fast mode, use valid enabled photo GPS to help choose candidate pairs. Requires global Use Photo GPS and enough complete photo positions. It does not georeference the result or impose control-marker constraints.

Key point limit

Maximum features per photo: SIFT starts at 24,000; zero means Engine default. More features can help detail-rich sets but increase memory and matching work. Neural choices set a smaller bound (default 2,048, maximum 4,096); review the field when changing methods.

Guided image matching

Use estimated pair geometry to search for extra correspondences. Consider it when overlap exists but matches are weak. It adds matching work and still depends on valid pair geometry.

Share calibration between compatible photos

Group compatible camera models, dimensions and initial calibration so they share intrinsics; camera poses stay independent. Usually useful for one fixed camera/lens. Do not force dissimilar lenses, zoom settings or image sizes into one calibration.

Review calibration groups

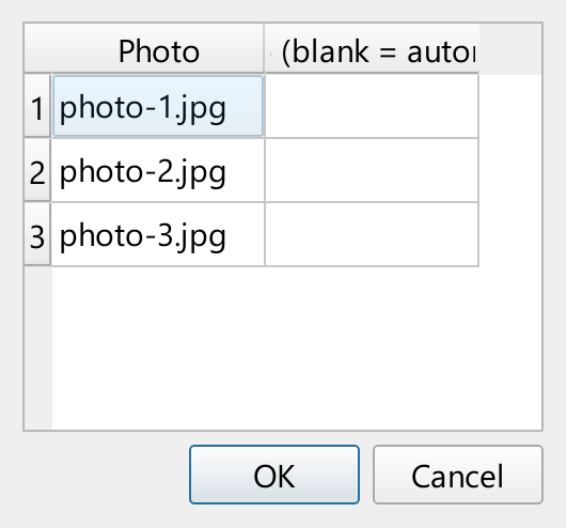
Open the per-photo group table. Photo is read-only; Group is editable. Blank uses automatic grouping; group labels provide deliberate grouping overrides. Keep genuinely incompatible cameras separate. OK uses the table for this alignment; Cancel discards the dialog edits.

Context/status note

Shows active photos, intended solver, working image size/stages and usable GPS count. Use it to check the actual requested workload. The number of aligned cameras alone is not an accuracy test.

OK / Cancel

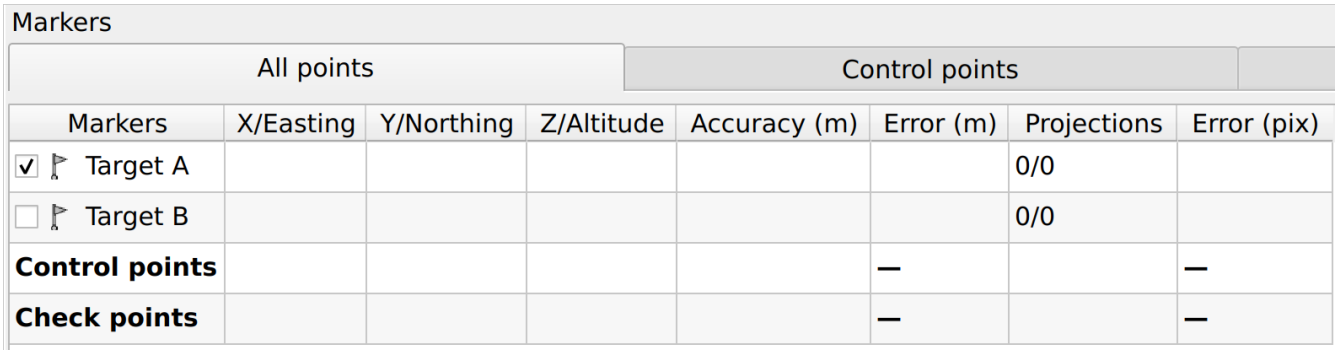
OK validates and starts the requested operation after prerequisite checks; Cancel leaves it unstarted. On a successful alignment, previous cameras/tie points are replaced transactionally; a failed run retains the previous valid result.



Additional interface detail: calibration groups. The control meanings and selection guidance are explained in this chapter.

Markers, control points and camera reference

A marker can be a named photo feature, a surveyed control or an independent check point. Source-photo observations and reference XYZ coordinates play different roles. Use measured data and confirm observations before allowing them to constrain geometry.



Actual interface reference: Markers, control points and camera reference. See the option explanations below; screenshot values are not universal recommendations.

Add and place a marker

Create an entry using Survey > Add Marker or right-click a source photo > Add Marker. On another photo, use Place Existing Marker and choose its name. Selecting a marker also enables Place Marker [name] Here. These operations link observations to the same marker identity.

Move or remove a photo observation

Drag a photo marker or place it at the corrected source-image position. Right-click its observation to remove it from that photo. This does not necessarily delete the marker from all photos. Review all linked images after an identity/position correction.

Green / blue / grey

Green represents a pinned measurement, blue a refined measurement, grey a prediction or ignored location. Predictions are display hints and do not enter adjustment/error calculations as measured observations until appropriately pinned/refined. Legacy predictions may need confirmation.

Markers column and checkbox

Double-click to edit a label. The checkbox selects Control (checked) or Check (unchecked); it is not a visibility toggle. Separately ignored/disabled markers are excluded. A marker without complete reference coordinates is not an independent XYZ control.

X/Easting, Y/Northing, Z/Altitude

Editable reference coordinates. Enter all three in the correct coordinate system. In WGS84 geographic input use longitude/latitude/height, not metric easting/northing. Do not copy arbitrary estimates back as supposedly measured ground truth.

Accuracy (m)

Optional marker-specific coordinate uncertainty; a single value or XY/Z pair can be entered. Smaller values give stronger weight. Set it from measurement quality, not the accuracy you hope the model will achieve.

Error (m)

Available 3D reference residual for the estimated marker versus supplied reference. A small control residual is not independent proof of accuracy. In an unscaled/local frame, interpret units carefully; missing/invalid estimates cannot produce a valid residual.

Projections

Shows measured/pinned-compatible observations versus all stored locations. A high total made only of grey predictions is not strong control. Useful viewpoint separation matters as much as observation count.

Error (pix)

Image reprojection residual from usable measured observations. It measures image consistency, not ground-distance accuracy. A marker can have a small pixel error yet poor depth precision if viewing rays are nearly parallel.

All points / Control points / Check points

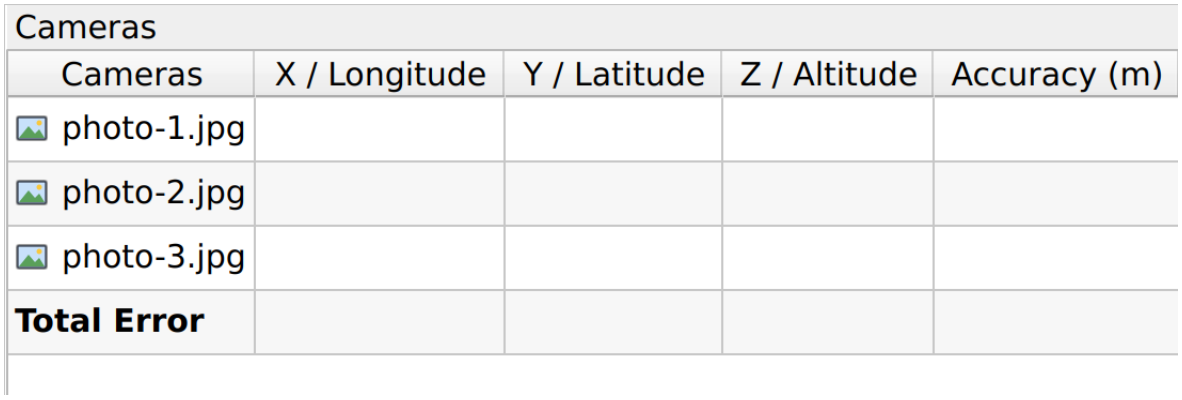
Tabs filter the marker table. Summary rows report available control/check coordinate RMSE and observation-weighted pixel RMSE, excluding ignored points. Blank values do not mean zero error.

Camera table

Cameras lists input X/longitude, Y/latitude, Z/altitude, accuracy, available error and estimated coordinates. It has no per-photo GPS checkbox. Use the project-wide GPS policy and the operation-specific reference option; camera visibility is unrelated.

3D/orthomosaic marker interaction

Markers on the model and orthomosaic are for display and selection. Correct observations on the actual source photos and reference values in the table. This avoids deriving control observations from an already imperfect reconstructed surface.



Additional interface detail: cameras reference detail. The control meanings and selection guidance are explained in this chapter.

Error (m)	X estimated	Y estimated	Z estimated
—			

Additional interface detail: cameras estimate detail. The control meanings and selection guidance are explained in this chapter.

Marker and photo context commands

Right-click commands depend on the item under the cursor and the current selection. If a command is unavailable, verify the selected item, photo alignment and required marker observations before trying again.

Add / Rename / Remove Markers

Marker/table/tree menus create entries, rename one identity across its links, or remove selected marker entries after confirmation. Removing a marker is broader than removing one photo observation.

Show Marker

Select/open the marker context so it can be reviewed. It is a navigation operation, not an optimizer. A marker must have usable linked photos/estimates for every view to be informative.

Filter Photos by Markers / Reset Filter

Restrict thumbnails to photos associated with the selected markers. Useful for correcting repeated observations efficiently. Reset Filter restores the full thumbnail set; filtering does not delete photos or change their enabled state.

Open Photo / Enable-Disable Camera

The Project camera context opens a photo or changes its processing inclusion. Display Show Cameras merely hides frusta; it does not disable the camera for processing.

Align Selected Photos from GCPs

Register selected new, unaligned photos into an existing aligned project using measured control correspondences. Supply sufficiently distributed, non-degenerate controls and a compatible calibrated camera. It is not a general first-time Align Photos command.

Calibration chooser for registration

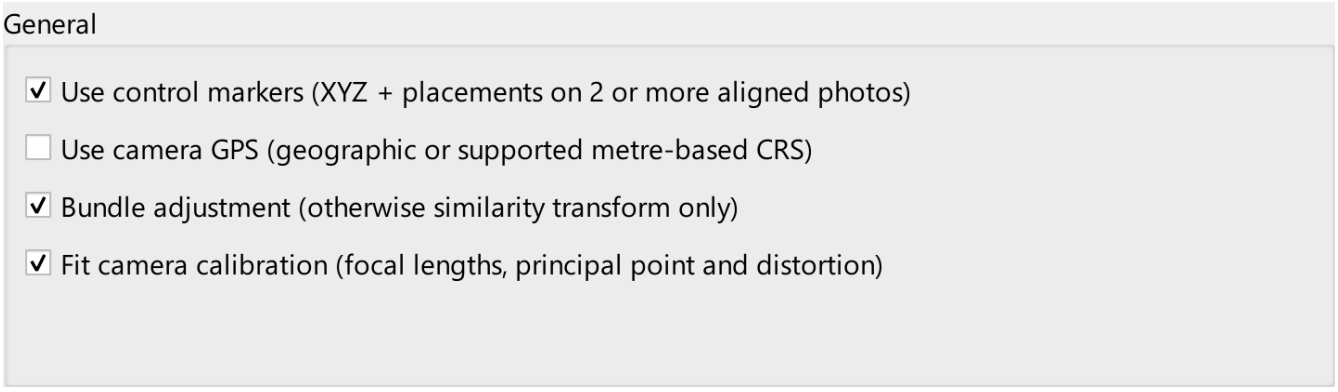
If more than one compatible calibration matches the new photo dimensions, select the appropriate existing calibrated camera ID. Cancel if no choice represents the actual lens/image configuration rather than forcing an incorrect calibration.

Show / layer and shape commands

Product context Show brings up that product. Shape context provides layer/shape style, export and removal. Remove Layer is different from toggling layer visibility. Use the appropriate scope before applying a style or deletion.

Optimize Cameras: choose the calculation

Alignment initially reconstructs from image matches. After alignment, Optimize Cameras can use reference data and refine geometry. Before optimizing, save a baseline, verify marker identities/coordinates and decide which references deserve weight.



Actual interface reference: Optimize Cameras: choose the calculation. See the option explanations below; screenshot values are not universal recommendations.

Use control markers

Include eligible control-marker coordinates and measured source-photo observations. Requires XYZ plus observations on at least two aligned photos. At least three non-collinear usable controls are needed to establish a reference frame. Unchecked excludes marker constraints even if markers remain displayed.

Use camera GPS

Include valid photo position references for the selected supported geographic/UTM workflow. Requires Use Photo GPS in Preferences. Choose it only when GPS has suitable quality and a height reference compatible with marker coordinates.

Both reference sources enabled

Eligible markers and GPS can both enter weighted adjustment. The initial frame uses sufficient non-collinear controls, otherwise usable GPS when sufficient. A conflicting weak reference still influences the answer according to its accuracy weight; validate before combining.

Neither / insufficient reference

A local image-based bundle adjustment may be offered when sufficient reference data are absent. This can improve image consistency but cannot create independent absolute scale or surveyed world coordinates.

Bundle adjustment

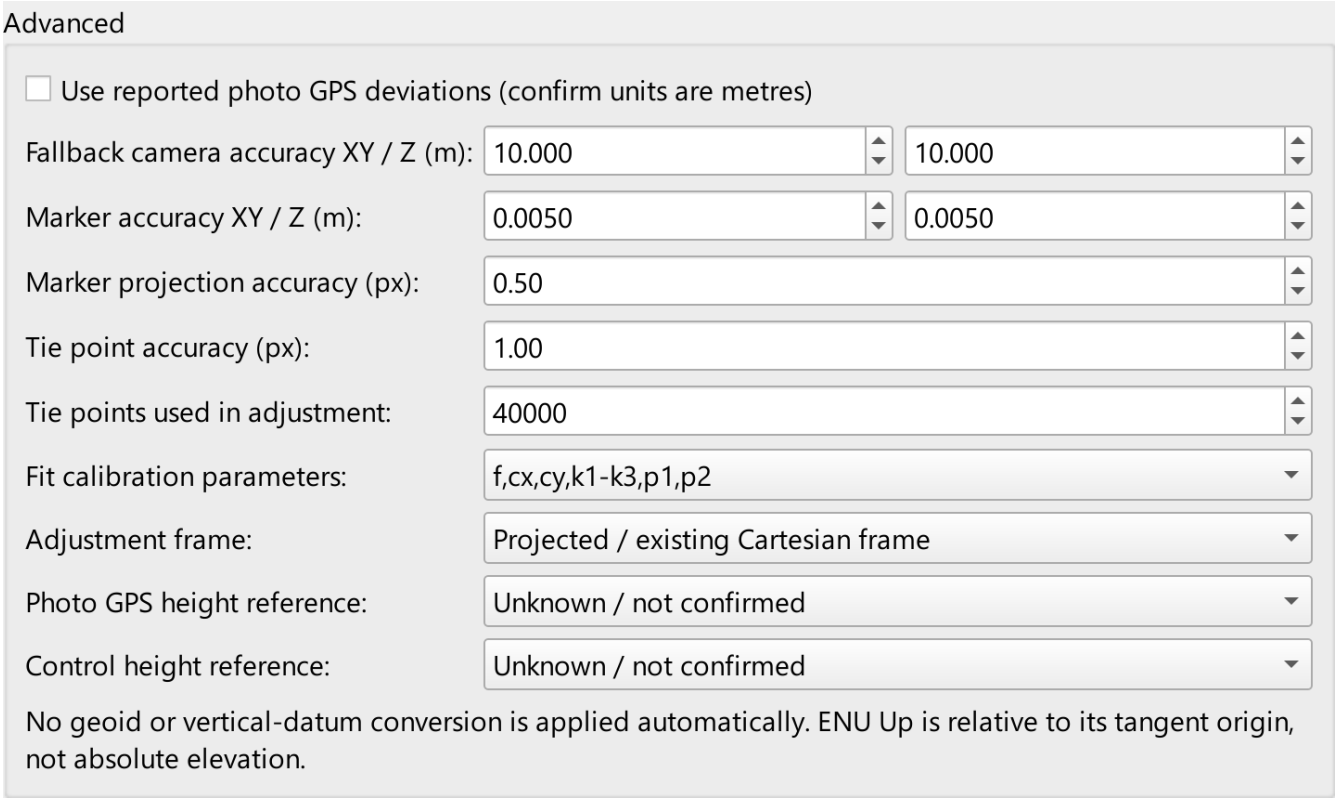
Checked refines camera/point geometry subject to the selected observations/reference. Unchecked requests a similarity transform only: scale, orientation and translation without the same nonrigid camera/point refinement.

Fit camera calibration

During bundle adjustment, refine supported intrinsics such as focal lengths, principal point and distortion while preserving the camera model. Disable to keep calibration fixed for a controlled comparison. Similarity-only mode does not perform calibration fitting; weak geometry cannot reliably constrain every parameter.

Optimize Cameras: weights and interpretation

Accuracy fields represent expected uncertainty: smaller numbers mean stronger constraints. They are not target output errors. Coordinate and image uncertainties are different quantities and use different units.



Actual interface reference: Optimize Cameras: weights and interpretation. See the option explanations below; screenshot values are not universal recommendations.

Camera accuracy XY / Z (m)

Separate horizontal and vertical GPS uncertainty; defaults are 10 m and 10 m. Increase uncertainty for unreliable positions instead of assigning implausibly strong weight. Vertical GPS uncertainty can differ substantially from horizontal accuracy.

Marker accuracy XY / Z (m)

Default coordinate uncertainties are 0.005 m horizontally and vertically. Appropriate only when independently measured controls justify that precision. Use larger honest values for rough measurements or disable unsuitable control constraints.

Marker projection accuracy (px)

Default 0.50 pixel expected photo-placement uncertainty. Sets the weight of marker image observations. Sharp well-defined targets and careful placement justify a different uncertainty than broad, ambiguous corners.

Tie point accuracy (px)

Default 1.00 pixel expected image uncertainty for automatically matched tie points. Changes their relative influence in bundle adjustment. Do not reduce it merely to force a smaller reported residual.

Tie points used in adjustment

Default cap 40,000, adjustable from 1,000 to 1,000,000. More tie points may improve coverage/constraint but cost time and memory. More points do not repair wrong matches or missing scene connections.

After optimizing

Inspect control and check residuals separately, look for spatial patterns, and inspect camera/surface shape. Markers clustered in one corner or nearly parallel rays may leave other parts weakly constrained. Rebuild affected derived products after accepting the new geometry.

Build Point Cloud: quality and depth maps

Process > Build Point Cloud creates dense depth estimates from aligned photos and fuses consistent estimates. It requires a compatible CUDA depth engine. Start with an accurate camera solution; increasing quality magnifies workload and may expose alignment errors without fixing them.



Actual interface reference: Build Point Cloud: quality and depth maps. See the option explanations below; screenshot values are not universal recommendations.

Source data: Depth maps (CUDA)

Read-only route description. The feature-extraction GPU checkbox is not a switch to a CPU dense-depth implementation. Confirm engine/hardware availability and the app preflight checks before starting.

Quality: Ultra high

Uses full photo dimensions for depth work (scale 1). Retains the smallest available texture details but has the highest memory/disk/runtime cost. Consider after validating a smaller-quality build and confirming available resources.

Quality: High

Uses half dimensions (scale 1/2), approximately one quarter the pixels per image. Often a useful detail/resource compromise. It does not imply half the coordinate accuracy.

Quality: Medium

Uses quarter dimensions (scale 1/4), approximately one sixteenth the full pixels. Default starting point for a test dense cloud. Inspect thin objects and low-texture areas before committing to higher resolution.

Quality: Low / Lowest

Use 1/8 and 1/16 dimensions, roughly 1/64 and 1/256 full pixels. Suitable for quick previews or very constrained resources when detail can be sacrificed. Small objects may disappear.

Depth filtering: Disabled

Do not apply the PatchMatch depth-filter preset. Necessary fusion consistency checks still occur. Use for a controlled comparison when suspecting the depth filter removed real detail; expect more unreliable depths/noise.

Depth filtering: Mild

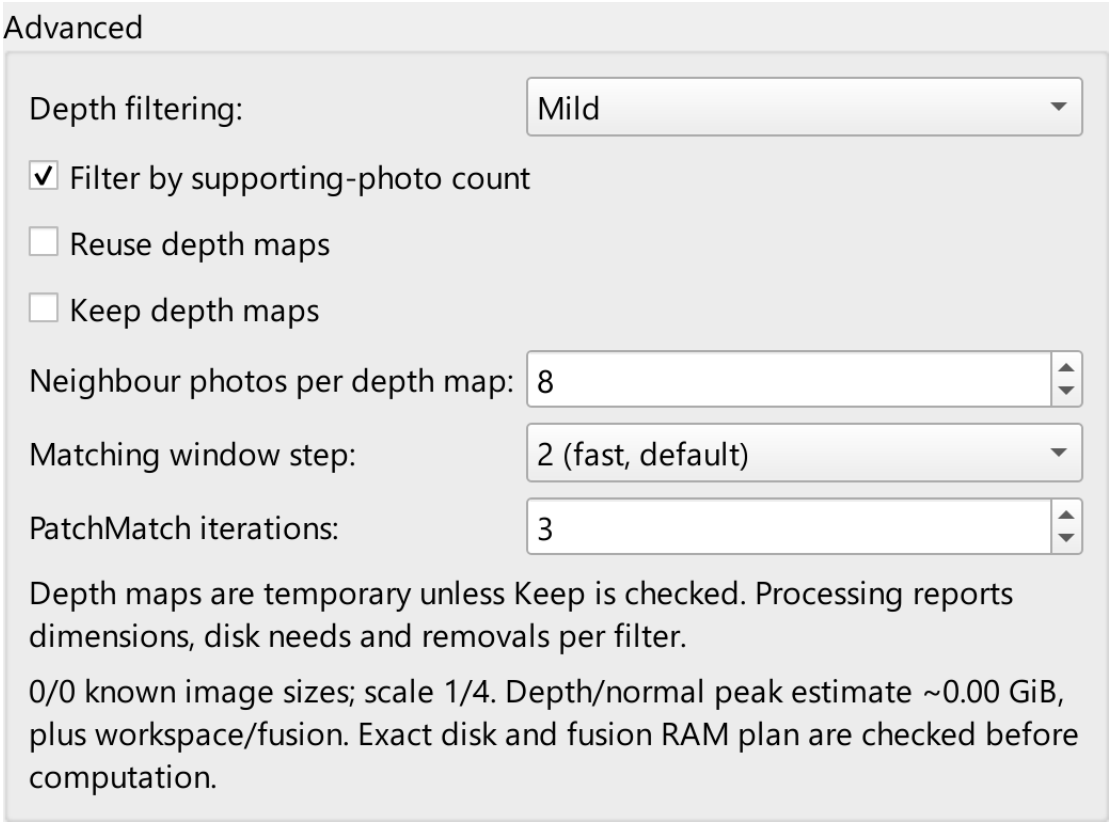
Least restrictive active preset, the default. A sensible initial comparison when thin structures or weak surfaces matter. Still cannot establish reliable depth on blank or reflective surfaces.

Depth filtering: Moderate

Apply a stronger depth rejection preset than Mild. Consider when a first cloud contains inconsistent depth/noisy patches and sufficient coverage remains. Compare loss of legitimate fine geometry.

Depth filtering: Aggressive

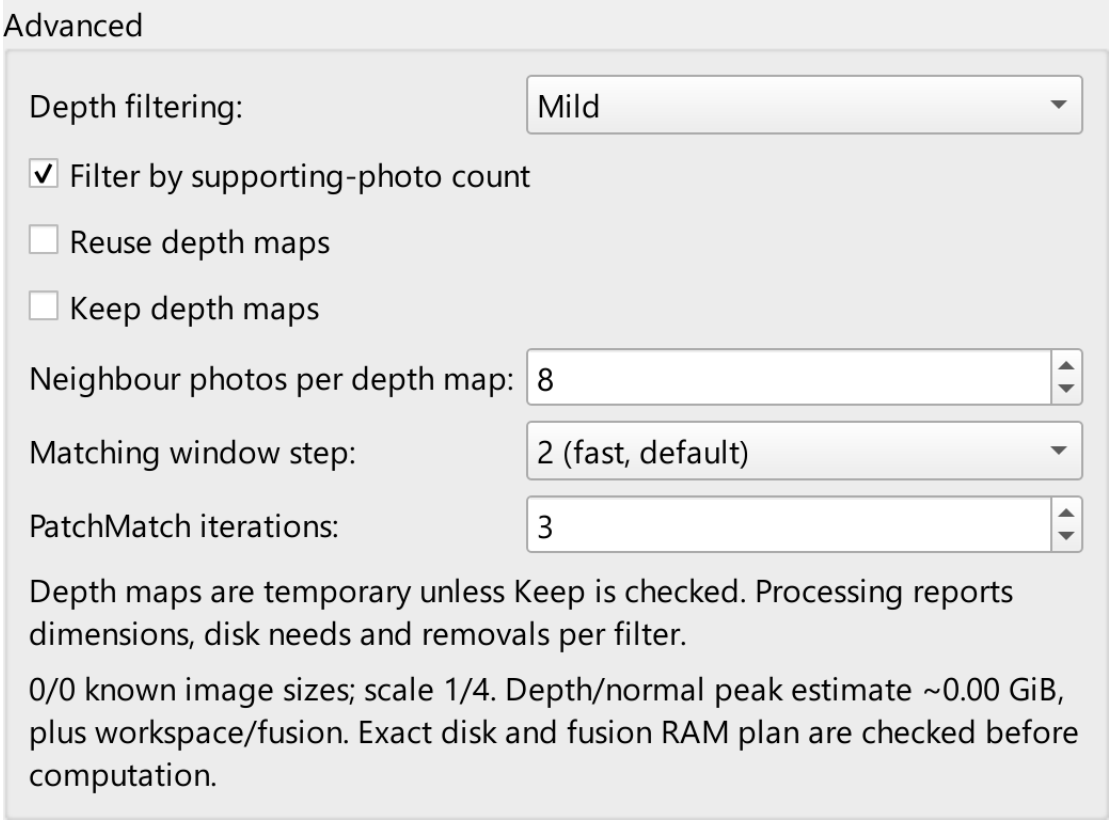
Strongest depth rejection preset. Can reduce noise at the cost of missing weak or thin surfaces. Avoid treating a cleaner-looking sparse result as proof of higher dimensional accuracy.



Additional interface detail: advanced. The control meanings and selection guidance are explained in this chapter.

Build Point Cloud: computation and reuse

These settings govern depth-map effort, storage and compatibility. They are separate from the post-fusion isolation filters. Change one group at a time when diagnosing missing geometry.



Actual interface reference: Build Point Cloud: computation and reuse. See the option explanations below; screenshot values are not universal recommendations.

Filter by supporting-photo count

Apply the minimum supporting-photo threshold to fused points. More independent supporting views can improve reliability. Disable only for a diagnostic comparison or deliberately low-support data, expecting more uncertain geometry.

Minimum supporting photos

Default 3. A higher minimum rejects more weakly supported points and can remove occluded/thin parts seen in few images. Lowering it may retain detail and noise. Zero or disabled support filtering does not remove all other fusion validity tests.

Neighbour photos per depth map

Default 8, range 2-30. Limits neighbouring source views considered for each depth map. More views increase work and may help coverage if they actually overlap well; unrelated/oblique views do not guarantee improvement.

Matching window step: 1

Slow, finest sampling of the matching window. Use when testing difficult fine texture and available processing time justifies it. This is a sampling control, not a replacement for well-exposed sharp source photos.

Matching window step: 2

Fast default with less sampling work. Start here, then compare step 1 on a manageable case if matching detail is the issue. Keep other settings fixed for a meaningful comparison.

PatchMatch iterations

Default 3, range 1-10. More iterations allow more hypothesis refinement and increase runtime. Returns can diminish; missing texture/coverage is not solved simply by maximizing iterations.

Keep depth maps

Retain generated depth/normal maps after a valid cloud is published so they can be reused later. Can consume large disk space. Unchecked removes temporary maps only after successful cloud publication.

Reuse depth maps

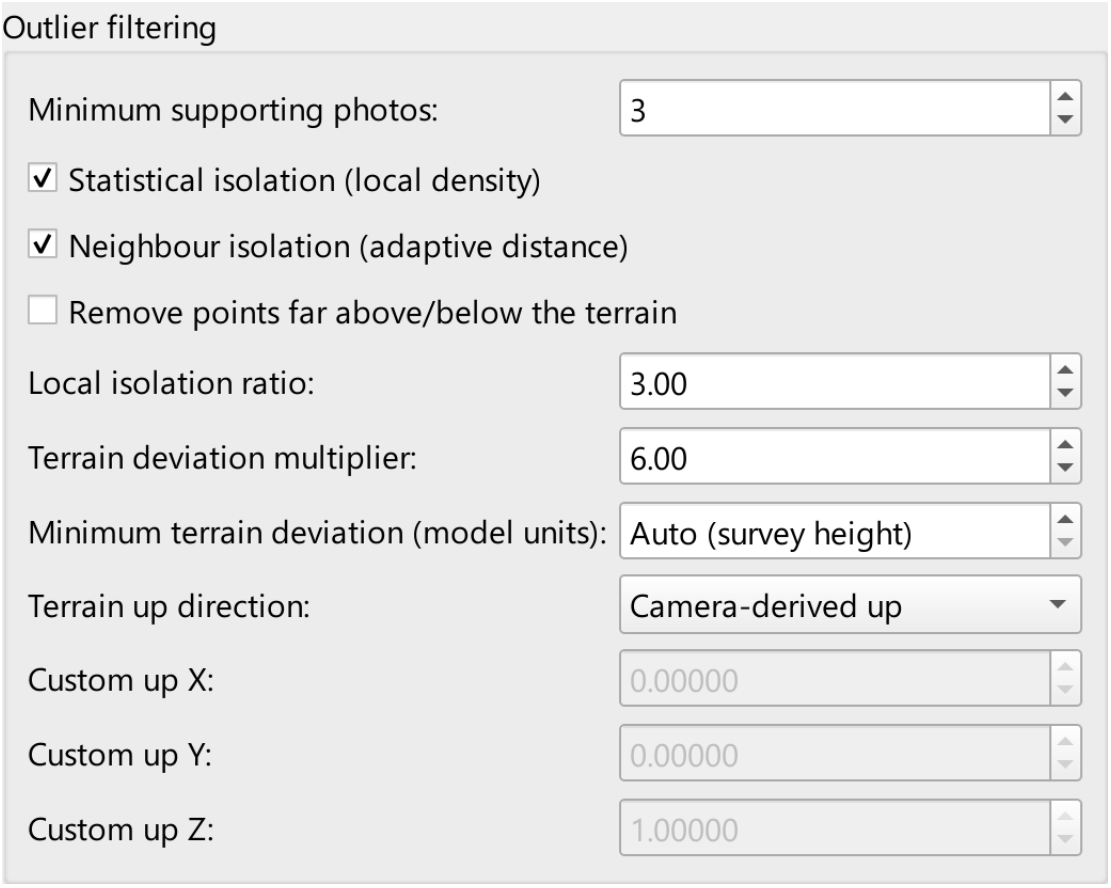
Use existing maps only if camera geometry, photo inputs and depth parameters are compatible. Camera/observation changes can require fresh computation. Keep must have preserved suitable maps; the app checks compatibility at start.

Memory/disk notes

The dialog reports image dimensions, scale and an estimate for depth/normal storage plus workspace/fusion needs. It is an estimate, not the whole process peak. Preflight performs further checks; preserve headroom for temporary and final files.

Build Point Cloud: post-fusion outlier filters

Outlier filtering can remove false points, but it can also discard legitimate low-support surfaces. For a room, a terrain assumption can be inappropriate. Read the per-filter removal counts in product metadata/logs rather than judging only the final point count.



Actual interface reference: Build Point Cloud: post-fusion outlier filters. See the option explanations below; screenshot values are not universal recommendations.

Statistical isolation (local density)

Reject locally isolated points using neighbourhood density. Useful for floating noise. A legitimate sparse region may look isolated too; compare with it disabled when real thin parts disappear.

Neighbour isolation (adaptive distance)

Reject points lacking suitable local neighbours using adaptive spacing. Useful for scattered speckles. It is separate from photo support and can remove isolated valid geometry as well.

Local isolation ratio

Default 3.00; range 1-20. Controls tolerance for local isolation; larger tolerance is generally less restrictive. Use a controlled comparison instead of assuming a universal threshold works for both dense and sparse regions.

Remove points far above/below the terrain

Optional and off by default. Applies a terrain-height sanity filter, not ground classification. It can remove buildings, steep slopes, room walls/ceilings or multiple levels. Leave off unless the scene fits a terrain assumption.

Terrain deviation multiplier

Default 6.00. Sets tolerance relative to robust local terrain variation. A smaller multiplier is more restrictive; increasing it preserves larger departures and may preserve noise too.

Minimum terrain deviation (model units)

Auto uses the available survey-height context; an explicit positive value provides an absolute tolerance floor in model units. Only meaningful relative to the established project scale. Do not enter metres into an arbitrary-scale model by assumption.

Terrain up direction: Camera-derived up

Estimate up from the camera arrangement. Suitable only if that arrangement reflects the intended terrain direction; unusual interior capture geometry may make this inappropriate.

Terrain up direction: Project Z

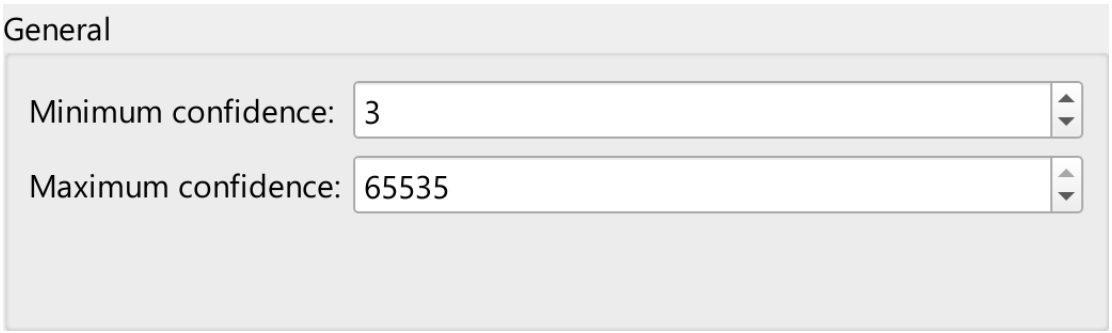
Use the project Z axis as vertical. Choose after verifying the model/reference frame is correctly oriented. Rotating the view to Top does not establish the project Z axis.

Terrain up direction: Custom; Custom up X/Y/Z

Enter a nonzero direction vector when the intended up axis differs. Custom values are enabled only for Custom. The vector defines direction, not a point or scale; the zero vector is invalid.

Filter an existing cloud

Process > Filter Point Cloud works on an existing cloud. It is distinct from computing depth maps or re-running all Build Point Cloud filters. Save a baseline and inspect geometry lost as well as noise removed.



Actual interface reference: Filter an existing cloud. See the option explanations below; screenshot values are not universal recommendations.

Minimum / Maximum confidence

Keep the intended range of supporting-photo counts. Defaults are 3 and 65,535. Increasing the minimum removes low-support points; reducing the maximum deliberately excludes high-support points, rarely needed for ordinary cleanup. An imported cloud may lack usable confidence values.

Filter by Confidence dialog

Shows the confidence range and suppresses other advanced filters. Use when support is the only variable to test. It does not require recomputing depth maps.

Statistical outlier removal

Toggle a k-neighbour distance test. Neighbours (k), default 16, defines the local sample. Std ratio, default 2.0, defines tolerance; smaller values reject more points. Large k can bridge different density/feature regions.

Radius outlier removal

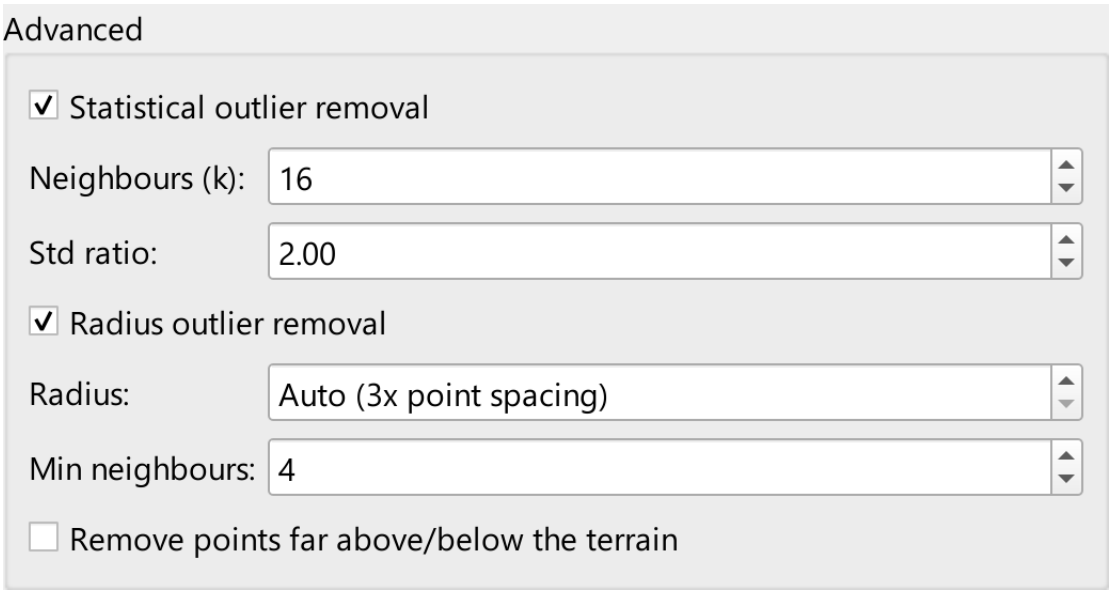
Toggle the radius/minimum-neighbours test. Radius zero means Auto, approximately 3x point spacing. A smaller radius or higher Min neighbours (default 4) is more restrictive. A fixed radius uses model units and needs known scale.

Terrain removal

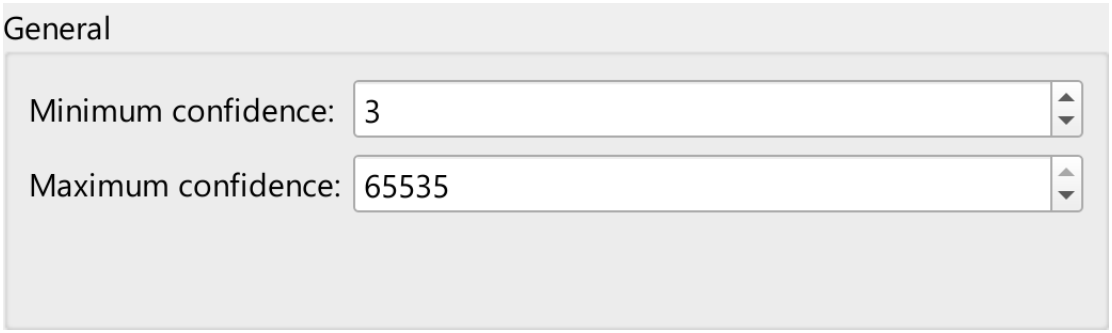
Optional terrain sanity filter. Use only when an up/terrain assumption fits the geometry; it is unsuitable as a general noise filter for an interior containing walls and ceilings.

Apply carefully

Confirm the chosen filters, run, then compare counts and inspect the affected surfaces. Filtering changes cloud geometry and may require rebuilding meshes/rasters that depended on it.



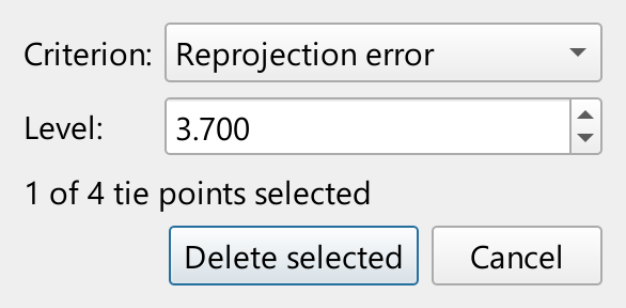
Additional interface detail: advanced. The control meanings and selection guidance are explained in this chapter.



Additional interface detail: general. The control meanings and selection guidance are explained in this chapter.

Clean Tie Points / Gradual Selection

This dialog removes selected sparse tie points using one criterion at a time. It is not the dense-cloud filter. Inspect the selected count before deleting; aggressive deletion can disconnect views or remove the very coverage needed to constrain the scene.



Actual interface reference: Clean Tie Points / Gradual Selection. See the option explanations below; screenshot values are not universal recommendations.

Reprojection error

Select points whose maximum available observation error exceeds Level. Units are pixels. Use to inspect unusually inconsistent tracks; a lower threshold selects more. It is an image residual, not a measured distance error.

Reconstruction uncertainty

Select points with poorer triangulation geometry above Level. The metric uses viewing-ray geometry; nearly parallel rays give weak depth constraints. Lowering the threshold removes more uncertain points but can reduce coverage.

Image count

Select points seen in fewer than Level images. Unlike the other criteria, low values are worse; raising Level selects more points. Default starting Level is 3. Good viewpoint separation still matters.

Projection accuracy

When source keypoint scales are available, select points with a larger average feature scale than Level. It describes image-feature scale, not independently measured world accuracy. This criterion may be absent if the required metric is unavailable.

Level and selected count

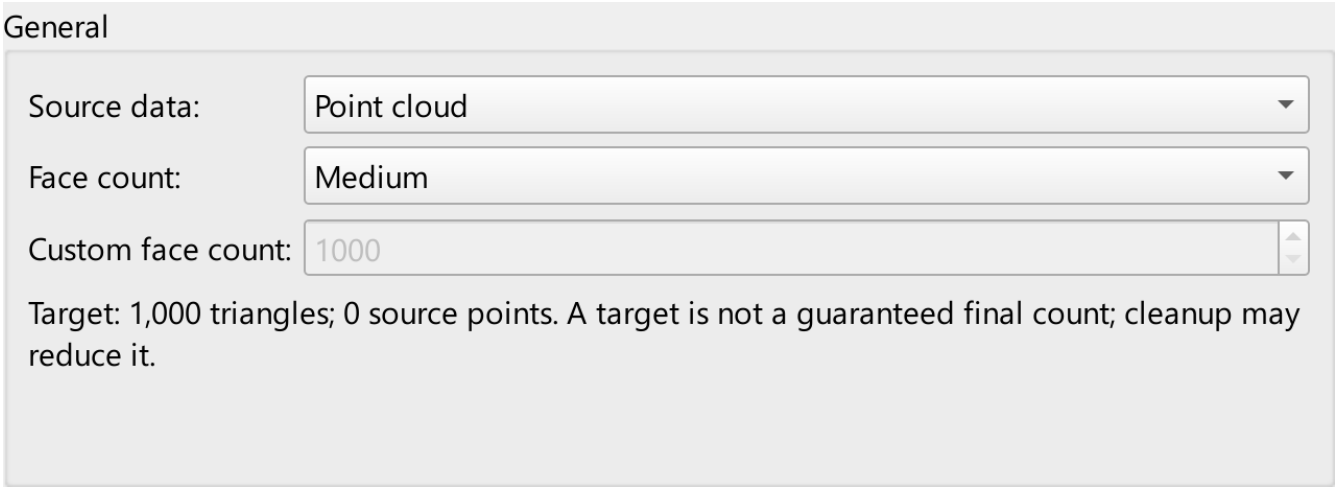
For most criteria the initial threshold is based on the current metric distribution. Read N of M tie points selected. There is no universal safe percentage to remove; preserve distributed scene constraints.

Delete selected / Cancel

Delete selected applies the removal; Cancel leaves the cloud unchanged. Save a baseline before experiments, inspect camera/scene connectivity afterward, and rebuild affected downstream products as needed.

Build Mesh: source and face target

The toolbar/menu says Build Mesh; the settings dialog title may say Build Model. It creates a triangle surface, then performs cleanup/decimation. A high face count cannot recover surfaces absent from the input cloud.



Actual interface reference: Build Mesh: source and face target. See the option explanations below; screenshot values are not universal recommendations.

Source: Point cloud

Screened-Poisson meshing of the current cloud, with downsampling/trimming controls. Useful for dense surface samples without needing the original depth-map visibility route. Check that the cloud is current and in the correct frame.

Source: Delaunay (fused cloud + visibility)

Uses the fused dense cloud and associated visibility from the depth workspace. This is the displayed name of the internal depth-map source route. It needs compatible fused/visibility data; an arbitrary imported PLY alone does not supply those inputs.

Face count: High

Targets approximately input point count divided by 5, with a minimum target of 1,000 triangles. Retains more geometry than the lower presets and increases memory/export/rendering cost.

Face count: Medium

Targets approximately point count divided by 15, minimum 1,000. Default balanced starting point. Inspect edges, thin objects and surface shape rather than selecting by count alone.

Face count: Low

Targets approximately point count divided by 45, minimum 1,000. Useful for fast previews or constrained downstream viewers. Small geometric details may be simplified away.

Face count: Custom / Custom face count

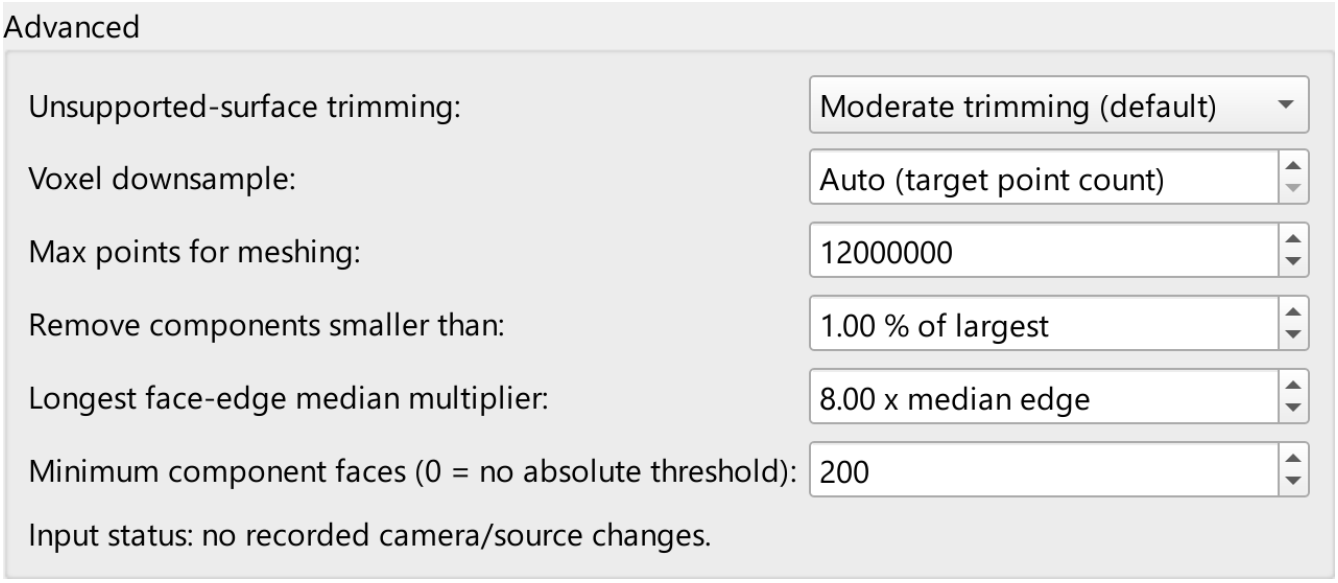
Enter the desired triangle target directly when Custom is selected. It is a target, not a promise: source detail, cleanup and component/edge filtering can reduce the final count. Do not inflate it to manufacture survey detail.

Source status / acknowledgment

Read the input source and any stale warning. When a stale-source acknowledgment appears, understand why the source is outdated before choosing to proceed. Rebuilding from an up-to-date source is needed for current geometry.

Build Mesh: trimming and cleanup

These controls balance surface retention, interpolation and memory. Some apply only to the Point cloud/Poisson source and become disabled for Delaunay. Inspect shaded and wireframe output for invented bridges as well as holes.



Actual interface reference: Build Mesh: trimming and cleanup. See the option explanations below; screenshot values are not universal recommendations.

Unsupported-surface trimming: Strong trimming

Remove more low-support Poisson surface (the strongest preset). Use for noisy boundaries when false surfaces are worse than holes. Can remove legitimate weakly supported geometry.

Moderate trimming (default)

Moderate removal of unsupported surface. A sensible initial setting. Compare with stronger trimming for noisy fringes or extrapolation only when consciously accepting inferred areas.

Allow extrapolation

Retain more extrapolated Poisson surface with no density-quantile trimming. May bridge holes or extend edges with unmeasured geometry. It is not a reliable recovery of missing walls/floors.

Voxel downsample

Zero/Auto lets the mesher target a manageable point count. A positive voxel size reduces samples spatially in model units (metres only when properly referenced/scaled). Larger voxels reduce memory and fine detail.

Max points for meshing

Default 12,000,000; cap the working point count through downsampling for the Poisson route. Raising it increases memory/work. Lower it when resources are limited, then compare detail loss.

Remove components smaller than

Default 1% of the largest connected component. Remove isolated surface fragments below the relative threshold. Reduce/disable when legitimate disconnected objects must remain; retain a baseline for comparison.

Minimum component faces

Default 200; absolute component-size threshold in addition to relative cleanup. Zero disables the absolute threshold. Small valid objects can be removed even when the main surface is good.

Longest face-edge median multiplier

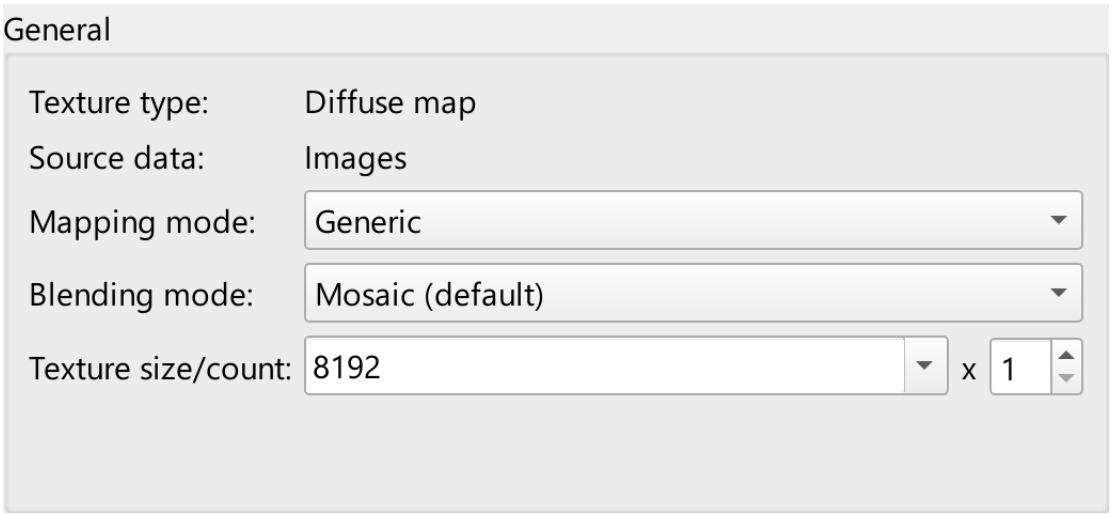
Default 8x median edge; zero is Off. Reject triangles with unusually long edges, useful against false bridges. Too restrictive can open gaps or remove valid coarse areas.

After meshing

Check final face/vertex counts, holes, disconnected pieces and boundary extrapolation. Build Texture only after the geometry is acceptable; texture colour can conceal a bad surface.

Build Texture: mapping choices

Build Texture samples the aligned source images onto the current mesh. Diffuse map and Images identify the supported texture type/source; there are no normal/displacement-map generation choices in this dialog.



Actual interface reference: Build Texture: mapping choices. See the option explanations below; screenshot values are not universal recommendations.

Generic

General camera-chart atlas suitable for varied surface orientations. Start here for objects, buildings or rooms with multiple sides and overlapping geometry.

Adaptive orthophoto

Uses an orthophoto-style approach with camera charts for steep faces. Consider mixed terrain and vertical structures where a single top-down projection is insufficient. Inspect seams and side coverage.

Orthophoto

Projects using an orthophoto-like plane. Useful for mainly terrain-like surfaces. Can collapse vertical or vertically overlapping geometry into the same texture locations; avoid as a universal room/object setting.

Keep uv

Available when compatible UV information exists. Reuse the mesh atlas instead of remapping. Page-count controls reflect the existing atlas and may be disabled. Do not assume unrelated UV files are compatible with a replaced mesh.

Texture size

Preset sides 1,024, 2,048, 4,096, 8,192 and 16,384 pixels; editable integer sizes are validated from 256 to 32,768. Larger pages improve potential colour sampling but consume memory/disk and cannot create source detail.

Texture count

Number of atlas pages, range 1-64. With Automatic page count off, this is a hard maximum. More pages allow more texel area at a given side length, but increase file count and downstream complexity.

Automatic page count

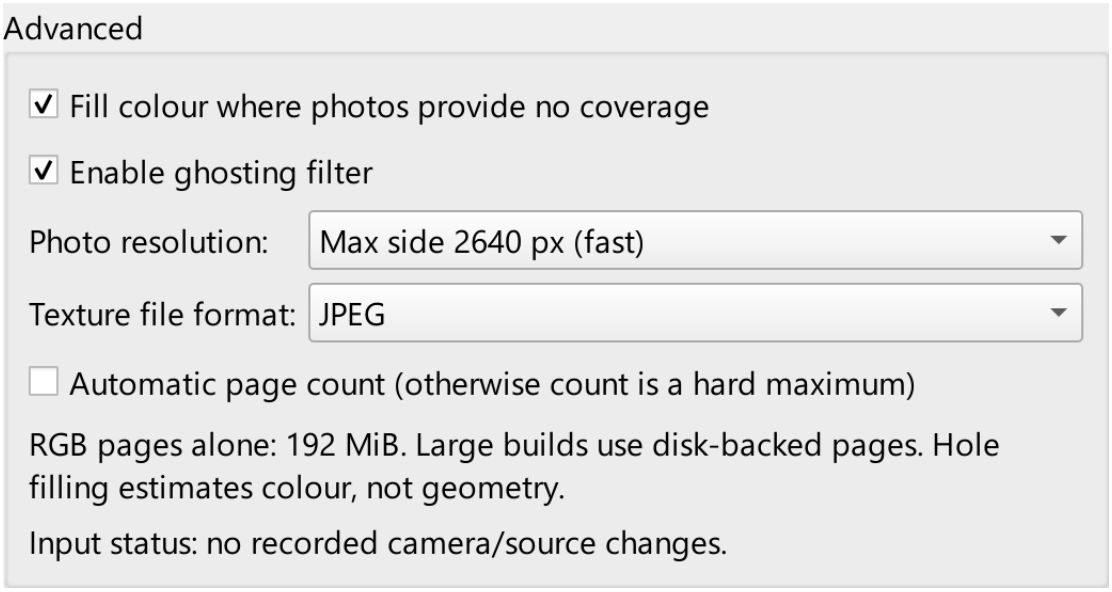
Allow the builder to choose additional pages rather than treating Count as a hard maximum. Use when coverage/detail needs vary across a large model. Keep uv follows the existing page structure instead.

Memory estimate

The dialog reports RGB page storage alone; other working arrays add overhead. Large builds can use disk-backed pages, so free disk space still matters. Test the export viewer limits before choosing huge atlases.

Build Texture: blending and image options

Blending combines valid source colours; it does not improve the mesh geometry. Inspect moving objects, illumination differences, boundaries and occluded faces after building.



Actual interface reference: Build Texture: blending and image options. See the option explanations below; screenshot values are not universal recommendations.

Mosaic (default)

Choose suitable source views with seam handling rather than simply averaging all colours. A general starting choice when detail sharpness and transitions matter.

Average

Average available source colours. Can reduce exposure differences but blur fine details or create ghosting if geometry, alignment or the scene differs between photographs.

Max intensity / Min intensity

Choose brighter/darker contributions respectively. Useful for deliberate appearance comparisons or special illumination cases; can bias colours. They are not a radiometric calibration or universally better texture mode.

Single photo (no colour blending)

Use a selected/best contributing photo without colour blending for a region. Can retain sharper source detail but produce more visible exposure transitions.

Fill colour where photos provide no coverage

Infer colour for unobserved regions. Use for presentation only when that inference is acceptable. It fills colour, not missing measured geometry; report unobserved/inferred areas honestly.

Enable ghosting filter

Reject inconsistent colour contributions; coarse colour-outlier detection needs at least four valid views. May help moving objects/colour conflicts but cannot repair wrong camera poses or guarantee clean dynamic scenes.

Photo resolution: Max side 2640 px (fast)

Use bounded working-resolution photos to reduce sampling work and memory. Good for previews and moderate texture needs; source detail beyond that working resolution is not retained in the same way.

Photo resolution: Full resolution

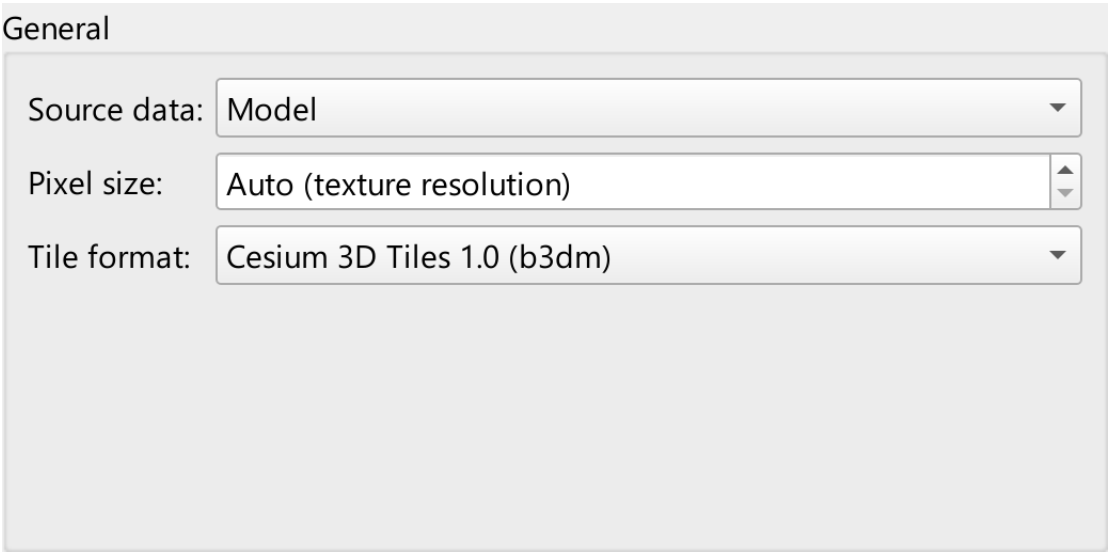
Use full source imagery for texture sampling. Choose after confirming geometry and sufficient atlas resolution; otherwise extra processing may not yield useful visible detail.

Texture file format: JPEG / PNG

JPEG gives smaller lossy colour images; PNG preserves pixels losslessly but can be larger. Use PNG when lossless colour storage is required, JPEG when delivery size matters and its artefacts are acceptable.

Build Tiled Model: mesh and point variants

A tileset partitions a product for scalable viewing; it does not change the original observations or prove accuracy. Choose formats that the intended downstream viewer actually supports.



Actual interface reference: Build Tiled Model: mesh and point variants. See the option explanations below; screenshot values are not universal recommendations.

Source data: Model

Build triangle tiles from the current mesh. Uses current texture when compatible; missing/stale texture can lead to vertex colours or grey. Inspect the input-status message before building.

Source data: Point cloud

Build point tiles from the cloud. Pixel-size and texture-size controls become inapplicable/disabled; points do not use a mesh texture atlas.

Pixel size: Auto

For mesh tiles, derive from source texture resolution or mesh-edge spacing when no texture exists. A manual positive value requests a minimum texel size in model/project units; effective values are reported, not blindly guaranteed.

Cesium 3D Tiles 1.0 (b3dm)

Mesh tiles packaged using the 1.0 b3dm route. Choose when required by the target viewer/workflow. Keep the tileset and referenced content together.

Cesium 3D Tiles 1.1 (glb)

Mesh content in the 1.1 glb route. Choose for compatible modern viewers. Changing container format is not a change to survey scale or geometric precision.

Point tiles (pnts)

The offered format for Point cloud source. There is no mesh texture in this mode. Export the complete hierarchy as a folder/ZIP.

Triangles / Points per tile

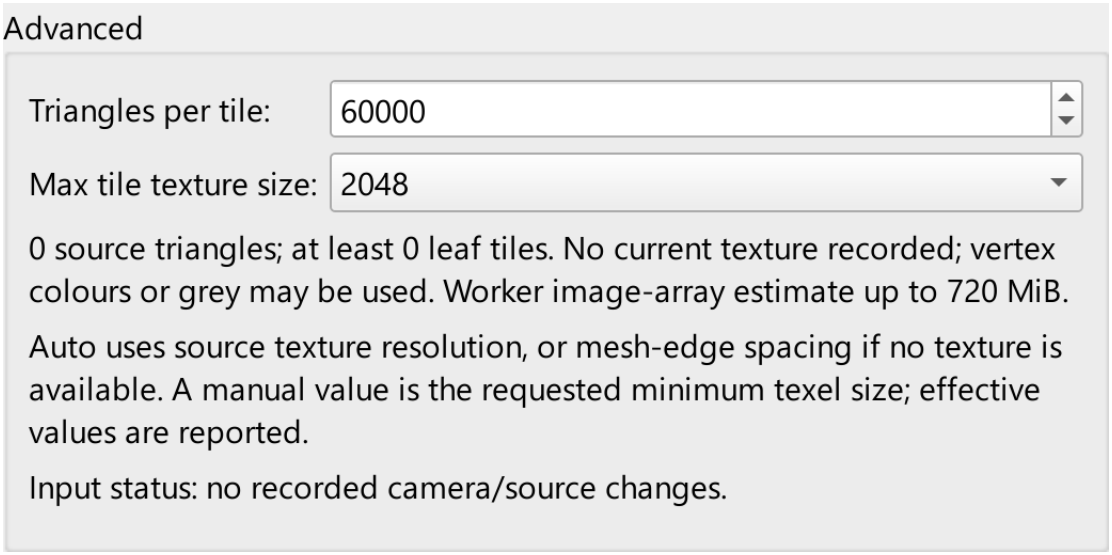
Default 60,000, range 2,000-1,000,000. Smaller limits make more tiles and finer loading units; larger limits make fewer heavier tiles. The label changes with source type.

Max tile texture size

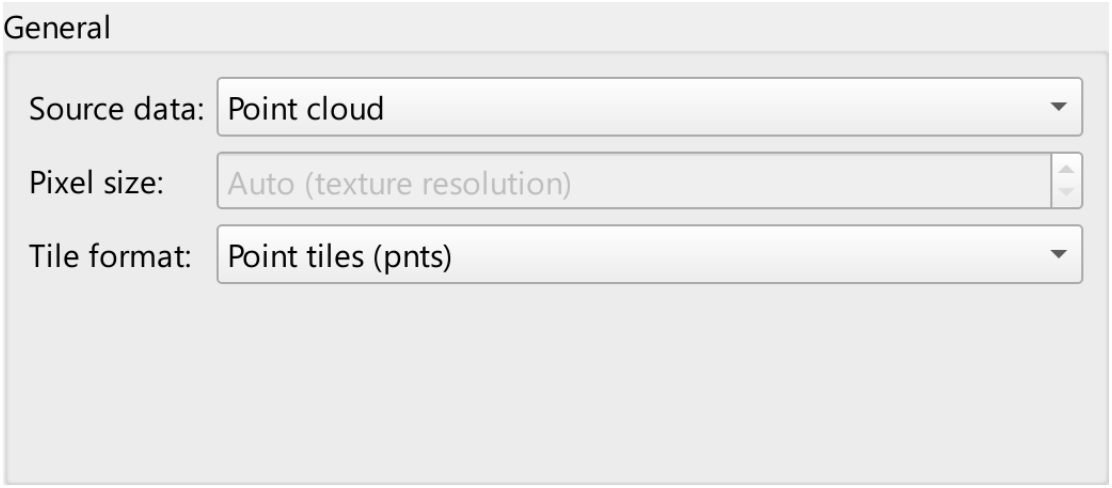
Mesh choices 512, 1,024, 2,048 or 4,096 pixels. Larger tile images increase potential detail and working memory. Disabled for point tiles; the on-screen worker estimate is not the entire machine memory requirement.

Output review

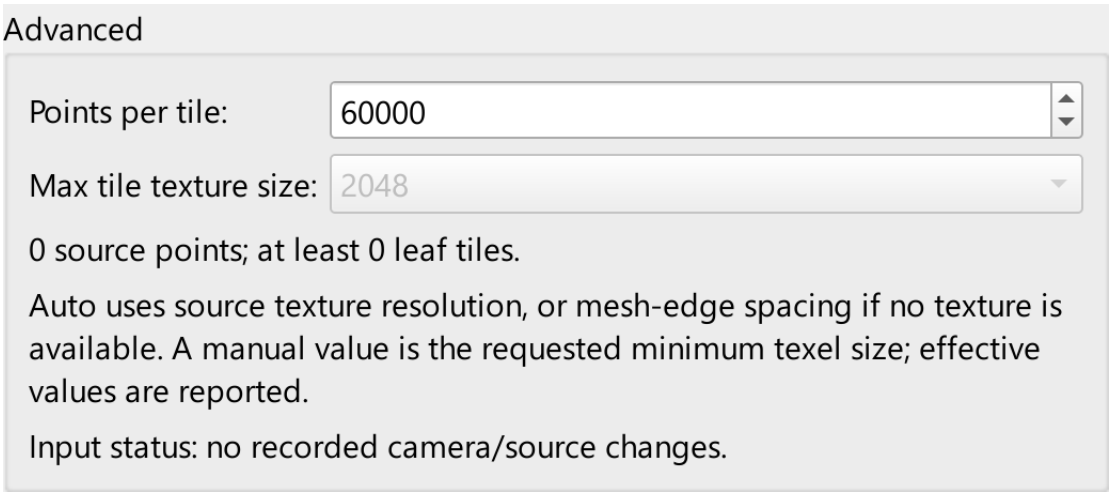
Check leaf count, format, texture currency and actual viewer behavior. Do not deliver only the top-level tileset index or call an untextured tile failure if the source contained no current texture.



Additional interface detail: advanced. The control meanings and selection guidance are explained in this chapter.



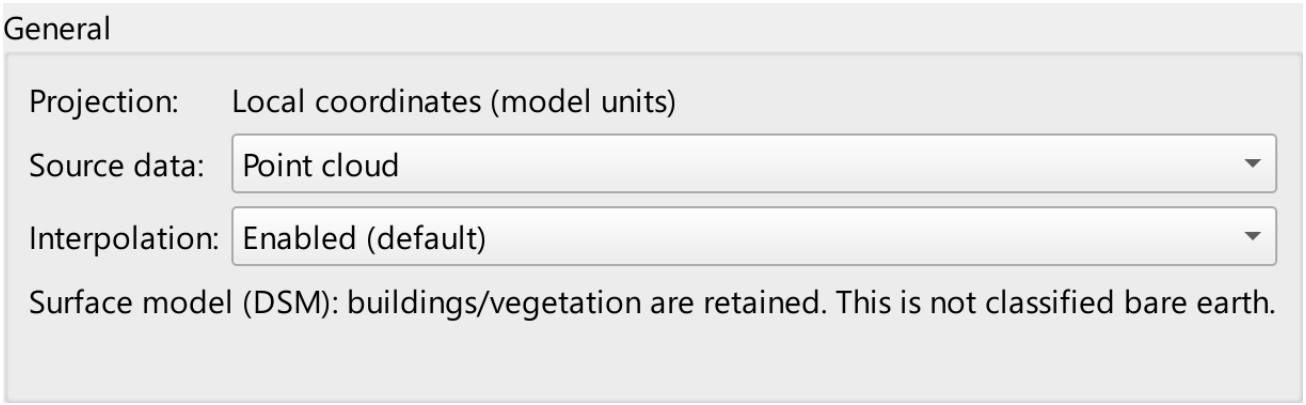
Additional interface detail: general. The control meanings and selection guidance are explained in this chapter.



Additional interface detail: advanced. The control meanings and selection guidance are explained in this chapter.

Build DEM: source, heights and interpolation

A DEM here is a digital surface model derived from mesh/cloud geometry. Buildings and vegetation are retained unless the input itself excludes them. It is not classified bare earth.



Actual interface reference: Build DEM: source, heights and interpolation. See the option explanations below; screenshot values are not universal recommendations.

Projection

Displays the project/output reference description. Local model units remain arbitrary until independently scaled. Verify CRS and vertical reference before interpreting raster elevations as survey heights.

Source: Point cloud / Mesh

Point cloud groups samples into raster cells; Mesh samples the triangle surface. Choose the current suitable product with needed coverage. Point-cell height aggregation applies to point clouds, not mesh sampling.

Interpolation: Disabled

Leave unfilled/missing cells instead of inventing values. Useful for accuracy-sensitive inspection of actual coverage. Expect gaps where no adequate source exists.

Interpolation: Enabled (default)

Fill suitable gaps within the selected fill-distance behavior. Provides a more continuous surface, but filled heights are inferred. Inspect voids and edges before use.

Interpolation: Extrapolated

Allow broader inferred extension beyond ordinary bounded filling. Can produce smooth-looking but weakly supported areas. Use deliberately for presentation/continuity, not as measured detail.

Resolution

Zero/Auto derives a cell size from point density/source context. A manual value is raster cell spacing in project/model units. Smaller spacing makes more cells and larger files, not automatically more accurate heights.

Point-cell height: Mean

Average point heights within each cell. Good for a representative surface but sensitive to outliers and mixed vertical surfaces in a cell.

Point-cell height: Median

Use a robust middle height. Often less sensitive to isolated high/low noise; still represents a surface statistic rather than automatically classifying ground.

Point-cell height: Highest

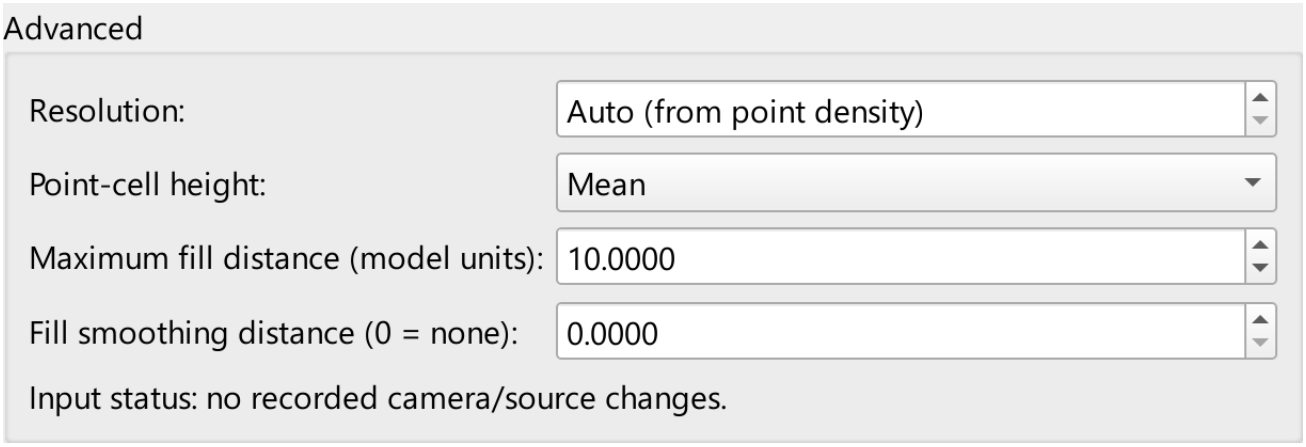
Use the upper height in a cell. Suitable when the top envelope is intended. Can preserve roofs/canopy and high outliers; unsuitable as an automatic bare-earth choice.

Maximum fill distance

Default 10 model units, used with bounded interpolation. A large distance can bridge real voids. Disabled for Disabled/Extrapolated where this bounded setting does not apply. Choose relative to actual scale and acceptable inference.

Fill smoothing distance

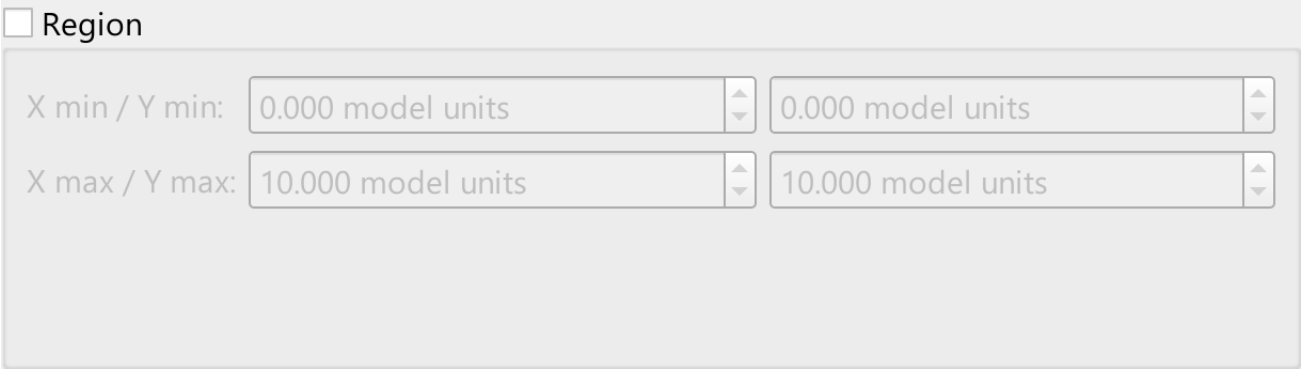
Default zero/none. Smooth interpolated fill over a chosen distance. Can reduce discontinuities but blur real surface changes; it is not an increase in measurement precision.



Additional interface detail: advanced. The control meanings and selection guidance are explained in this chapter.

Output Region and source warnings

DEM and Orthomosaic include a checkable Region group. Region limits output extent; it does not fix a distorted model or add missing surface detail. Read whether the displayed values are in world or local model coordinates.



Actual interface reference: Output Region and source warnings. See the option explanations below; screenshot values are not universal recommendations.

Region unchecked

Use the available computed source extent rather than an explicit user bounding rectangle. Useful for an initial full-coverage preview.

Region checked

Enable a specific output rectangle. Set X min/Y min and X max/Y max in the project coordinate system. Maximums must exceed minimums and the region must intersect the available surface.

X min / Y min; X max / Y max

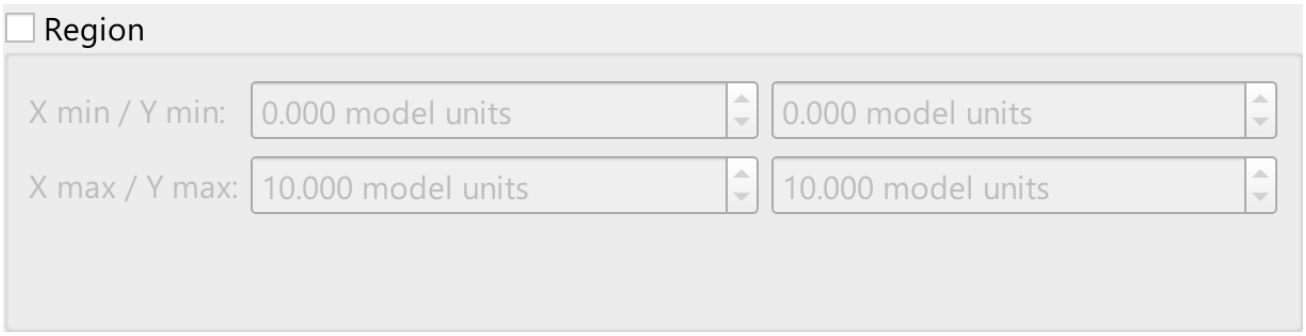
The southwest/lower and northeast/upper coordinate bounds of the desired XY rectangle. Use an established frame and correct units. A display grid or arbitrary view zoom is not this numeric rectangle.

Estimated grid / Total size (pix)

Read width, height, pixel spacing and memory notes after changing resolution, surface or bounds. An unexpectedly huge grid often indicates units/CRS/resolution problems. Increase pixel size or reduce region when the output is too large.

Source changed / stale acknowledgment

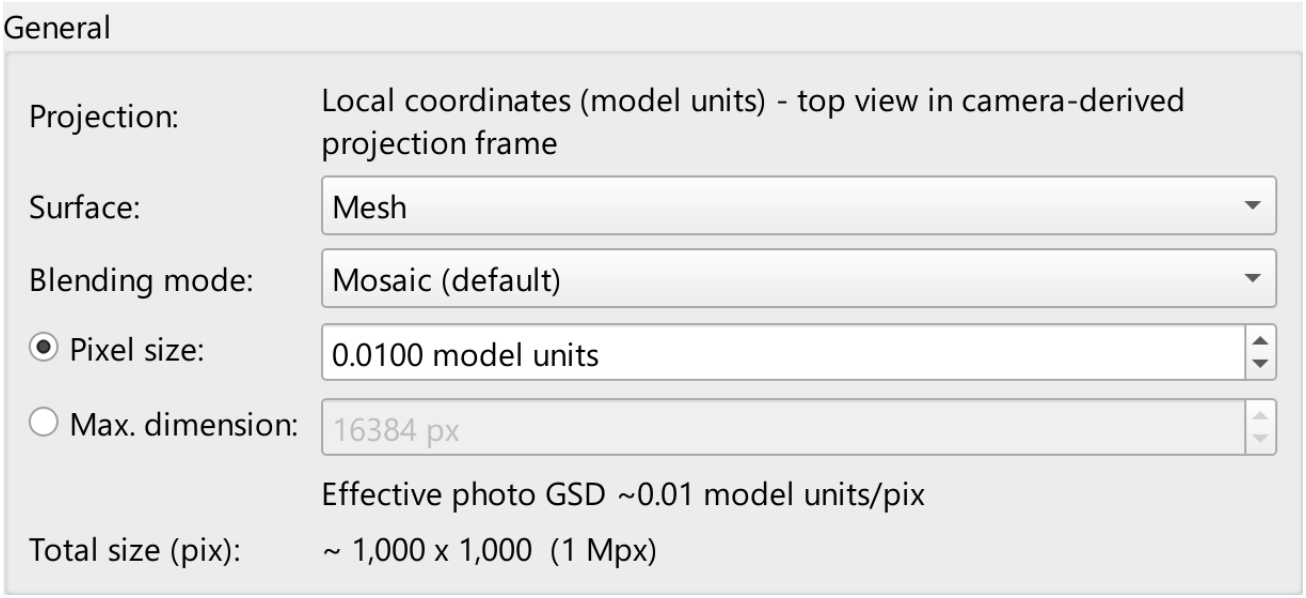
Some dialogs show a warning requiring explicit acknowledgment when source products are stale. Understand and rebuild the source if it no longer matches cameras/markers. Acknowledging permits use; it does not update the old geometry.



Additional interface detail: region. The control meanings and selection guidance are explained in this chapter.

Build Orthomosaic: surface and output sampling

Orthomosaic uses camera images and a mesh or DEM. It is a plan projection, not simply stitching arbitrary photos. A correctly aligned and referenced surface is essential for useful coordinates.



Actual interface reference: Build Orthomosaic: surface and output sampling. See the option explanations below; screenshot values are not universal recommendations.

Projection

Shows the output frame. An unreferenced project uses a local camera-derived projection frame. Geographic/world placement requires valid reference information; a nice plan image alone does not establish absolute accuracy.

Surface: Mesh / DEM

Choose the available up-to-date surface to project photos onto. Mesh can represent triangles directly; DEM is a height surface and cannot fully describe overhangs or multiple heights at one XY location.

Pixel size

Select this radio option to request explicit output spacing. Auto uses estimated photo GSD. Smaller values produce more pixels; they cannot create image detail or geometric accuracy beyond the sources.

Max. dimension

Alternative radio option caps the raster longest dimension (default 16,384 pixels). Useful to control file/memory size for previews or delivery limits. It selects an effective pixel size from the extent.

Estimated photo GSD / Total size

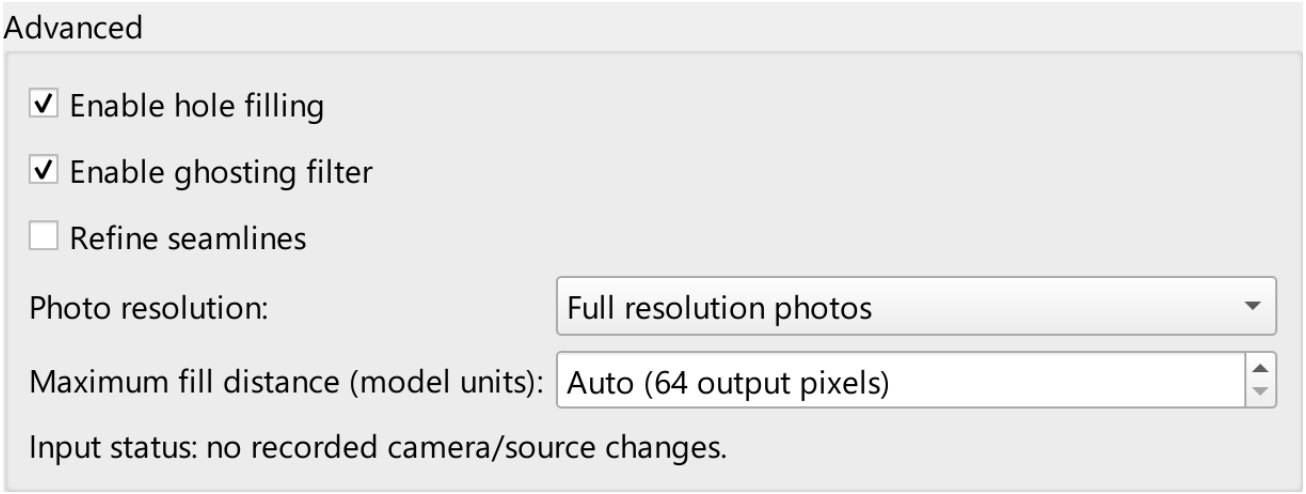
Review effective source resolution and predicted raster dimensions. Half-resolution photos increase effective photo GSD. Warnings identify outputs too large or finer than justified by the imagery.

Region

Optional bounding rectangle described in the Region chapter. Useful for excluding irrelevant margins or reducing a large product. It does not create missing coverage inside the rectangle.

Build Orthomosaic: blending and gap controls

Use blending to manage transitions between source views. It cannot remove incorrect geometry, reflections or movement from the underlying acquisition.



Actual interface reference: Build Orthomosaic: blending and gap controls. See the option explanations below; screenshot values are not universal recommendations.

Mosaic (default)

Select suitable views and mosaic their contributions, preserving detail with seam handling. General starting point. Refine seamlines is available for this mode.

Average

Average multiple colour contributions. Can smooth exposure differences while producing blur/ghosting where cameras, surfaces or objects do not agree.

Single photo (no colour blending)

Use a source contribution without colour mixing. Can preserve source sharpness but show stronger illumination transitions. This is not a guarantee that the entire output comes from only one photograph.

Enable hole filling

Infer short missing gaps rather than leaving all of them empty. Treat resulting colours/heights as inferred areas, not new measured details.

Maximum fill distance

Zero/Auto uses approximately 64 output pixels. A positive value bounds filling in model units. A very large distance can hide missing coverage by creating broad inferred patches; use only as far as acceptable.

Enable ghosting filter

Reject coarse inconsistent colour contributions where sufficient valid views exist (at least four for that test). Can help conflicts but cannot guarantee removal of moving objects or incorrect registration.

Refine seamlines

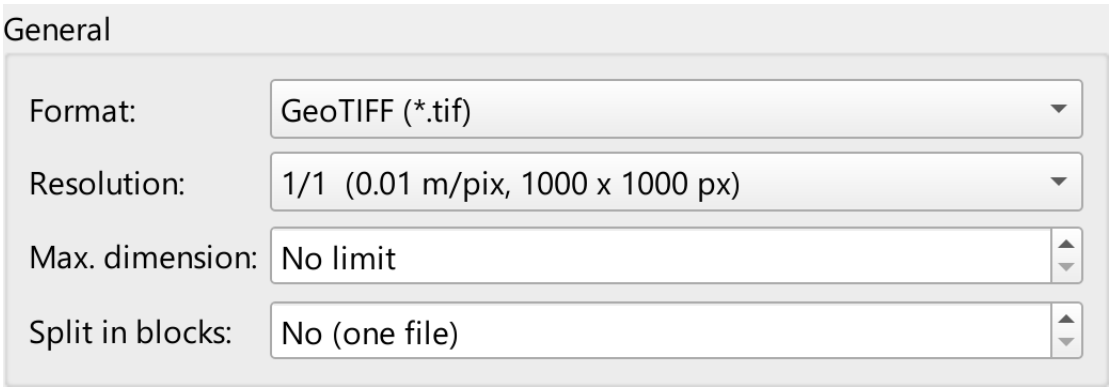
Additional seam adjustment for Mosaic blending. It can improve transitions at extra cost. Disabled for other blending modes; it is an appearance operation, not camera refinement.

Photo resolution: Full resolution photos / Half resolution of each photo

Full samples full-resolution imagery; Half uses half dimensions to reduce work and changes effective GSD. Compare output at its intended display/export scale before spending more resources.

Raster export: format and resolution

Export > Orthomosaic or DEM opens a dedicated export dialog. Building a raster and exporting it are separate operations: exporting at a smaller scale does not rerun reconstruction. Verify coordinate reference and keep associated files.



Actual interface reference: Raster export: format and resolution. See the option explanations below; screenshot values are not universal recommendations.

Orthomosaic format: GeoTIFF

Stores raster colour with geospatial metadata in a common GIS container. Use for GIS delivery where georeferencing is needed. Sidecar placement files may also be written.

Orthomosaic format: JPEG + world file

Smaller lossy colour image plus .jgw placement. Suitable for compatible CAD/GIS or presentation workflows that accept JPEG. Preserve the world file; the picture alone does not carry the full placement.

Orthomosaic format: PNG + world file

Lossless colour/transparency-capable image plus .pgw placement. Useful when JPEG artefacts are unacceptable. File size can be larger; preserve georeference sidecars.

DEM format: GeoTIFF

The elevation export choice preserves numerical height data. Do not replace it with a screenshot of an elevation colour ramp when the recipient needs actual heights.

Resolution: 1/1, 1/2, 1/4, 1/8, 1/16, 1/32

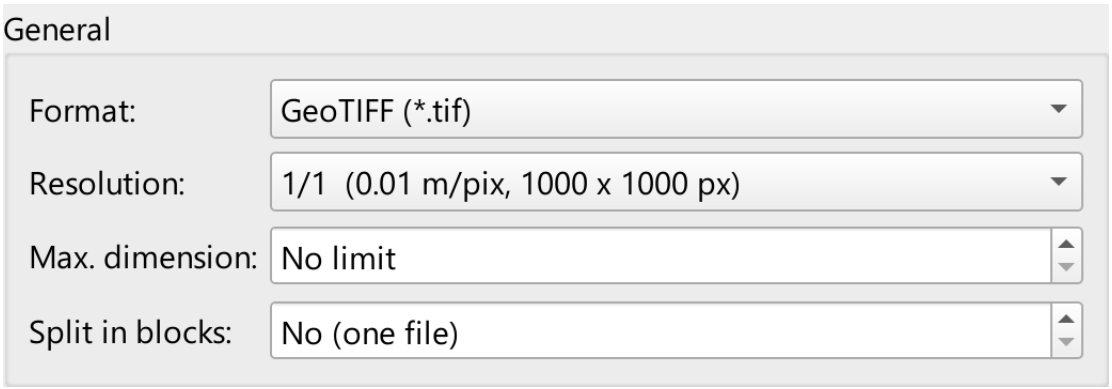
Choose full output or an available lower-resolution level; the list includes only levels available for the actual product, up to six. Each doubled denominator doubles pixel spacing and roughly quarters pixel count. Inspect dimensions and actual coordinate units, not only a suffix label.

Max. dimension

Zero means No limit. A positive cap limits the longest output side and can coarsen effective output resolution. Use for delivery-size limits; it does not improve source geometry.

Split in blocks

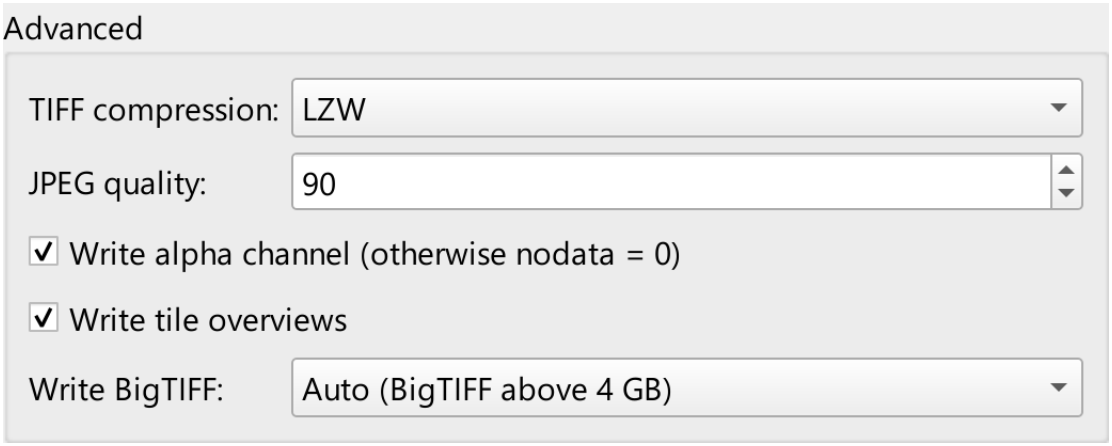
Zero means one file. A positive pixel block side divides large output into pieces. Useful for downstream size limits. Deliver the complete block set and associated placement files.



Additional interface detail: general. The control meanings and selection guidance are explained in this chapter.

Raster export: compression and sidecars

Compression affects storage and viewer compatibility. Pixel spacing and coordinate metadata determine placement; compression cannot repair an incorrect CRS or height reference.



Actual interface reference: Raster export: compression and sidecars. See the option explanations below; screenshot values are not universal recommendations.

TIFF compression: LZW

Lossless compression, a useful general GeoTIFF default. Preserves stored numerical/colour values while often reducing file size.

TIFF compression: Deflate

Another lossless compression method. Choose when supported by the receiving software and compare file/runtime size if needed. It does not change the raster spatial resolution.

TIFF compression: None

Write uncompressed TIFF data. Larger files, but potentially useful for a restrictive reader or controlled compatibility test.

TIFF compression: JPEG

Available for colour orthomosaic export, not numerical DEM compression. Lossy; use only when acceptable to the destination and intended colour quality.

JPEG quality

Range 10-100, default 90. Higher values reduce visible loss and usually increase size. Relevant to JPEG output/compression; it is not a DEM height precision setting.

Write alpha channel

Orthomosaic only. Alpha preserves an explicit transparency mask where supported. Otherwise nodata may be written as zero, which can be confused with real black pixels in some viewers.

Write tile overviews

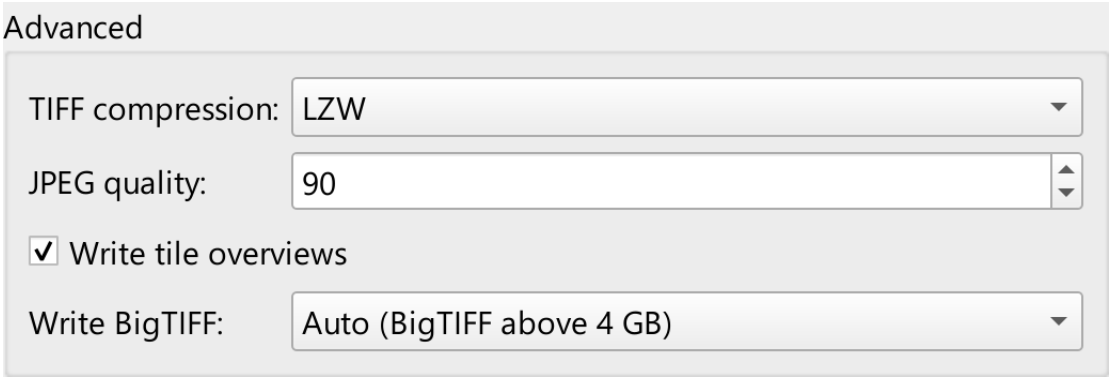
Include lower-resolution preview levels for efficient raster viewing. Improves navigation in compatible software; it does not add measured information.

Write BigTIFF: Auto / Yes / No

Auto uses BigTIFF for output beyond the classic approximately 4 GB limit. Yes forces it; No favors classic TIFF but cannot make oversized classic output valid. Confirm the recipient supports BigTIFF before delivery.

World/projection/placement files

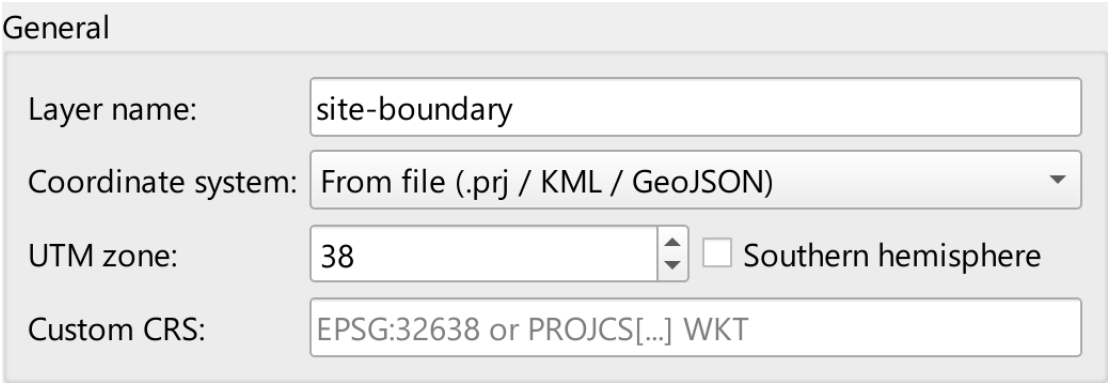
Coordinates include the project world offset. Keep .tfw/.jgw/.pgw and any CRS/placement notes/scripts alongside the raster. Local/arbitrary coordinates remain local; an export unit label alone does not establish metres.



Additional interface detail: advanced. The control meanings and selection guidance are explained in this chapter.

Import Shapes: coordinate choices

Use Project > Import Data > Import Shapes for vector overlays. A correct coordinate interpretation matters more than whether the shape looks close at one zoom level. Review the detected file CRS and project CRS shown at the top.



Actual interface reference: Import Shapes: coordinate choices. See the option explanations below; screenshot values are not universal recommendations.

Layer name

Name the imported layer so it is identifiable in the Project tree and style/export choices. It does not change feature coordinates or attribute values.

From file (.prj / KML / GeoJSON)

Use the declared source coordinate reference where supported. Appropriate when accompanying metadata is valid. Do not assume a DXF without CRS metadata is automatically in project coordinates.

Project coordinate system (world)

Treat the file coordinates as already in the full project world frame. Use for vectors exported in the same established CRS, including its full easting/northing coordinates.

Project local coordinates (add offset)

Treat coordinates as the internal local frame and add the project world offset when needed. Useful for compatible local engineering exports, but choosing it for already-world coordinates can produce a large shift.

WGS 84 (EPSG::4326)

Interpret source coordinates as geographic longitude/latitude. Verify axis order and degree values. Not appropriate for a metric CAD drawing unless it has been transformed beforehand.

WGS 84 / UTM zone

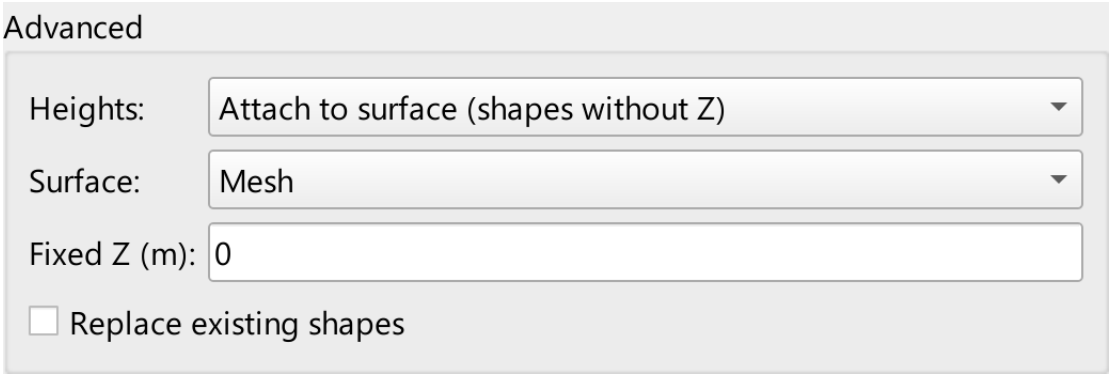
Interpret in the selected UTM zone and hemisphere. Zone ranges 1-60; Southern hemisphere changes hemisphere. Choose the actual source zone rather than a zone merely near the project.

Custom (WKT / EPSG)

Enter an explicit supported CRS description, for example an EPSG code or WKT. This declares the source interpretation; it is not a free-form image alignment adjustment. Unsupported CRS transformations can be rejected.

Import Shapes: heights and replacement

Choose how imported vectors receive Z before styling or measuring them. Draping onto an imperfect surface transfers that surface error to the shapes; it is not an independent height measurement.



Actual interface reference: Import Shapes: heights and replacement. See the option explanations below; screenshot values are not universal recommendations.

Attach to surface (shapes without Z)

Drape only shapes missing height values. Preserves existing valid Z values while assigning surface heights to 2D shapes.

Attach all shapes to surface

Replace height placement with the chosen surface for every shape. Useful for intentional surface overlays; can discard meaningful heights already present in the file.

Keep Z from file

Retain source height values. Use when they already represent the intended vertical reference. Ensure the source and project datums/units are compatible.

Fixed Z for shapes without Z

Assign a constant height only where Z is absent. Fixed Z is the chosen value in the relevant project/model units; a field label in metres is not proof of established scale.

Surface

Choose an available Mesh/DEM surface for draping. Check currency and coverage. A missing/insufficient surface can prevent sensible attachment; do not accept distant misplaced geometry merely because the file imports.

Replace existing shapes

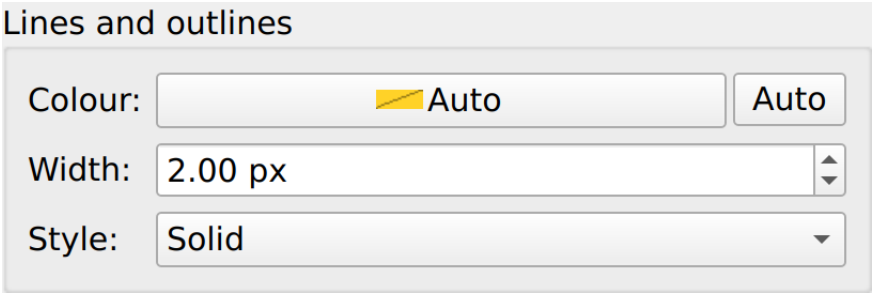
Remove the previous shape set in favour of this import when enabled/checked. Leave unchecked to retain existing shapes alongside the import. Review scope before confirming.

Warnings about distant geometry

A large separation may indicate wrong CRS, units, zone or local/world-offset choice. Correct those causes before choosing to keep or transform the imported geometry.

Shape Settings: scope, lines and polygons

Survey > Shape Settings, or the layer/shape context menu, edits how imported vectors appear. It does not move their vertices or change measurement coordinates. The dialog previews style changes.



Actual interface reference: Shape Settings: scope, lines and polygons. See the option explanations below; screenshot values are not universal recommendations.

Layer / Apply to

Choose the layer, then Layer for all its shapes or Selected shape (override) for one feature when available. A feature override can differ from the layer. Verify scope before applying a batch style.

Lines and outlines: Colour

Choose an explicit line colour or automatic colour inherited from the file/layer. Use contrast with the underlying surface; a style colour is not a classification of reconstructed accuracy.

Width / Style

Width controls screen line thickness (0.5-20 px); Solid or Dashed controls appearance. Wider lines help visibility but can hide small geometry in a screenshot. These are display pixels, not real-world line widths.

Points: Size / Symbol

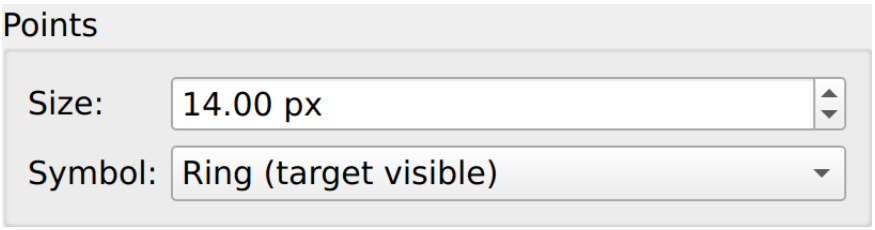
Screen marker size is 2-60 px. Choose Ring (target visible), Filled circle, Square, Diamond, Triangle or Cross to distinguish overlays. Ring leaves the target centre visible; Filled circle covers its centre. It changes drawing only, not point coordinates.

Polygons: Fill colour / Fill opacity

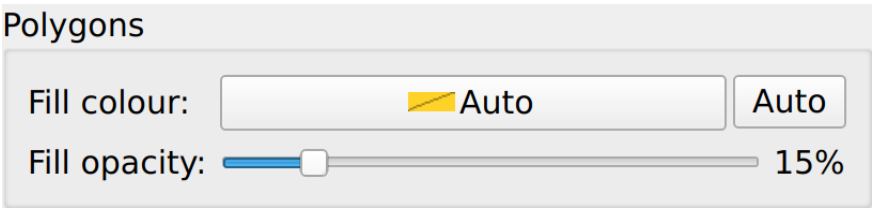
Choose fill colour, or inherited line colour, and opacity from 0-100%. Low opacity reveals the model beneath. Filling a polygon visually does not create a mesh surface.

OK / Apply / Cancel

Apply commits the current style and establishes a new dialog baseline. OK applies and closes. Cancel restores changes since opening or the last Apply. Reset Defaults restores layer defaults, or removes the selected-shape override so it inherits its layer style.



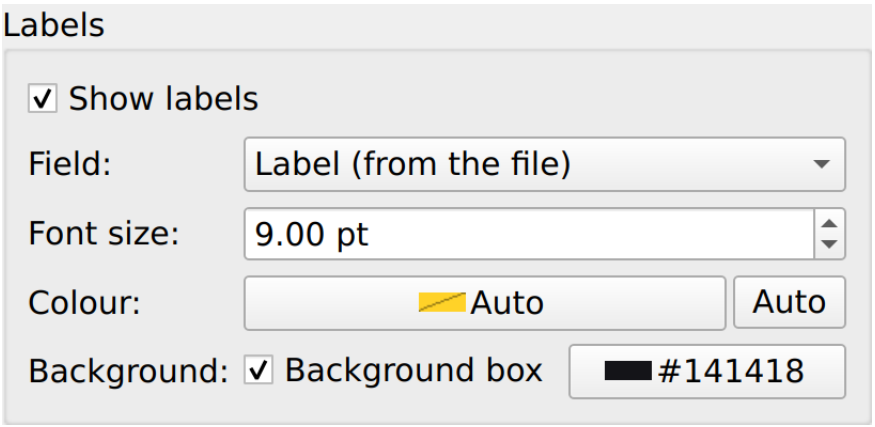
Additional interface detail: points. The control meanings and selection guidance are explained in this chapter.



Additional interface detail: polygons. The control meanings and selection guidance are explained in this chapter.

Shape Settings: labels and depth behavior

Readable labels and deliberate occlusion settings make imported geometry easier to interpret. These choices describe presentation, not whether a feature is geometrically above or below the model in the data.



Actual interface reference: Shape Settings: labels and depth behavior. See the option explanations below; screenshot values are not universal recommendations.

Show labels / Field

Enable labels. Label (from the file) uses its imported label; Shape name uses the application feature name; bracketed choices such as [name] use that attribute column. Attribute choices depend on the import. Pick a meaningful identifier rather than assuming every dataset has identical fields.

Font size / Colour

Choose label text size (5-48 pt) and colour. Use sufficient contrast at the intended viewing scale. Label size does not alter output coordinate units.

Background box / background colour

Add a solid label backdrop for contrast over complex imagery. A useful readability choice; reduce clutter by hiding labels when they obscure inspection.

Draw on top of the model

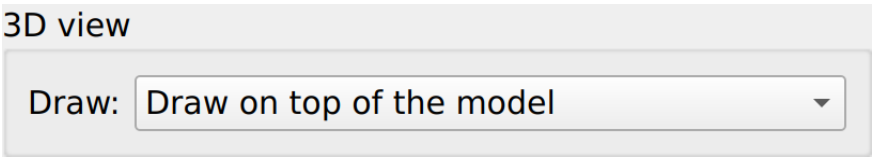
Always draw the overlay above the mesh visually, including otherwise hidden parts. Useful to locate all features, but can obscure the actual depth relationship.

Depth-tested (hidden parts dashed)

Let the model occlude features; hidden parts are shown dashed/faded. Useful for depth relationships. Transparent View draws shapes on top regardless of this mode.

Export and styling

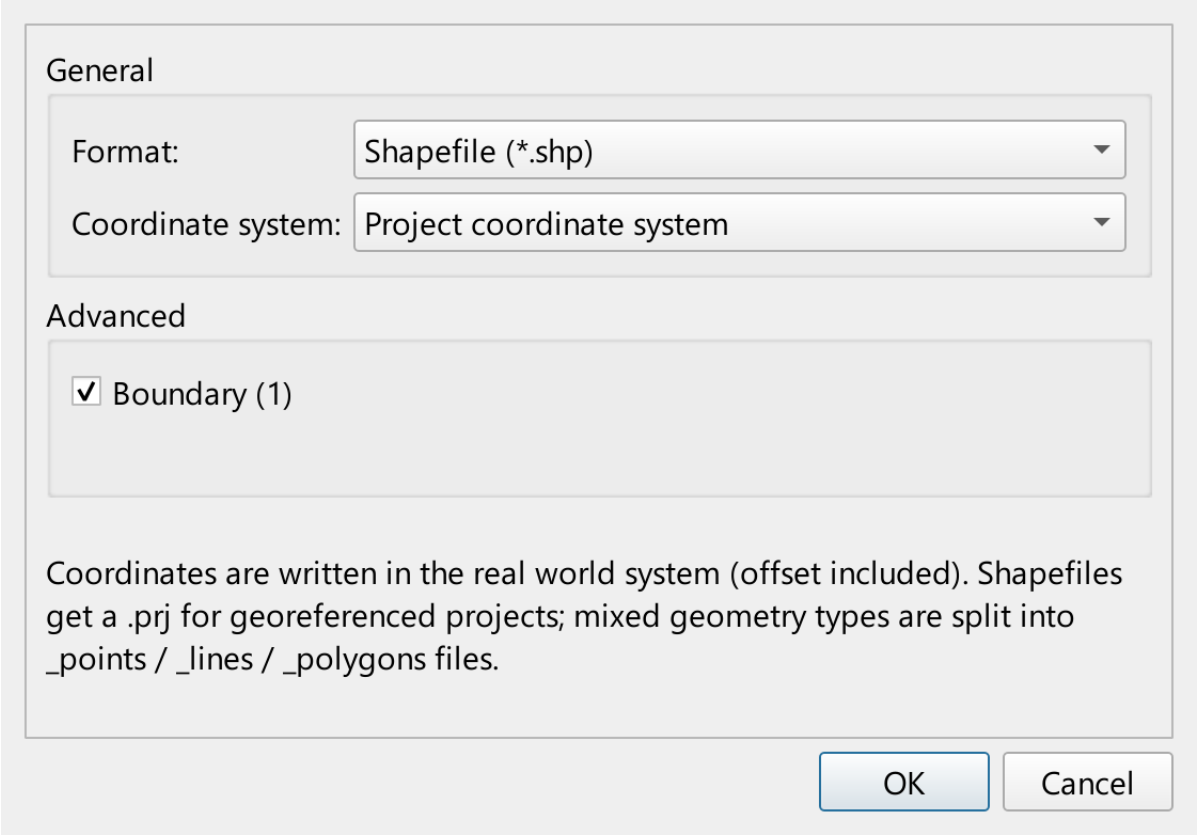
Use Export Shapes for actual vector data. Do not assume every destination format/viewer preserves all application-specific screen styling; inspect the delivered file in the receiving software.



Additional interface detail: 3d view. The control meanings and selection guidance are explained in this chapter.

Export Shapes

Choose the output format, coordinate interpretation and included layers. Mixed feature types and sidecar requirements differ between formats; keep the entire export package.



Actual interface reference: Export Shapes. See the option explanations below; screenshot values are not universal recommendations.

Shapefile (*.shp)

Common GIS format with multiple required components, including .shp/.shx/.dbf and a .prj for a georeferenced project. Mixed points, lines and polygons can be split into separate suffixed files. Deliver all components.

AutoCAD DXF R12 (*.dxf)

CAD vector exchange. Verify units and world/local frame on import into CAD. A DXF alone does not guarantee every receiver interprets a geographic CRS automatically.

GeoJSON (*.geojson)

Text-based vector exchange with attributes. Choose an appropriate offered output coordinate system and verify the intended receiving application; do not silently treat local units as longitude/latitude.

Coordinate system

Select from the systems available for the actual project. These choices depend on its reference. World export includes the required offset; local output is useful only when the receiver expects the same local frame.

Layer checkboxes

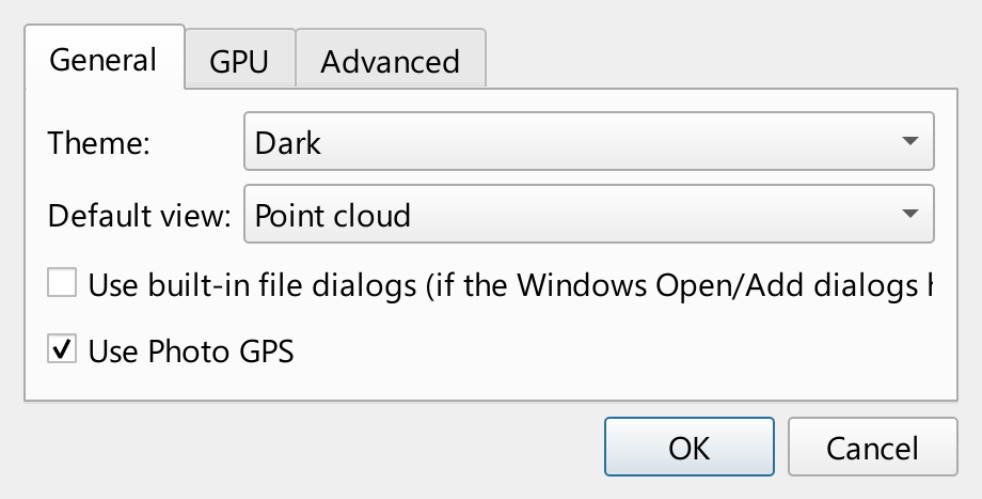
Checked layers are exported; unchecked layers are excluded from this export. This is an export selection, separate from hiding layers in the current view.

After export

Open the result in the target CAD/GIS viewer. Check coordinates, elevations, attribute fields and multipart geometry. Style and geometry interoperability are separate checks.

Preferences: General, GPU and Advanced

Project > Preferences and the Settings toolbar button open the same dialog. General appearance settings are stored between sessions; photo-GPS policy belongs to the project reference behavior.



Actual interface reference: Preferences: General, GPU and Advanced. See the option explanations below; screenshot values are not universal recommendations.

Theme: Dark / Light

Select the interface palette for comfortable reading. This affects widget/view appearance, not processing maths, photo colours or exported coordinates.

Default view: Tie points / Point cloud / Model

Choose the preferred product display when appropriate/available. A default view cannot display a product that has not been built or loaded.

Use built-in file dialogs

Use Qt file dialogs instead of native platform dialogs, especially if Windows Open/Add dialogs hang. The file chooser appearance changes; supported processing products do not.

Use Photo GPS

Global project policy for using valid photo GPS in pair preselection and camera references. Off does not remove photographs. Adding markers does not automatically switch it off. Operation-specific choices still determine whether eligible GPS is used for that step.

Legacy-project GPS note

Some older projects preserved per-camera reference exclusions; the app can show a note and keep GPS off until deliberately enabled. Review the project policy before enabling all valid photo positions.

Use GPU for feature extraction/matching

Requests supported GPU feature work. If no suitable NVIDIA driver/engine is found, CPU alignment remains available. Neural features have their own compatible runtime/model requirements. This checkbox does not enable CPU dense depth reconstruction.

COLMAP executable

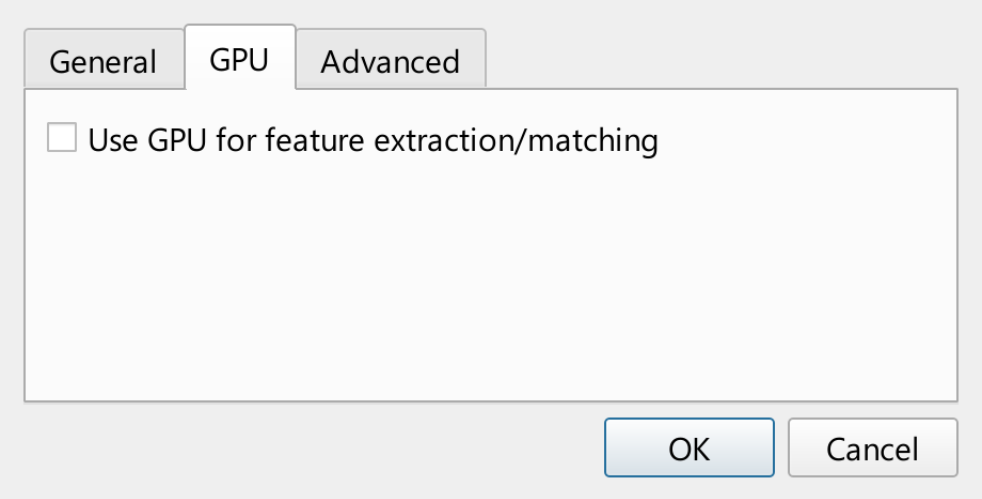
Optional explicit reconstruction-engine path. Browse selects an executable; blank permits auto-detection. Choose a compatible engine, not an arbitrary similarly named program. Wrong paths can cause missing-engine or unsupported-option errors.

Detected

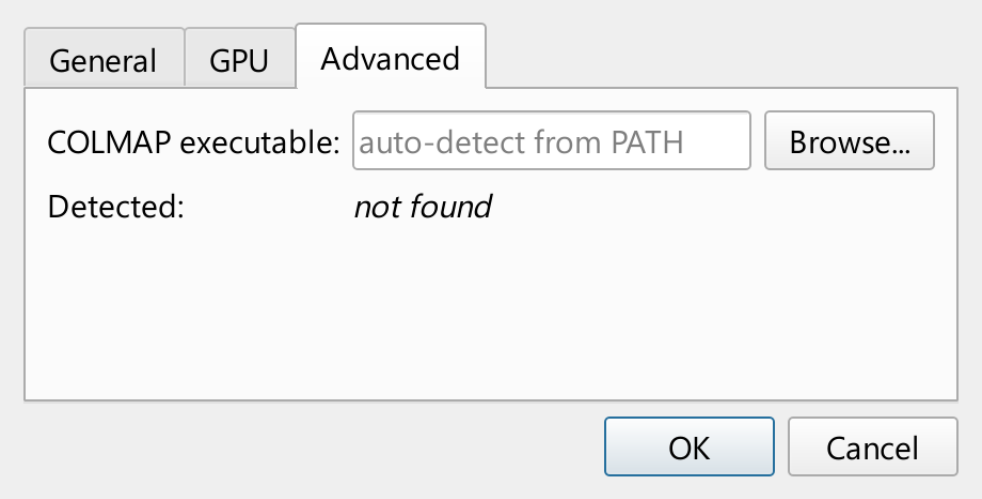
Read-only discovered executable location. The manual screenshots are from remote Linux; Windows installation paths differ. Inspect this when troubleshooting engine discovery.

OK / Cancel

OK stores the selected settings and applies theme/GPS changes as appropriate. Cancel closes without accepting this dialog. Re-running a processing operation is still needed to apply calculation changes to previously built products.



Additional interface detail: preferences gpu. The control meanings and selection guidance are explained in this chapter.



Additional interface detail: preferences advanced. The control meanings and selection guidance are explained in this chapter.

Export formats, reports and packaging

Use Export for deliverables and Capture View for pictures. Format availability follows the current product: a mesh without a compatible texture does not offer the same textured choices as a finished textured model.

Point cloud: PLY / DXF

PLY stores compact geometry and supported attributes. DXF writes POINT entities and can be very large. LAS and compressed LAZ retain supported attributes/classes with explicit coordinate quantization. Eligible exports can use sparse tie points if no dense cloud exists. See the LAS/LAZ reference.

Mesh: Stanford PLY

Export triangle geometry, or a compatible textured PLY when offered. Check whether the recipient supports the texture conventions and referenced image files. Geometry-only PLY is different from a texture atlas package.

Mesh: AutoCAD DXF

CAD mesh export. Inspect scale/orientation in CAD. Large meshes can overwhelm a CAD viewer; simplify intentionally through the mesh target rather than expecting DXF to optimize the geometry.

Mesh: OBJ with texture

Writes model geometry plus material/texture dependencies. Deliver the OBJ together with its MTL and referenced images, preserving relative paths. Moving only the OBJ commonly loses colour.

Mesh: GLB with texture

Binary glTF packaging useful for 3D viewers. The app keeps local float32 vertices and world placement in a root translation. Confirm that the receiving viewer honors the transform and expected up direction.

Mesh Contours: interval

Choose a strictly positive spacing in world/model Z units. Major every five intervals, minor between. Too small an interval can create a heavy DXF and imply unsupported precision. Contours follow the actual mesh including noise and inferred geometry.

Tiled Model: ZIP / Folder

ZIP packages the tile tree; Folder copies it. Retain all indices/content. Inspect the chosen 3D Tiles version in the target viewer; a successfully written ZIP is not proof of viewer compatibility.

Cameras CSV / Reference CSV

Cameras exports estimated aligned camera information with the appropriate offset; Reference exports input photo longitude/latitude/altitude. Do not confuse estimated coordinates with independently measured input reference.

Markers TXT / XML

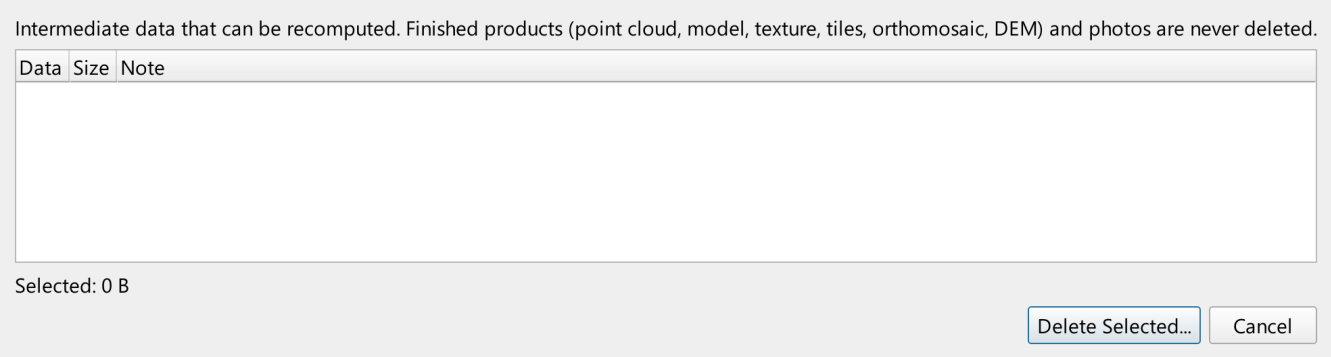
Text contains reference/residual information available at export time. XML preserves marker structure and linked image observations for compatible workflows. Incomplete/unchecked markers require interpretation; blank error fields are not zero.

Generate Report: HTML + PNG

The report uses project name, camera statuses, counts, a view image and processing log. Deliver the companion image with the HTML. It is not a comprehensive certified numerical accuracy report or a PDF export option.

Cleanup, progress, logs and support

Resource estimates and progress help manage a run, but elapsed/remaining time is approximate. Keep the full error message and relevant log when an operation fails; a short description such as file missing is insufficient to identify the failed component.



Actual interface reference: Cleanup, progress, logs and support. See the option explanations below; screenshot values are not universal recommendations.

Clean Up Intermediate Data window

Data, Size and Note describe removable recomputable categories. Check the intended items and review Selected total. Delete Selected opens the deletion flow; Cancel leaves the files. Finished cloud/model/texture/tiles/ortho/DEM and source photos are protected by this operation.

Progress window

Read operation/stage text, overall and stage progress where shown, log output and approximate remaining time. Some stages lack exact work counts. A stationary percentage alone does not prove a hang; examine stage activity/logs.

Cancel

Request cancellation and allow the worker to stop/clean up. Not every external step stops instantly. Verify whether a previous valid product was retained before restarting; do not manually delete in-use workspace files.

Jobs and Console

Jobs records operation entries; Console holds useful messages and permits Python commands. The console is an advanced interface: only execute code you understand. Neither panel is a substitute for saving the project.

Report a Problem: description

Explain the action, selected settings, expected result and actual error. Include application version and relevant hardware. Review the generated report for paths or private information before sharing.

Details / Copy report / Open report folder

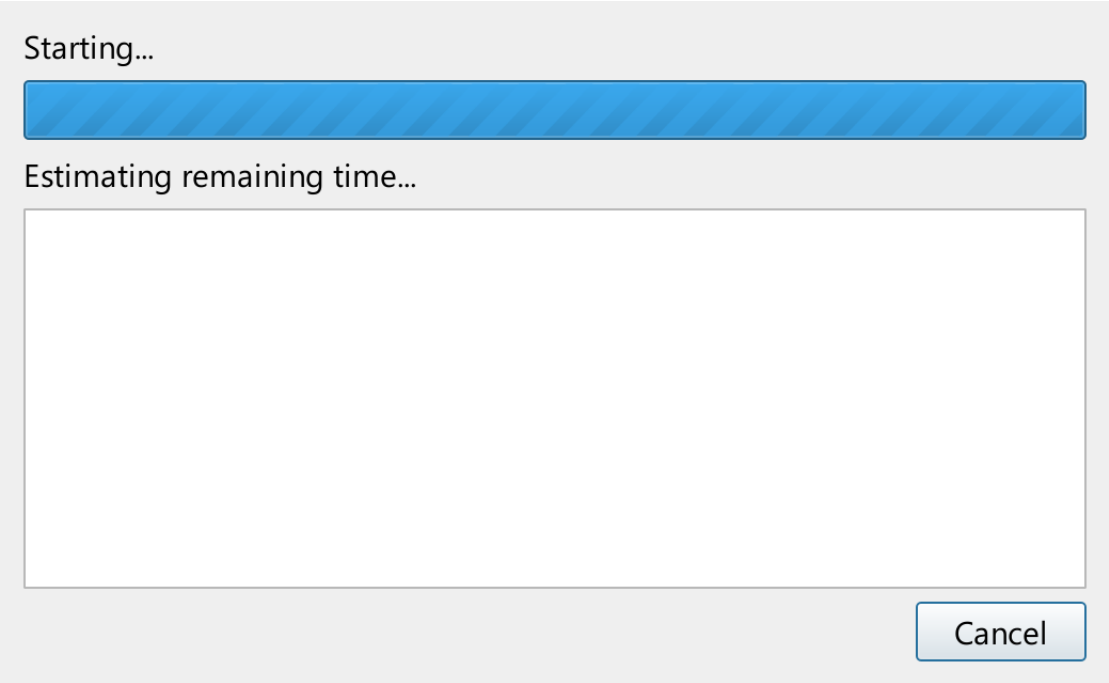
Details reveals saved technical text. Copy report copies it for review. Open report folder shows the saved report/ZIP for later sharing. The report remains available when offline.

Report to GitHub / Close

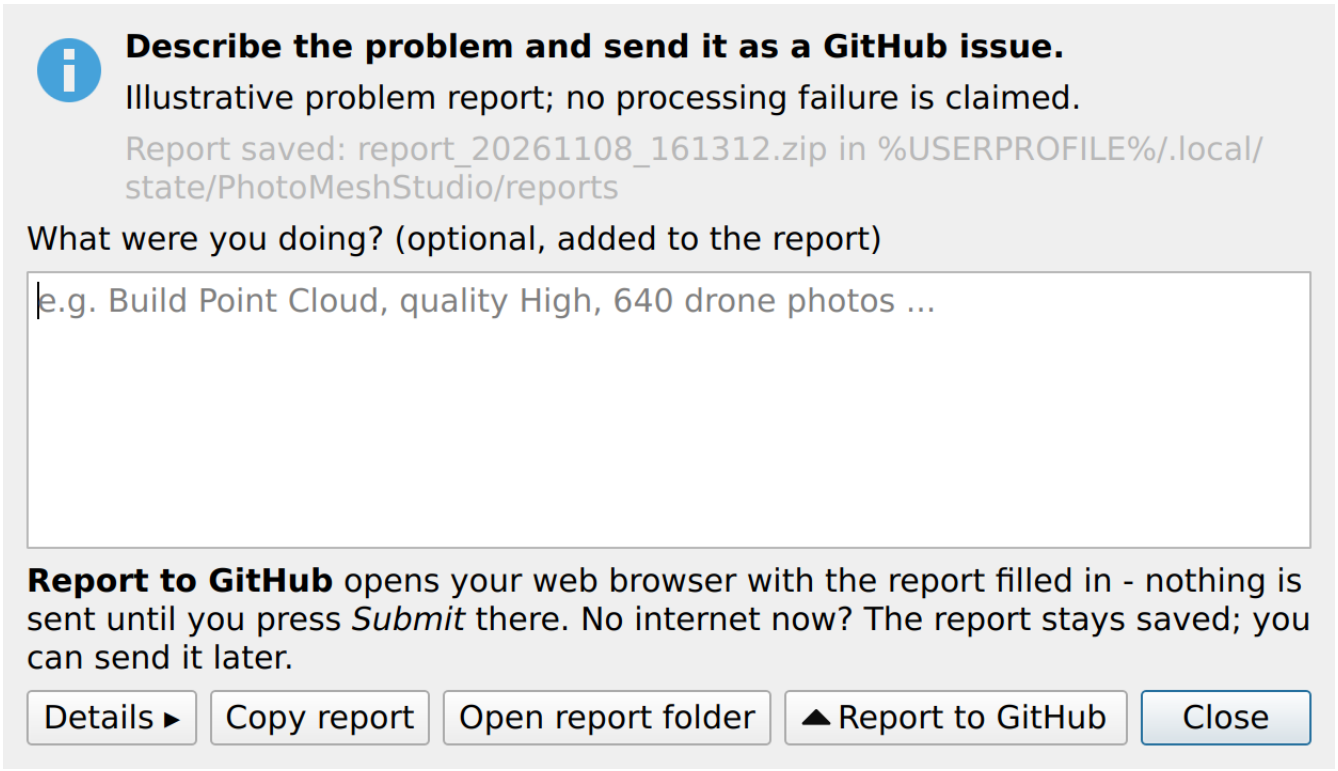
Report to GitHub opens a browser issue form; you still review and submit it yourself. Close dismisses the report dialog. This manual does not send diagnostic reports automatically.

About / View full license

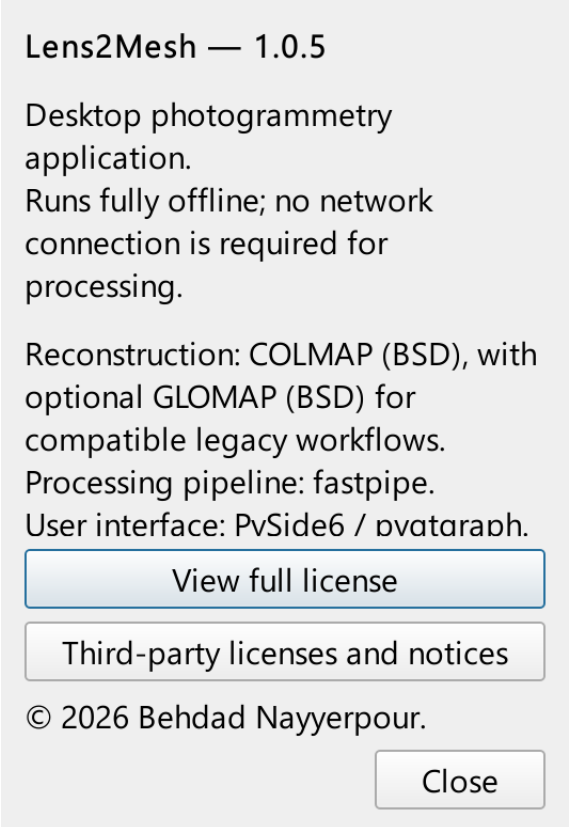
Verify version 1.0.5 and consult the actual licence. Personal and educational non-commercial use is permitted; commercial use, including evaluation for company workflows, requires prior written permission. Third-party components retain their own licences.



Additional interface detail: progress. The control meanings and selection guidance are explained in this chapter.



Additional interface detail: report problem. The control meanings and selection guidance are explained in this chapter.



Additional interface detail: about. The control meanings and selection guidance are explained in this chapter.

Choose settings for common situations

These are starting strategies, not guaranteed presets. Test a small baseline, change one relevant group of settings, compare geometry and resource cost, and keep the better validated result.

A folded room or split reconstruction

First inspect sharpness, repeated patterns and views linking corners. Compare Incremental against the prior solver; try Exhaustive on a manageable set to test missed pair links. Do not proceed to a high-quality dense build while camera geometry is visibly wrong.

Missing blank walls or floors

Inspect whether source photos contain enough distinct texture and translated viewpoints. Compare support/isolation filters if removal is suspected. Keep the terrain filter off for interiors. Ultra high cannot manufacture matches on an untextured surface; additional suitable capture may be necessary.

Too much floating dense noise

Check alignment first. Compare Mild versus Moderate depth filtering and support/isolation settings in a saved baseline. Track which filter removes points and inspect legitimate fine features before choosing stronger cleanup.

Memory or disk limits

Reduce dense quality, use realistic neighbour counts, limit meshing working points, choose modest texture pages and reduce output raster extent/resolution. Decide whether depth reuse justifies disk retention. Work from estimates with headroom, not a promise of exact peak usage.

A mesh has bridges or islands

Compare trimming and edge multiplier settings and component thresholds. Stronger cleanup removes false surfaces but may discard genuine disconnected objects. Inspect a nontextured view to judge shape.

A texture looks blurry or ghosted

Check camera geometry, source image sharpness and movement. Compare appropriate mapping/blending, full source resolution and adequate atlas size. Increasing atlas pixels alone does not correct misregistered sources.

Reference residuals look contradictory

Check marker identity, XYZ axis order, units, height references and observation confirmation. Separate control and independent checks. Do not assign small accuracy weights to approximate coordinates taken from the same model.

An export appears shifted or colourless

Check local/world frame and offset interpretation, CRS/units, root transforms and required sidecars/MTL/images. Open the complete package in a compatible viewer before concluding the reconstruction is wrong.

A complete production workflow

Use the sequence as a working checklist. Stop at the product you need; a point-cloud delivery does not require texture, and a 3D viewing model does not automatically require a DEM.

1. Capture and organize

Acquire sharp photos with substantial overlap, translated viewpoints and stable camera settings. Preserve originals and capture order. Measure control/check coordinates independently when accuracy or metric scale matters.

2. Create, add and save

Create a project, add the intended photos and save to a local folder. Inspect photo count, dimensions and paths. Verify the global GPS policy before alignment.

3. Align and inspect

Choose matching/mapper/resolution suitable to the dataset. Inspect registered cameras, sparse shape and missing photos. Correct input/reference problems before dense work.

4. Observe markers and optimize

Place identical markers precisely on source photos. Import measured reference with the correct CRS, reserve independent checks and set honest uncertainties. Optimize with only the reference sources you intend.

5. Build dense geometry

Choose a manageable dense quality and filter settings. Confirm CUDA/memory/disk preflight. Inspect the cloud and removal counts. Keep/reuse depth maps only where useful and compatible.

6. Build the required surface

Mesh for 3D or a suitable DEM source for surface heights. Select face targets, trimming and interpolation to suit the deliverable. Inspect gaps, bridges and inferred areas.

7. Build colour/raster/tile products

Texture a validated mesh; orthomosaic uses a suitable mesh/DEM; tiled model packages an available source. Match resolution to actual source information and viewer/resource limits.

8. Validate and deliver

Inspect independent checks and product coverage, rebuild stale dependencies, export the intended formats, open them in the receiving software, package all companions and save the final project.

Version scope and reading this manual

This edition updates the approved complete 1.0.4 manual for Lens2Mesh 1.0.5. Changed menus, reference/calibration controls, terrain products and section tools are checked against release-candidate source. Updated controls are native high-DPI captures of the actual Windows Qt application with isolated settings. Unchanged explanations/screenshots are retained from the approved edition. UI demonstration entries are not measured survey results.

Screenshots are UI references

The settings screens explain controls; they are not claims that every product was computed on the screenshot project. Disabled entries can reflect the current prerequisites, source choices or hardware. All dropdown choices are documented in the text, even when only one is visible in a screenshot.

Defaults and persisted values

Some dialogs remember prior selections. Examples shown in screenshots need not be the defaults on every existing project. Read the option explanations and the displayed context before accepting settings.

Source-verified terminology

The menu uses Build Mesh and Clean Tie Points while some dialog/internal terminology uses Build Model and Gradual Selection. This manual describes the visible 1.0.5 options, including dynamically added controls, rather than future planned features.

Accuracy and limitations

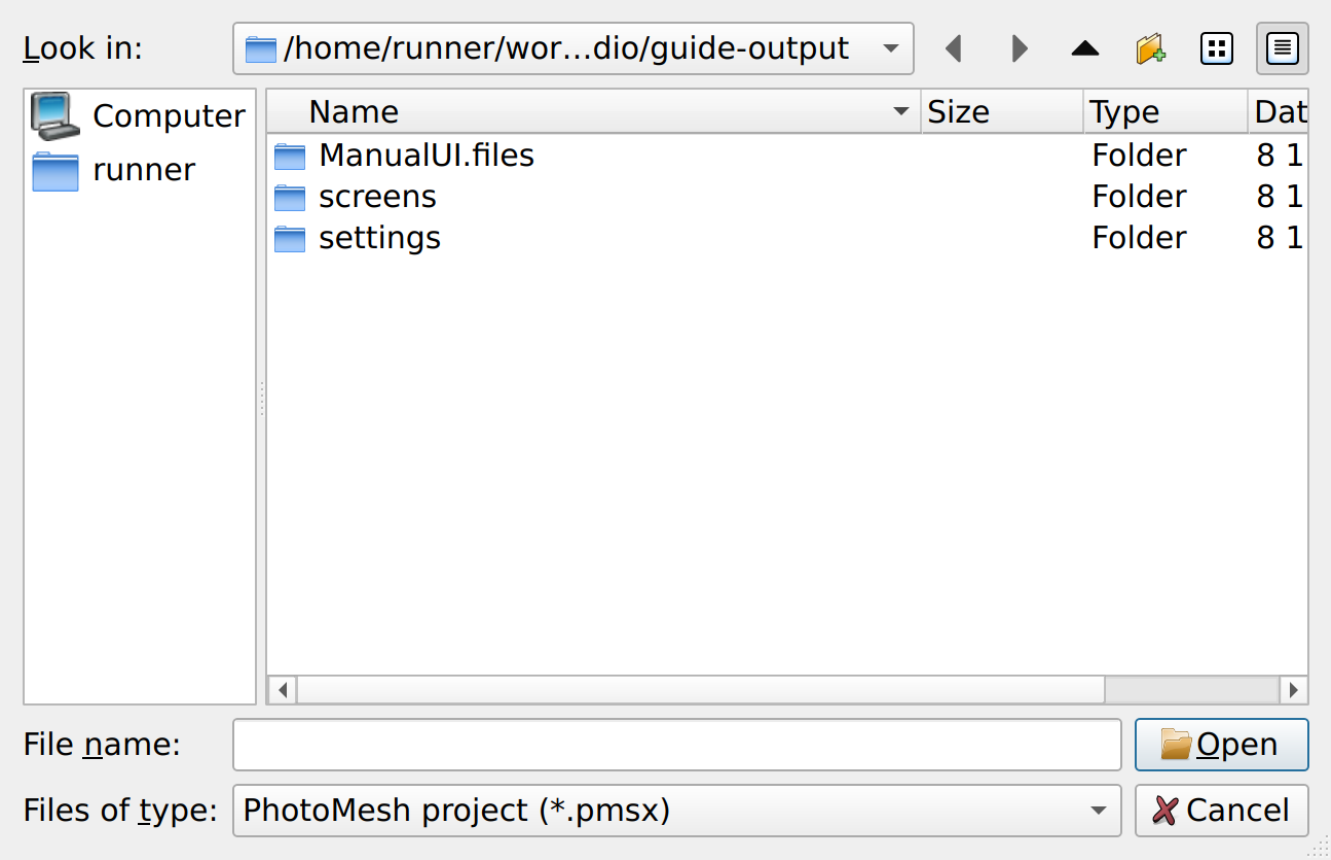
Reprojection RMS, GSD, cell spacing, texture size and control residuals measure different things. None alone proves absolute accuracy. Independent checks, appropriate geometry and correct coordinate/height references are necessary for a defensible survey result.

Reference sources

The versioned application menus, dialogs, workflow controls, processing handlers and export implementations were checked together. The PDF menu index is generated from the actual reachable menu registry, with a coverage check for every leaf command.

File pickers, colours and confirmation dialogs

Several commands use shared operating-system or Qt dialogs instead of a unique PhotoMesh settings window. Their visual layout differs across Windows/Linux, but file type, destination and confirmation choices still matter.



Actual interface reference: File pickers, colours and confirmation dialogs. See the option explanations below; screenshot values are not universal recommendations.

Open / Add Photos

Navigate to the intended folder, select the correct files and file-type filter, then Open. Multi-select only files belonging to the survey. Cancel leaves the command unstarted. The app supports common JPG/JPEG, PNG, TIF/TIFF, BMP and WebP photo extensions; actual decoding still depends on valid image content.

Save / export destination

Choose a filename, folder and offered format. The format filter can determine export behavior; do not merely rename a file extension and assume its content converted. Confirm replacement only when intending to overwrite the prior deliverable.

Folder selection

Used by Add Photo Folder, relocation and compatible directory-based imports. Select the folder containing the expected inputs, not a parent full of unrelated surveys. Relocate Photos searches subfolders; Add Photo Folder is not the same operation.

Colour picker

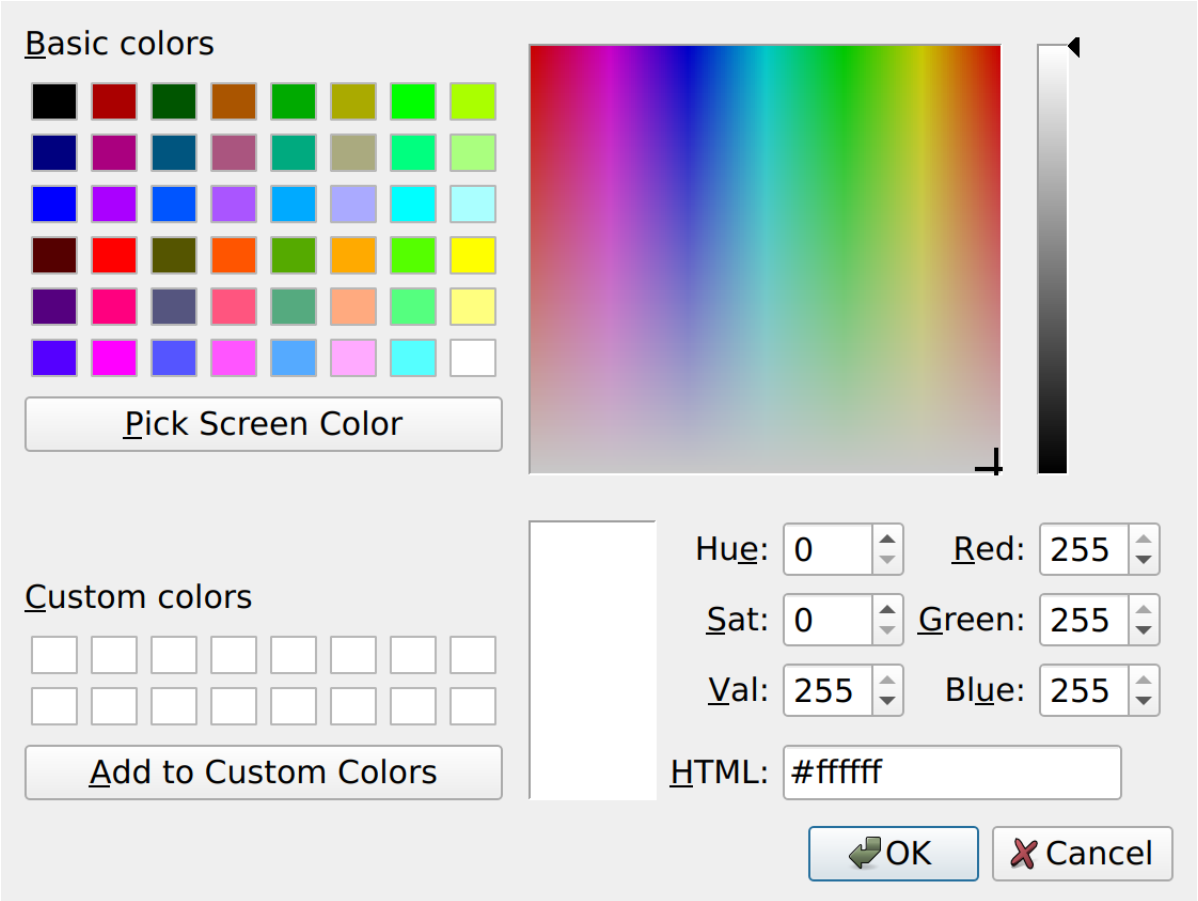
Basic colours selects a preset; Custom colours stores your own swatches. Hue/Saturation/Value and Red/Green/Blue are alternative ways to specify a colour; HTML uses a hexadecimal colour value where shown. Add to Custom Colours stores the chosen swatch. Confirm to use it; Cancel retains the prior colour. Shape colour Auto restores inheritance, rather than choosing black or transparency.

Yes / No / Cancel / Save / Discard

Read the operation named in the prompt. Yes confirms that specific operation, No declines it, Cancel abandons it, Save preserves current project changes and Discard abandons unsaved metadata edits. These are not interchangeable global approval choices.

Validation message

Invalid bounds, unavailable engines, zero up vectors or unsuitable input references can keep a dialog open. Correct the highlighted cause instead of repeatedly clicking OK; the app does not silently turn every invalid setting into a safe default.



Additional interface detail: colour picker. The control meanings and selection guidance are explained in this chapter.

Photo marker actions: pin, move, ignore and delete

In a source photo, right-click near a visible marker to act on the marker under the cursor. These are distinct from the table Control/Check checkbox. In 3D or raster views, marker actions are limited to selection/display.

Delete Marker

Remove the marker identity, not just its location on the current image. Use Remove Marker from this Photo when only one observation is wrong. Read the confirmation and verify linked photos before deleting.

Pin Marker

Confirm the displayed photo location as a pinned measurement. Use only after verifying its precise physical feature, especially if it was predicted. A pinned wrong point is a stronger mistake, not a better observation.

Ignore Marker

Exclude the marker using its separate disabled/ignored state. It remains distinguishable for review; this is not the same as making it a Check point. Ignored markers are excluded from the relevant calculations/summaries.

Use Marker

Clear the ignored state so the marker can participate according to its Control/Check status and eligible measurements. It does not create missing XYZ coordinates or photo observations.

Move Marker...

Enter a crosshair placement mode, then click the corrected position on the source photo. Escape cancels. Confirm the physical feature and related observations afterward; the operation does not rewrite surveyed XYZ.

Place Existing Marker

Choose a listed marker identity and pin it at the clicked pixel, without first selecting its table row. Use it to efficiently add the same target across images. Choosing the wrong name links unrelated physical features.

Remove Marker from this Photo

Remove that photo measurement while keeping the marker entry and its observations elsewhere. Useful for occluded or misidentified features; it differs from Delete Marker.

Ctrl+click and dragging

The photo view also supports a shortcut for adding a new pinned marker and interaction with visible marker handles. Use the explicit context commands when uncertain about the active marker, and verify the resulting observation state.

Import reference files and preserve photo identity

Use Project > Import Data > Import Markers for supported XML/TXT/CSV reference data. File structure and declared coordinate system determine how values are interpreted; a column labelled X does not establish units by itself.

Simple coordinates

A plain list can contain label, X, Y, Z and optional accuracy, separated by comma, semicolon, tab or spaces as supported. For example, a local measured row may read GCP_A,0.000,0.000,0.000,0.010. This is a syntax illustration, not suggested coordinates for your survey.

Coordinate-system header

Supported text reference files can declare a # CoordinateSystem: header and named columns. Use an explicit correct reference when available rather than relying on numerical guessing. Verify geographic versus projected axis order after importing.

Complete measurements

Supply valid X, Y and Z deliberately; do not rely on a blank or default height being meaningful. State uncertainty from actual measurement quality and keep separate independent checks.

XML with observations

Supported marker/reference XML can carry photo projections, pinned flags and coordinates. Verify that photo names/IDs resolve uniquely to the intended original images. Repeated basenames or edited/resized originals can invalidate an apparently plausible link.

Merge by label

Import merges matching labels and adds new markers. Existing observations can be updated; a reused label must refer to the same physical target. Save a baseline before importing revised reference data.

Reference is not a full reconstruction

Importing coordinates does not initially align all photos from marker XYZ alone. Perform Align Photos for the image network, then place/confirm markers and Optimize Cameras. Registering selected new photos from GCPs is a separate existing-network workflow.

Move/rename originals carefully

Relocate Photos repairs missing links by matching names/trailing paths under the selected folder. Review unresolved matches and save afterward. It is not a safe way to substitute cropped/rotated/resized image content under the same filename.

Project-data copy and cleanup choices

These prompts manage storage dependencies. They are separate from cloud filtering, photo enable/disable and hiding a layer. Keep a baseline and understand which files a future rebuild will need.

Copy results only

Copy results, databases and other needed data while omitting recomputable depth maps when that choice is offered. Reduces transfer/storage cost, but later depth reuse requires maps to be regenerated.

Copy everything / Copy

Copy all offered data, including depth maps where present. Needs enough free destination disk space and time. Original non-temporary data are kept; migration of temporary unsaved-project data follows the app save flow.

Do not copy (keep using the data where it is)

Leave links to the existing data location. Faster and smaller, but moving only the new .pmsx does not create a portable complete project. Keep the referenced location reachable.

Cancel project copy

Abort the save/copy flow rather than accepting an incomplete migration. If a copy is interrupted, the app can later identify unfinished unused copies and offer cleanup.

Cleanup: Depth maps

Recomputable maps used for Reuse depth maps; selected by default when present and eligible. Removing them saves space but prevents immediate depth-map reuse.

Cleanup: Undistorted photos (dense workspace)

Working photo copies used for dense processing, not the original source photos. Recreated by Build Point Cloud. Review the category note and size before deletion.

Cleanup: Feature matching database

Feature/matching databases used for alignment work. Recomputed by a later Align Photos run. Removing intermediates can prevent reuse of associated work even when the current finished products remain.

Cleanup: Unfinished copy

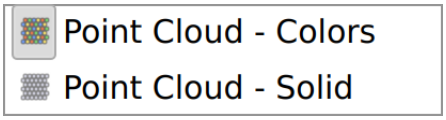
Unused leftovers from an interrupted/failed Save As may be offered separately. Confirm they are not the referenced current data. If files are in use or protected, the app reports what could not be removed.

Synced-folder warning

A cloud-synced folder can upload very large intermediates and lock files during saving/processing. Choose a local project folder when possible, or deliberately accept the displayed limitations; this is a storage warning, not an accuracy setting.

Toolbar popups and detailed controls

Toolbar popups share the corresponding menu actions. Use the primary part of a split button for visibility and its arrow for a style or setting. Small window widths can move commands into an overflow menu; those commands remain available in the main menus.



Actual interface reference: Toolbar popups and detailed controls. See the option explanations below; screenshot values are not universal recommendations.

Cloud style arrow

Colors displays recorded RGB; Solid displays a uniform appearance. The main Cloud button still controls visibility. Neither choice filters the cloud or recalculates points.

Mesh style arrow

Shaded reveals surface form with lighting; Solid uses a uniform appearance; Textured uses the current atlas; Wireframe reveals triangle connectivity. Inspect multiple styles before judging mesh quality.

Layers popup

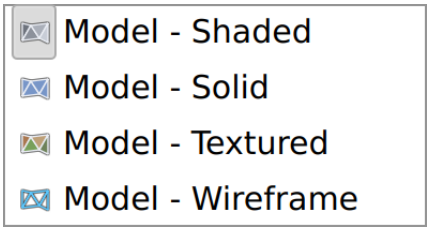
Additional tiled model, camera/marker label, trackball and projection/display controls. Tiled model display offers Textured, Solid, Wireframe and Elevation. These use the same actions documented in Display > Rendering Styles.

Surface opacity popup

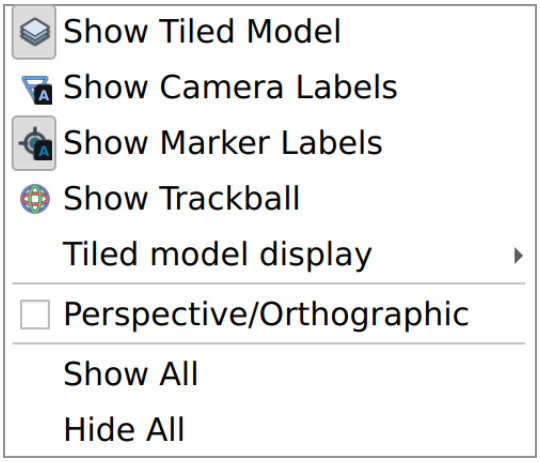
Slider range 5-100%, default 40%. Lower opacity reveals underlying shapes; 100% is opaque. It affects display while Transparent View is active and is stored between sessions. It is not a texture alpha-generation or output compression setting.

Outdated input checkbox

Use this outdated source anyway appears only when a source has a recorded stale reason. Rebuilding is recommended. Checking it explicitly permits that old source for the requested build; it does not reconcile its geometry with current cameras.



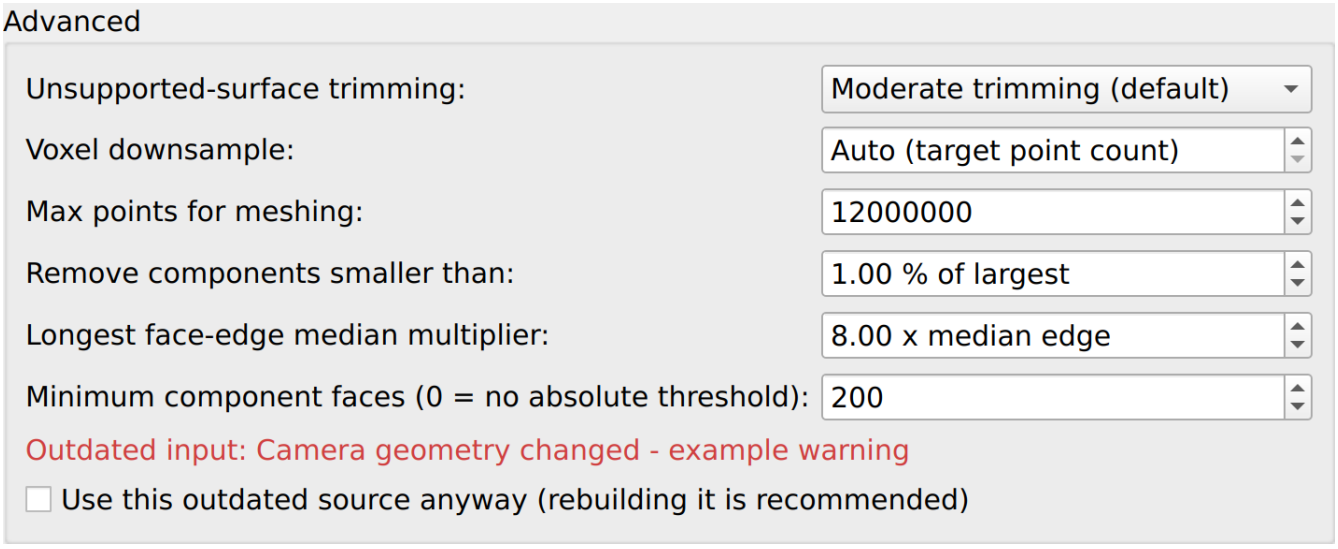
Additional interface detail: menu toolbar mesh style. The control meanings and selection guidance are explained in this chapter.



Additional interface detail: menu toolbar layers. The control meanings and selection guidance are explained in this chapter.



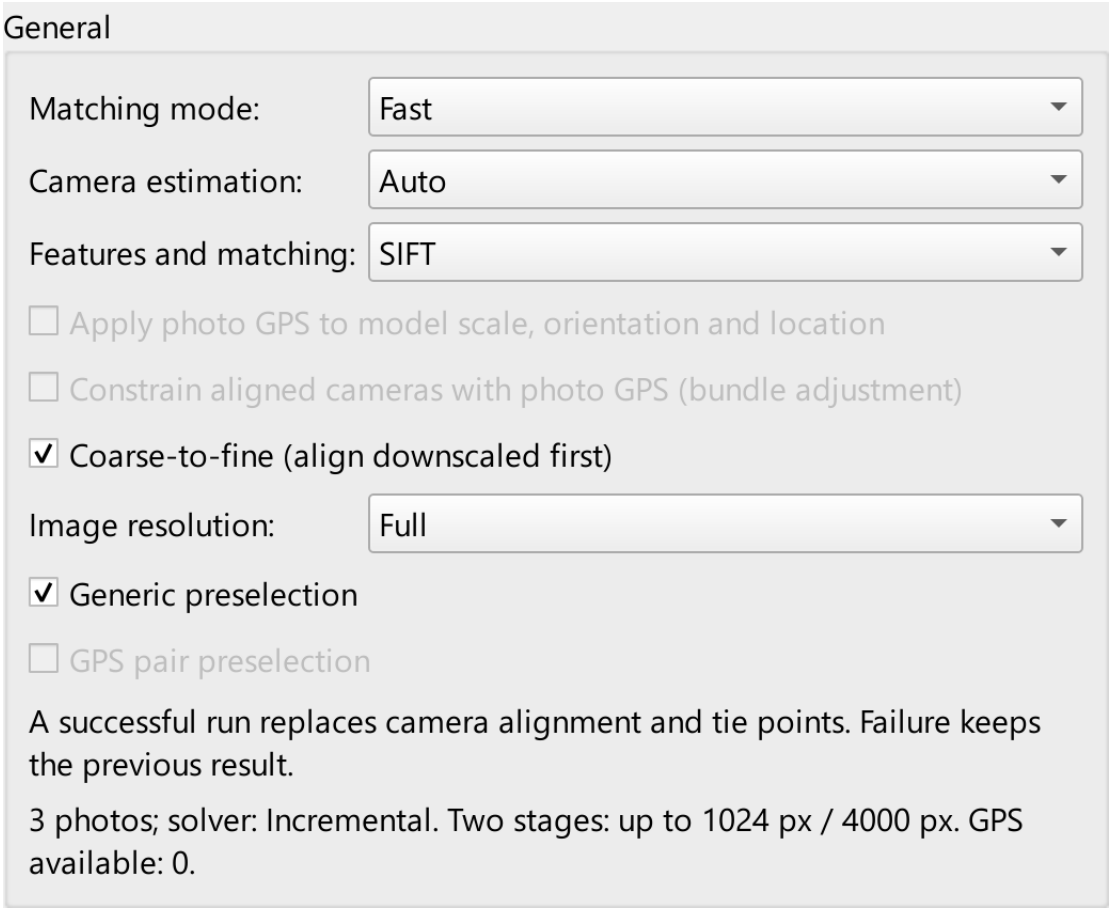
Additional interface detail: menu toolbar transparency. The control meanings and selection guidance are explained in this chapter.



Additional interface detail: advanced. The control meanings and selection guidance are explained in this chapter.

Photo GPS and calibration: new alignment controls

Image reconstruction, GPS pair selection, frame placement and weighted adjustment are separate operations. Control markers are not silently added as initial-alignment constraints.



Actual interface reference: Photo GPS and calibration: new alignment controls. See the option explanations below; screenshot values are not universal recommendations.

Constrain aligned cameras with photo GPS (bundle adjustment)

Optional camera-position constraints after image-based reconstruction. Requires the GPS frame option and project GPS policy. Leave off for unsuitable receiver positions. This differs from GPS pair preselection.

Use reported photo GPS deviations

Opt-in use of valid DJI XMP E/N/height deviations with metre units confirmed. Missing/invalid values use project fallback accuracy. RTK status alone does not establish accuracy. No vertical datum is converted automatically.

Refresh original metadata

Survey > Refresh Photo GPS Metadata rereads source metadata, including fields absent from older projects. It does not relocate the model by itself. Inspect references before optimizing.

Camera model: Engine default

Preserves the existing engine selection behavior. Review actual lens assumptions and calibration groups before overriding it.

SIMPLE_PINHOLE / PINHOLE

No distortion. SIMPLE_PINHOLE shares one focal value; PINHOLE has separate horizontal/vertical focal values. Use for genuinely corrected imagery or a known suitable calibration.

SIMPLE_RADIAL / RADIAL

One or two radial coefficients with a shared focal value. Restrained models for a perspective lens when coverage supports estimation.

OPENCV

Separate focal values and radial/tangential distortion for a compatible perspective lens. More terms need stronger image geometry.

OPENCV_FISHEYE

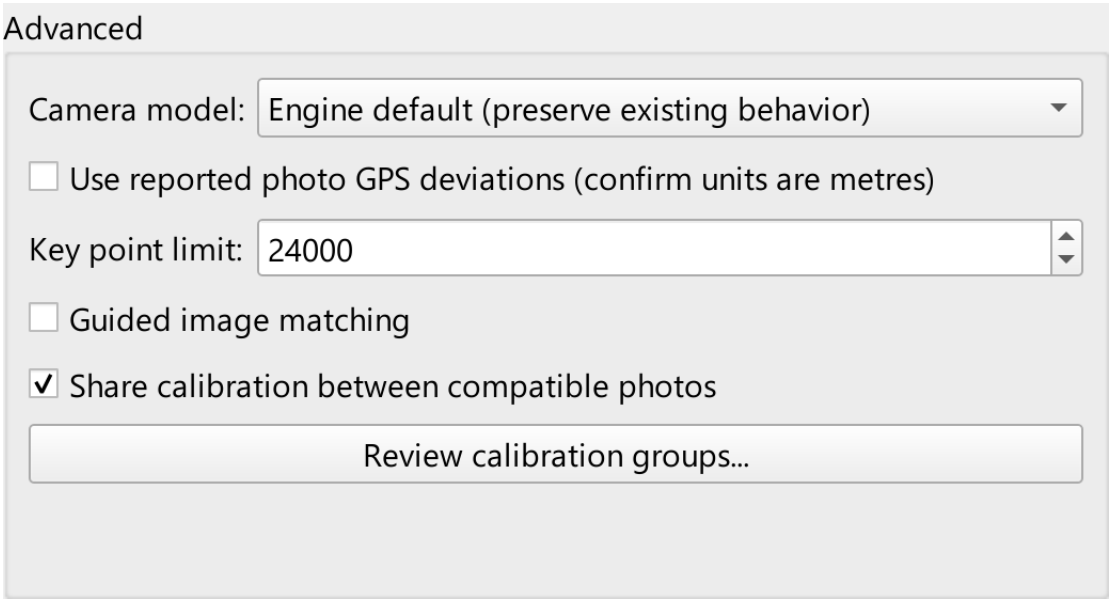
Fisheye projection and supported distortion terms. Choose for a fisheye lens, not merely because a wide-angle lens looks distorted.

FULL_OPENCV

A larger perspective distortion model. Additional coefficients need suitable geometry and calibration; they do not guarantee better accuracy.

Neural methods and GPU

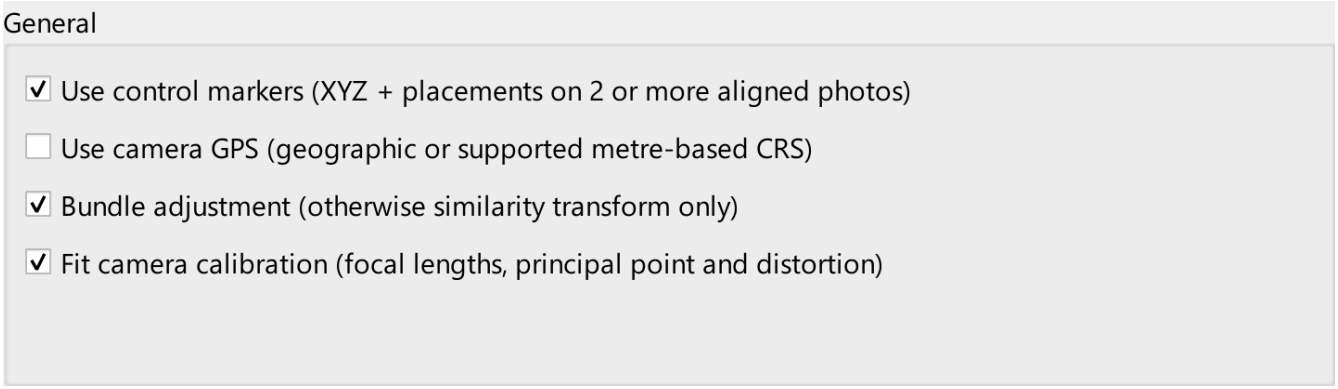
A native compatibility probe runs before survey processing. Failed GPU probing retries the same method on CPU, without silently switching to SIFT. Dense CUDA and neural GPU support are separate capabilities. CPU neural matching can be slow.



Additional interface detail: advanced. The control meanings and selection guidance are explained in this chapter.

Adjustment frames, height references and fitted calibration

Use independently measured coordinates. Confirm horizontal units and both height references before combining GPS and controls. Small control residuals alone cannot validate absolute accuracy.



Actual interface reference: Adjustment frames, height references and fitted calibration. See the option explanations below; screenshot values are not universal recommendations.

Projected / existing Cartesian frame

Adjust in the supported projected/existing Cartesian frame. Turning off a reference does not erase a frame already established by an earlier operation.

WGS84 local ENU

Local east/north/up tangent coordinates requiring confirmed WGS84 ellipsoidal heights. Up is relative to the tangent origin, not orthometric elevation. Export retains the actual local frame.

Unknown / not confirmed

No verified height convention. Resolve it before combining reference heights. Unsupported/incompatible mixed-reference adjustments are rejected rather than converted by guesswork.

WGS84 ellipsoidal

Heights relative to the WGS84 ellipsoid. Select only when confirmed by the measurement source; this label does not request conversion.

Orthometric / geoid-based

Geoid-based heights. Confirm the actual vertical datum; no geoid correction is downloaded or applied automatically.

Local height datum

Local measured heights for a compatible local/project workflow. This does not automatically make ellipsoidal photo GPS compatible.

Fit calibration parameters: f

Fit supported focal parameters only and hold the other terms fixed.

f,cx,cy,k1

Fit supported focal, principal-point and first radial terms. Unsupported parameters are not invented.

f,cx,cy,k1-k3

Also fit radial terms through k3 where supported. Requires sufficient geometry; other terms stay fixed.

f,cx,cy,k1-k3,p1,p2

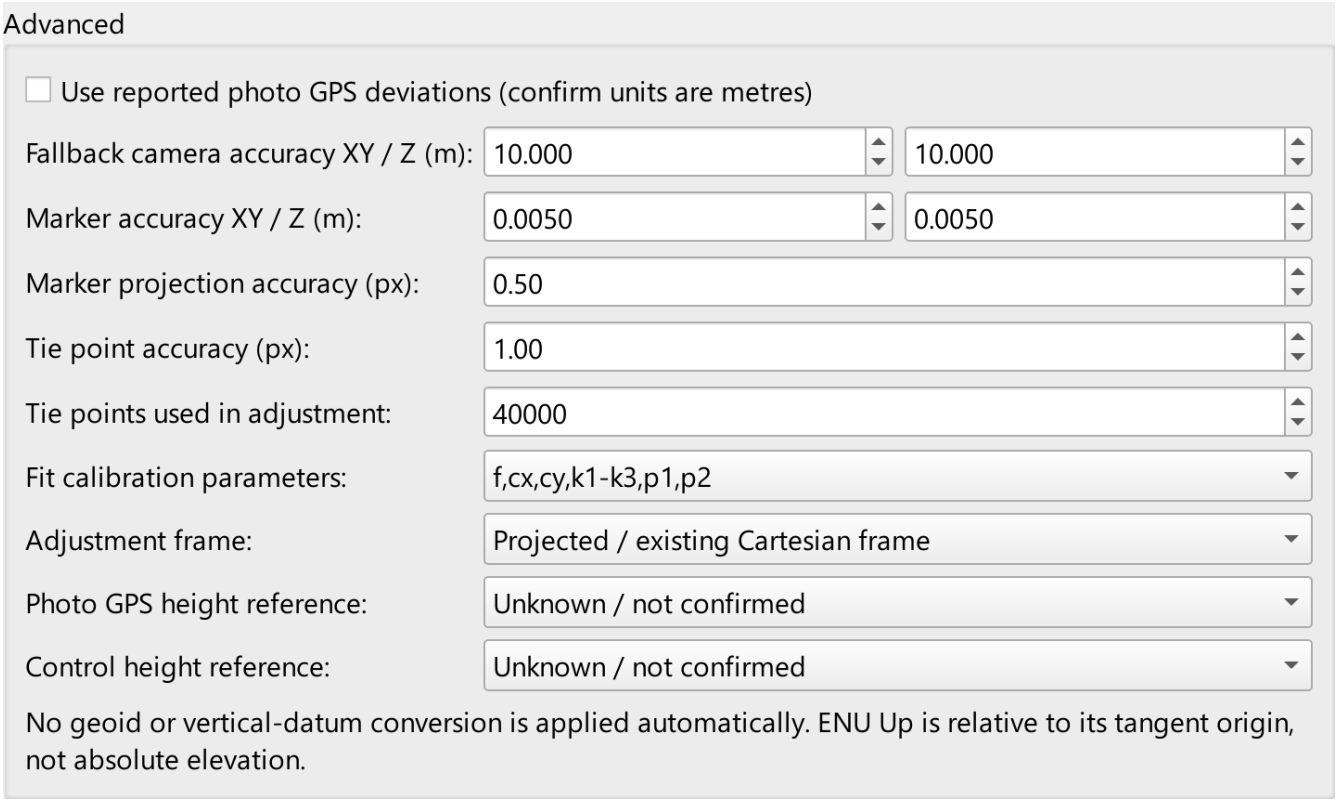
Also fit supported tangential terms. Coefficients outside the named set remain fixed.

All supported coefficients

Fit every implemented coefficient of the existing camera model. Broad fitting can overfit weak geometry; compare independent checks and plausible calibration.

Reported / fallback GPS uncertainty

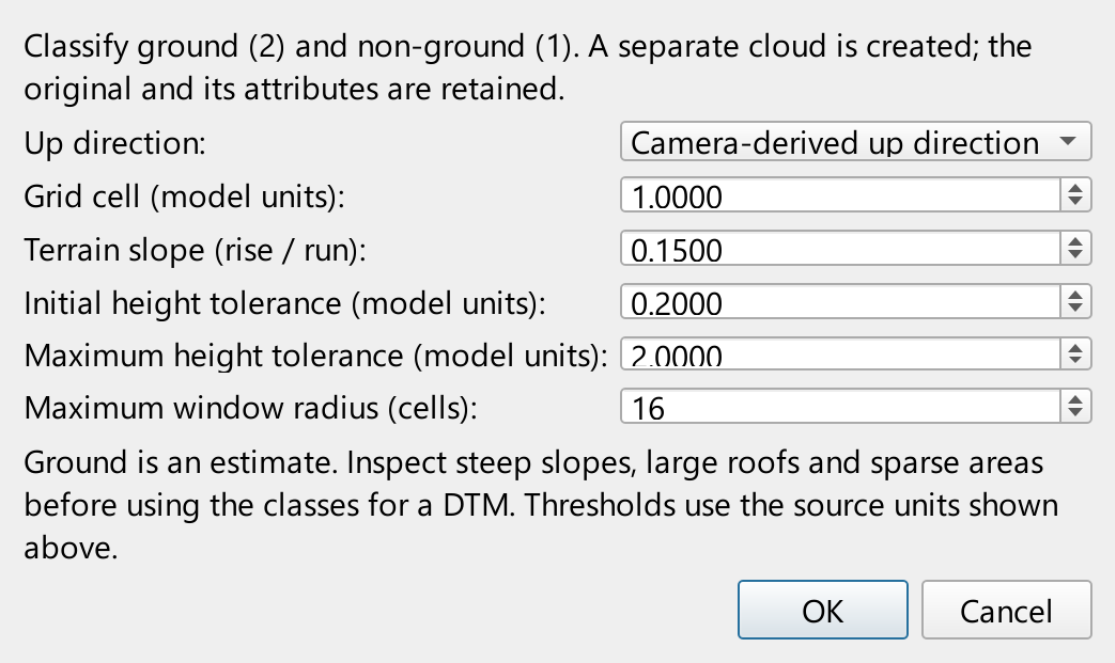
Valid per-photo deviations override fallback XY/Z uncertainty only when explicitly enabled. Smaller uncertainty gives stronger influence, not assured output accuracy.



Additional interface detail: advanced. The control meanings and selection guidance are explained in this chapter.

Point Cloud Classification: every setting

Process > Point Cloud Classification uses a progressive morphological lower envelope to estimate ground (2) and non-ground (1). It creates a separate cloud and retains the original. It does not automatically separate buildings, trees and other non-ground categories.



Actual interface reference: Point Cloud Classification: every setting. See the option explanations below; screenshot values are not universal recommendations.

Up direction: Project Z axis

Use a known vertical source axis. Georeferenced data uses project Z. An incorrect up direction changes the meaning of ground.

Up direction: Camera-derived up direction

Available for local reconstructions with aligned cameras. Estimates up from camera geometry; check orientation for interiors and oblique surveys.

Grid cell

Horizontal grid spacing in displayed source units. Smaller cells retain detail but need sufficient density/memory; larger cells smooth the envelope.

Terrain slope (rise / run)

Terrain inclination as a ratio, not degrees. Larger values tolerate steeper ground but may admit elevated objects.

Initial height tolerance

Permitted small-scale separation from the initial envelope. Too small can reject uneven ground; too large can admit low objects.

Maximum height tolerance

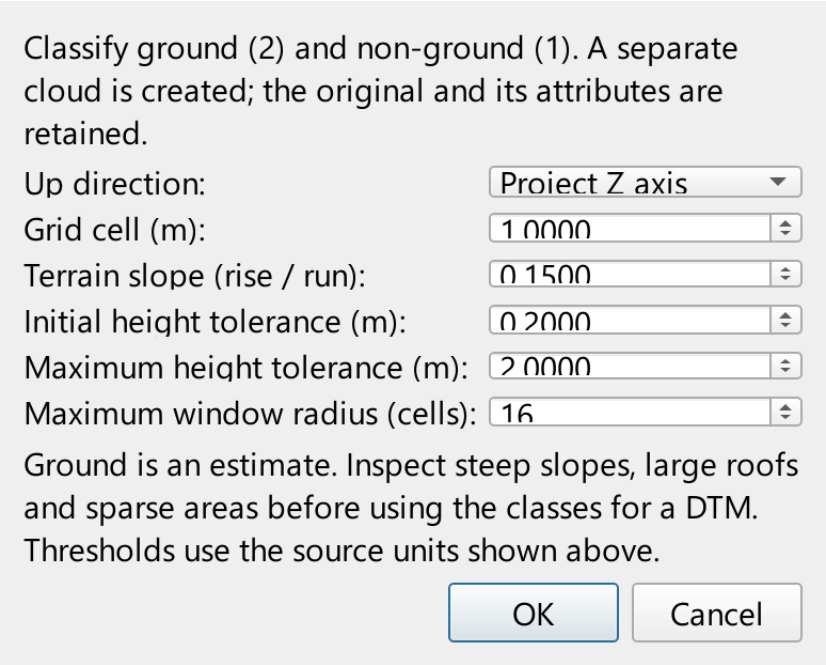
Upper permitted separation as the window grows. Choose for the terrain/object scale, not a desired retained point count.

Maximum window radius (cells)

Largest morphological radius. Physical reach depends on cell size and should span relevant objects. Large roofs, cliffs and sparse coverage can defeat the terrain assumption.

Inspect and compare

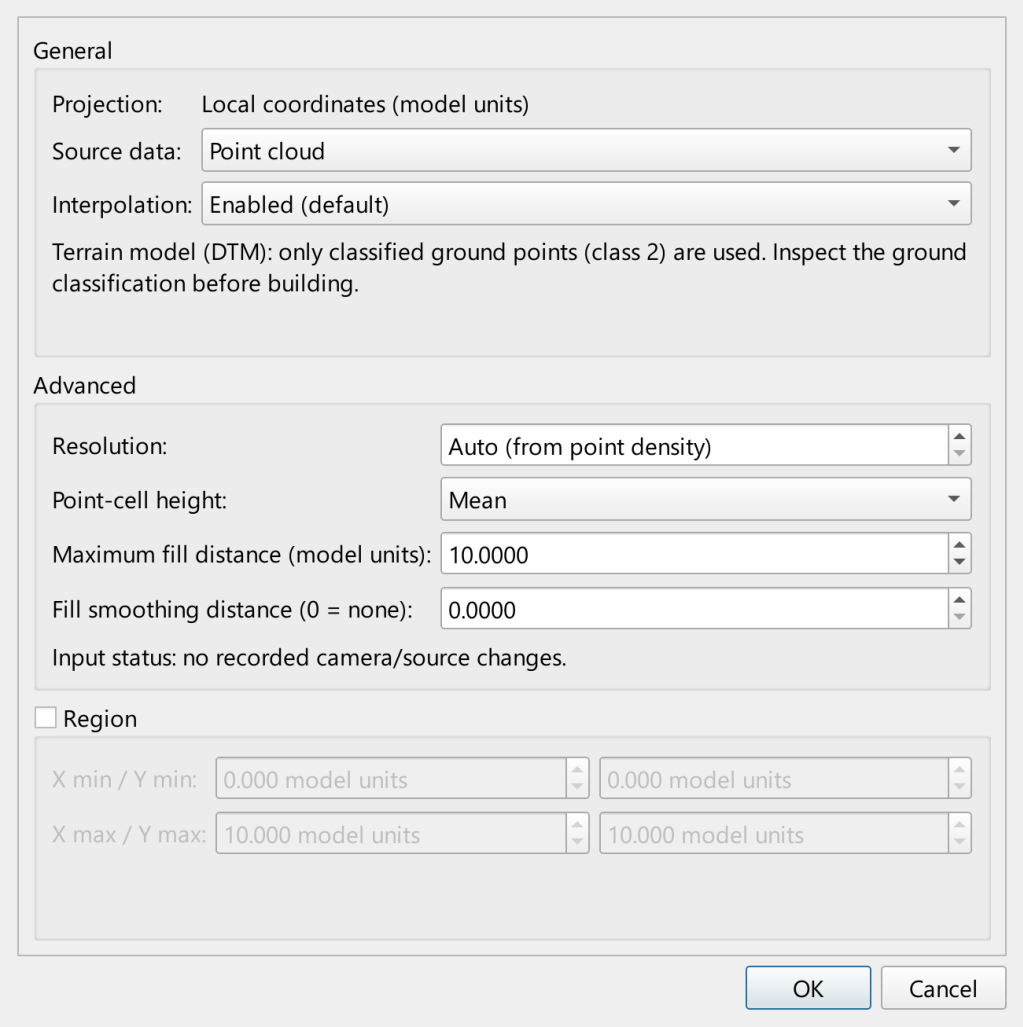
Classification colours and Show Ground / Show Non-ground / Show All change display only. Inspect terrain errors before DTM generation. Reclassification marks an older DTM stale.



Additional interface detail: classification georeferenced. The control meanings and selection guidance are explained in this chapter.

Build DTM and export ground products

DTM uses classified ground points only. Build DEM retains the DSM workflow, including buildings/vegetation. They are stored separately. Classification is an estimate, not a survey accuracy certificate.



Actual interface reference: Build DTM and export ground products. See the option explanations below; screenshot values are not universal recommendations.

Source data

Build/classify a cloud first. DTM uses class-2 points, not a mesh or the entire cloud. Without ground classes, a ground-only product cannot be built.

Resolution / Output Region

Choose spacing and extent with the shared raster controls. Auto uses applicable source density. Finer spacing does not add measured ground between observations.

Point-cell height: Mean

Mean of contributing ground heights per cell; a useful default for sufficiently dense, correctly classified ground.

Point-cell height: Median

Median of ground heights per cell, more resistant to isolated extremes. Coverage and classification errors still matter.

Point-cell height: Highest

Highest contributing ground height per cell. This statistic does not correct roofs wrongly classified as ground.

Interpolation: Disabled

Leave unsupported cells empty. Useful for inspecting measured coverage without inferring a surface through gaps.

Interpolation: Enabled

Fill eligible gaps within the allowed distance. Filled cells are inferred rather than measured terrain.

Interpolation: Extrapolated

Allow inference beyond measured coverage. A visually continuous surface does not establish observations in missing areas.

Maximum fill distance / Fill smoothing distance

Limit gap filling in displayed units where applicable. Zero smoothing keeps fill unsmoothed. These options do not repair classification errors.

Display and export

Successful builds activate their result. DTM - Elevation and Export > DTM use the terrain raster; DEM remains the DSM. Check CRS, actual frame, units and nodata in receiving software.

LAS and LAZ export: precision and attributes

Export > Point Cloud offers PLY, DXF, LAS and compressed LAZ. Interchange format and storage precision do not change reconstruction accuracy.

LAS / Compressed LAZ

LAS is uncompressed; LAZ losslessly compresses the encoded LAS records. Both preserve supported source attributes/classes. Export is cancellable and processed in bounded blocks.

Coordinate quantization

Coordinate storage step in displayed source units, initially 0.001. Smaller increments must fit the LAS integer range; unsupported combinations are rejected. Storage precision does not prove measurement accuracy.

Coordinates and CRS

The exporter applies the actual offset/frame. Local/ENU Cartesian results must not be labelled as ordinary projected eastings/northings. Inspect horizontal and vertical units independently.

Classes and colours

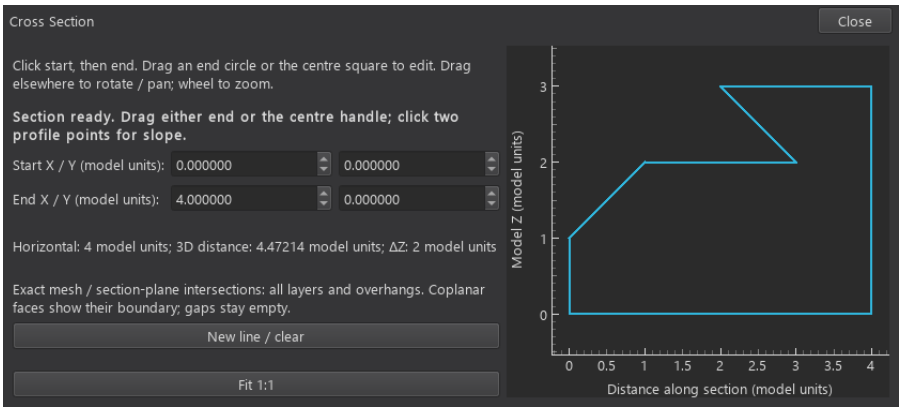
Supported attributes/classes are retained when present. Ground/non-ground estimates do not acquire building/vegetation categories by export.

Verify delivery

Reopen the export in compatible software; compare coordinates, extent, point count, classes and colours. Retain the source project and original images.

Cross Section: drawing, handles and measurement

Survey > Cross Section opens a closable bottom panel combining distance and mesh profile. The separate measurement command and Pick Start / Pick End buttons are removed.



Actual interface reference: Cross Section: drawing, handles and measurement. See the option explanations below; screenshot values are not universal recommendations.

Click start, preview, click end

Click the start, move the mouse to preview, then click the endpoint to calculate. Heavy profile geometry is not recalculated for every preview movement.

Navigate while drawing

Drag away from handles to rotate/pan and use the wheel to zoom. Navigation remains available during drawing/editing.

Endpoint circles

Drag either endpoint to change that end. Release to commit and recalculate.

Centre square

Drag the middle handle to translate both endpoints together. In 3D it sits on the actual section surface near the middle. Release to recalculate; the section line stays on the mesh while rotating the view.

Start X/Y / End X/Y

Enter coordinates manually in displayed units. An unscaled local model cannot provide assured metre distances.

3D View / orthophoto / photo

Requires a current mesh; photos additionally need alignment/calibration and mesh intersection. Orthophotos need a compatible mesh/frame relationship. Missing geometry cannot provide reliable section endpoints.

Orthophoto warning

Follow the specific diagnostic. Rebuild a genuinely stale/incompatible raster from the current mesh where requested. Do not force an arbitrary imported raster to use an unrelated frame.

Profile and length

Shows exact intersections of the mesh with the vertical section plane between the endpoints. All intersected layers, floors, vertical walls and overhangs are retained. Coplanar faces show their boundary; mesh gaps stay empty. Read distance/elevation units; a reconstructed surface does not establish survey accuracy.

Fit 1:1

Fit the entire profile with equal horizontal and vertical scale: one distance unit and one elevation unit occupy the same screen length. This is an axis ratio, not a paper printing scale. The equal-scale lock remains active when navigating the plot.

Slope between profile points

Click two points on the profile to read their slope. Picks follow the nearest actual segment, including lower layers. Separate disconnected components cannot be joined for a slope measurement.

Busy / Cancel calculation

Large-mesh calculation displays busy feedback and blocks competing line edits. Cancel stops pending work. Closing restores normal interactions.

New line / clear / Close

Clear starts a new line; Close dismisses the tool. Photo context/marker edits are blocked while the panel is open to prevent accidental control changes, then restored.

Observed progress, completed products and offline help

Progress uses reported work. Stages have different costs, so percentage is not a promise of elapsed-time proportion.

Known / unknown work

Completed files/views/items advance available counters. Unknown totals remain indeterminate; unknown inner work is not shown as a second precise percentage.

Remaining time

ETA needs enough stable measured samples. It can disappear after a stage/rate change or pause. Short tasks may have no ETA rather than an invented estimate.

Completion and activation

Successful tie-point, cloud, mesh and raster builds activate their result. Failed, cancelled or stale results are not published into another active project.

User Manual (PDF)

Help opens this complete bundled reference offline through the system PDF reader. Documentation and third-party notices have separate entries.

Point selection/deletion

Direct tie-point/cloud selection and deletion is deferred to 1.0.6. Class visibility filters are not deletion.

Final delivery checklist

Geometry

The cameras and surface are coherent, coverage is adequate, and missing/inferred regions are understood.

Reference

Coordinate system, axis order, scale and vertical reference are verified. Independent check information is distinguished from fitted control residuals.

Current products

No unintended stale cloud, mesh, texture or raster is being delivered after camera/marker changes.

Export package

The intended formats were reopened in the destination software. Texture, material, projection, world-file and tile companions remain together.

Project archive

The .pmsx, required .files products and reachable original photos are preserved. Important settings and limitations are recorded.

Licence

Use complies with the application and third-party licences; commercial/company-evaluation use requires the applicable prior written permission.

Version 1.0.5 release

Application source for the 1.0.5 release candidate